# Testing SWINGERsim

Part I: Validation against the literature

2026-11-06

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**Abstract.** SWINGERsim simulates hierarchical multiple stars – triples, quadruples and larger nested systems – together with the evolution of their stars, tides, mass transfer, eclipses, light curves, eclipse timings, fits to observations and ensembles of possible systems. This first of four test reports asks whether every physical module of SWINGERsim reproduces independent published results. The evidence has four levels: analytic results (A), published numerical results (B), other codes run on the same input (C) and observed systems (D). SWINGERsim carries 219 registered benchmarks, each with its source, its

expected value and a tolerance fixed before the comparison. With the final code all 219 were run: 205 lie within their tolerance and 14 lie outside it for a documented reason that is not an error of SWINGERsim (a slip or an inconsistency in the paper, a different definition, the length of the published runs, or a limitation shared by all published models). No benchmark fails. Three quarters of the benchmarks (165 of 219) test code written for SWINGERsim itself – the secular backends, the coupling of stellar evolution, tides and mass transfer to the orbits, the light curves on spheres and surface meshes, the eclipse timings, the fits and the ensembles – and not only the external N-body integrator it uses. Beyond the benchmarks, SWINGERsim reproduces the published model of the massive quadruple QZ Car, the dynamics of the SQUADRA candidates CzeV4315 and ASASSN-V J075203, 23 published multiple stars across the Hertzsprung–Russell diagram and the stability scores of a published neural classifier. The report ends with the problems met during testing, written as properties of the software that a user has to know, and with a list of what is validated, to which accuracy, and what is not.

# Chapter 1

# Introduction

## 1.1 Aim of this report

A simulation code is useful to astronomers only if its physics is right. That a code runs without errors, or that it agrees with itself, proves little: a wrong force can conserve energy perfectly. The aim of this report is therefore narrow and strict. For every physical effect in SWINGERsim we ask: **does SWINGERsim reproduce an independent, published result for it, and to what accuracy?** Where it does not, we ask why, and we trace the difference to its source.

Part I covers the physics modules one by one (Chapters 2–4), then whole published systems: the massive quadruple QZ Car (Chapter 5), the SQUADRA candidates (Chapter 6), a catalogue of 23 published multiple stars (Chapter 7), examples that follow a multiple star through a phase of its life (Chapter 8) and the stability tools (Chapter 9). Chapter 10 lists the problems that the tests revealed and how SWINGERsim handles them, Chapter 11 summarises what is validated and what is not, and Appendix A lists all 219 benchmarks with their results.

## 1.2 What SWINGERsim is, and which parts are its own code

SWINGERsim follows hierarchical multiple stars: systems of three to about ten stars in which close pairs orbit each other and the pairs orbit common centres of mass on wider orbits. It integrates the orbits with one of three **backends** (the engines that advance the system in time):

- the **N-body backend** integrates the equations of motion of all stars directly. It uses the integrator IAS15 of the REBOUND library (Rein & Liu 2012; Rein & Spiegel 2015) and three effects of the REBOUNDx library (Tamayo et al. 2020): `gr_full` (first post-Newtonian relativity for all bodies), `tides_spin` (the equilibrium tide with spins, Eggleton et al. 1998; Lu et al. 2023) and `modify_mass` (masses that change with time);
- the **secular backends** average the motion over the orbital periods and follow only the slow changes of the orbits (precession, ZLK cycles, defined in Chapter 2). They are written for SWINGERsim: a quadrupole–octupole code for triples, a 2+2 code after Vokrouhlický (2016) and a nested code for any hierarchy after Hamers et al. (2015) and Hamers & Portegies Zwart (2016);
- the **evolving backend** couples a secular backend to evolving stars: it cuts the run into segments within which no stellar radius changes by more than 1 % and applies winds, mass transfer, common envelopes, tides and magnetic braking between them. It is written for SWINGERsim.

The question “which code does a benchmark test?” matters for the weight of the evidence. A benchmark of the N-body backend that reproduces, say, the relativistic perihelion advance

of Mercury mainly confirms that REBOUNDx is used correctly – the force law itself is REBOUNDx's. A benchmark of the secular backend, of the mass-transfer coupling or of the light-curve model tests equations and numerics that exist only in SWINGERsim; it is the stronger evidence, because nothing outside SWINGERsim could have made it pass. We read the source code (`src/swingersim/`) and classify every benchmark into one of four classes (Table 1). The class is given for every benchmark in the tables of Chapters 2–9 and in Appendix A.

**Table 1.** Classes of benchmarks by the code they test. Column 1 is the label used in all later tables; column 2 says which code computes the tested number; column 3 lists the main modules; the last column gives how many of the 219 benchmarks belong to the class.

| Class | What computes the tested number | Modules (examples) | Benchmarks |
|---|---|---|---|
| own | equations and numerics written for SWINGERsim | secular backends, evolving backend, mass transfer and common envelope, tides and apsidal formulae, inertial waves, magnetic braking, Roche geometry and surface meshes, light curves, dust discs, eclipse timings, pulsation timing, fits of minima, ensembles, stability criteria, escape theory, gravitational waves, Galactic tide and passing stars | 126 |
| own/tab | SWINGERsim's interpolation and use of published tables | MIST and SSE tracks, Claret (2004, 2023) structure constants, Claret & Bloemen (2011) and TLUSTY limb darkening | 39 |
| R | REBOUND IAS15 with REBOUNDx effects, driven by SWINGERsim's set-up | hierarchical initialisation, element conversion, collision and hierarchy detection, mass-loss schedule, time-lag convention, operator splitting of SWINGERsim's own forces | 50 |
| ext | an external library through a SWINGERsim wrapper | PHOEBE 2.5 hybrid light curves, the `sbi` package, the scikit-learn neural network trained on SWINGERsim's own training sets | 4 |

Even an "R" benchmark is not a test of REBOUND alone: SWINGERsim builds the initial positions and velocities from the hierarchical orbital elements of the configuration, converts the integrated state back to elements, detects collisions and changes of the hierarchy, and sets the parameters of the REBOUNDx effects (for example the factor between the time lag of SWINGERsim and the `tau` of `tides_spin`). A benchmark of class R therefore tests this set-up and the external integrator together. Where a benchmark compares a SWINGERsim backend with an N-body run of the same system (for example the secular backend against N-body), the class names the tested side, which is SWINGERsim's own code.

Of the 219 benchmarks, 165 (75 %) test SWINGERsim's own code (classes own and own/tab), 50 the N-body set-up around REBOUND, and 4 an external library through a wrapper.

## 1.3 The validation rule

Every physical effect in SWINGERsim satisfies five rules.

1. **An independent reference.** The effect is checked against a result that SWINGERsim did not produce: an equation from the literature, a figure or table of a paper, another code, or an observed system. Internal checks (round trips of coordinate conversions,

energy conservation) are part of the automatic tests too, but they are not counted as validation.

2. **A recorded source.** Each benchmark stores the paper and the equation, figure or table it reproduces, the input values taken from it, the expected value and the tolerance. The command `swingersim validate` prints the source next to every result.
3. **A physical tolerance.** The tolerance follows from the approximation behind the reference (for example the neglected order of a series, the accuracy of a fitting formula, the error of a measurement or the statistical error of an ensemble). It is fixed before the comparison and never widened to make a benchmark pass.
4. **Every difference is explained or it is an error.** A result outside its tolerance is either an error of SWINGERsim or it has a documented reason. The reason is stored with the benchmark and printed with the result.
5. **Results are reproducible.** Every benchmark runs with one command, and its number in this report is the number printed by that command with the final code.

## 1.4 Levels of evidence and verdicts

The references fall into four **levels of evidence** (Table 2). The levels answer different questions. Level A tests the physics in limits in which the exact answer is known. Levels B and C test regimes in which no formula exists – chaotic dynamics, coupled evolution of stars and orbits. Level D tests whether the model as a whole describes real stars, including everything that the model leaves out.

**Table 2.** Levels of evidence. Column 2 says what the reference is, column 3 gives typical examples, column 4 the number of benchmarks of that level.

| Level | Reference | Examples | Benchmarks |
|---|---|---|---|
| A | analytic result from the literature | ZLK maximum eccentricity, relativistic perihelion advance, Sterne's apsidal formula, Roche-lobe radius, closed forms of mass transfer | 93 |
| B | published numerical result (a figure or table of a paper) | orbital flips of Naoz et al. (2013), mutual inclinations of Vokrouhlický (2016), scattering cross-sections of Antognini & Thompson (2016) | 60 |
| C | another code on the same input | MSE, BSE in COSMIC, PHOEBE 2.5, batman, ellc, the Xitau model of QZ Car, Geneva Cepheid models | 31 |
| D | observed system | V994 Her, CzeV343, QZ Car, the Sun, M67, white dwarfs, PSR B1913+16, ε Aur, HD 98800 | 35 |

Every benchmark ends with one of four **verdicts**:

- **PASS** – the result lies within the tolerance;
- **EXPLAINED** – the result lies outside the tolerance for a documented reason that is not an error of SWINGERsim: the input or the definition used by the paper, published numbers that contradict each other, the length or sampling of the published runs, or a stated limitation of SWINGERsim's physical model that is an open point rather than a coding error. The tolerance is not changed to hide the difference;
- **FAIL** – outside the tolerance without such a reason; this would be an error of SWINGERsim;
- **SKIP** – an optional package that the benchmark needs is not installed.

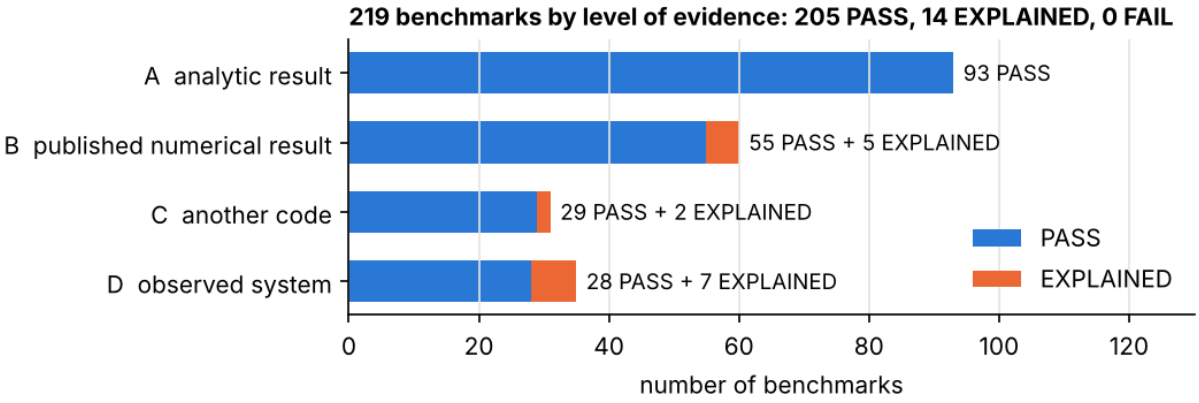



**Figure 1.** The 219 benchmarks of SWINGERsim by level of evidence (rows A–D, see Table 2) with the final code. The length of a bar is the number of benchmarks of that level; the blue part counts PASS, the orange part EXPLAINED. The numbers at the end of each bar repeat the counts. There is no FAIL and no SKIP.

Figure 1 summarises the result. All 93 analytic benchmarks (level A) pass. The 14 explained differences lie in levels B (5), C (2) and D (7) – that is, where the reference is a published simulation, another code or a real star. This is what one expects if SWINGERsim's physics is right: an analytic limit has no input to disagree about, whereas a paper can quote a number from a different definition, and a real star contains physics that no model has. Each of the 14 is discussed where its physics is treated, and they are collected in Chapter 11.

## 1.5 Where the numbers come from

The benchmarks are part of the software. `swingersim validate` runs the fast ones (seconds each), `swingersim validate --all` adds the slow ones (minutes each, for example ensembles of hundreds of scattering experiments) and `swingersim validate <id>` runs one benchmark by name. Nine *manual* benchmarks run only when named: they need hours of computing (the N-body ensemble of Pejcha et al. 2013), training sets of $10^4$ N-body runs (the stability classifier of Vynatheya et al. 2023), or an optional package (PHOEBE, `sbi`).

```
swingersim validate                 # the 184 fast benchmarks
swingersim validate --all           # all 210 automatic benchmarks
swingersim validate gr-mercury      # one benchmark by name
swingersim validate pejcha13-F6-fA  # a manual benchmark (hours), runs only when named
```

Every number in this report was computed with the final code of SWINGERsim: the fast benchmarks in the last run of the suite, the slow benchmarks in a full run on the production machine (a 12-core workstation), and the manual benchmarks in separate runs of the same code. The command also writes an HTML report with the table of all results and the reasons of the EXPLAINED ones.

The studies of whole systems (Chapters 5–9) are simulations with the configuration files of the SWINGERsim distribution; their numbers come from the outputs of these runs. Where a study is an ensemble, its size is given with the result.

## 1.6 How to read this report

Each section of Chapters 2–4 follows the same order: the physical effect and its reference, the input, the result (a table, often a figure), and the agreement stated as a number. The tables of benchmarks have the same columns throughout:

- **Benchmark** – the identifier by which `swingersim validate` runs it;
- **Class** – which code it tests (Table 1);

- **Expected** – the reference value (analytic, published or from another code);
- **Computed** – the value SWINGERsim computes;
- **Difference** – SWINGERsim minus expected, either relative (in %) or absolute (in the unit of the quantity), as defined by the benchmark;
- **Tolerance** – the largest difference that counts as agreement, in the same form;
- **Verdict** – PASS or EXPLAINED.

Terms are defined at first use. The notation follows the other SWINGERsim documents: $P$ is a period, $a$ a semi-major axis, $e$ an eccentricity, $i$ an inclination to the sky, $i_{\mathrm{mut}}$ the mutual inclination of two orbits, $\omega$ the argument of periastron and $\Omega$ the longitude of the node. The orbits of a 2+2 quadruple are called A and B (the two pairs) and "out" (their mutual orbit); in a 3+1 they are "inner", "mid" and "out". The equations behind each module are derived in *SWINGERsim – Physics and Methods*; this report quotes only what is needed to judge a comparison.

## 1.7 Parts II–IV

Part I asks whether the physics of SWINGERsim reproduces the literature. The other three parts ask whether SWINGERsim, used as an astronomer would use it, reproduces real observations of systems studied by the SQUADRA group:

- **Part II: Three published multiple stars** – the 2+2 quadruple BU CMi, the 3+1 quadruple TIC 120362137 and the sextuple V994 Her: their published solutions integrated forward and backward, synthetic light curves and eclipse times against TESS and ground-based data, and joint fits of all data;
- **Part III: Eclipsing binaries with a changing inclination** – V907 Sco, QX Cas and AH Cep, whose eclipses appear, deepen or vanish because a third body tilts their orbits;
- **Part IV: Special cases** – the 2+2 quadruples CzeV343 and CzeV4315, which test the limits of the models (resonances, unstable configurations, chaotic members of ensembles).

# Chapter 2

# Dynamics: orbits, secular evolution and stability

This chapter tests how SWINGERsim moves the stars: the N-body backend with relativity, the secular backends, the detection of instability, and the reproduction of four classic studies of hierarchical systems. It also tests hierarchies of more than four stars, scattering experiments and the influence of the environment (gas, the Galactic tide, passing stars, gravitational waves).

Two effects recur in the whole report. **Apsidal precession** is the slow turning of an elliptical orbit in its own plane (the argument of periastron $\omega$ grows); its causes are the other stars, general relativity and the tidal and rotational bulges of the stars. **Nodal precession** is the turning of the orbital plane around the total angular momentum of the system; it changes the inclination to the sky and so makes eclipses appear and disappear. When the mutual inclination of an inner and an outer orbit is large (roughly 40°–140°), the outer body exchanges angular momentum with the inner orbit and pumps its eccentricity up and down periodically: the **von Zeipel–Lidov–Kozai (ZLK) cycles** (Kozai 1962; Lidov 1962; Naoz 2016). The characteristic time of these cycles is the **Kozai time** $t_K$, of order $P_{\rm out}^2/P_{\rm in}$ times a mass ratio.

## 2.1 Relativity, ZLK cycles and nodal precession in the N-body backend

**Reference and input.** Three analytic limits test the N-body backend. (i) The first post-Newtonian (1PN) perihelion advance of Mercury, 42.98″ per century for the real orbit ($a$ = 0.387 au, $e$ = 0.206), one of the classic tests of general relativity (Will 2014). (ii) The ZLK cycles of a test particle at 1 au around a 1 $M_\odot$ star, perturbed by a 1 $M_\odot$ star on a circular orbit at 20 au. In this quadrupole, test-particle limit the theory gives $e_{\rm max} = \sqrt{1 - \frac{5}{3}\cos^2 i_0}$ for an initial mutual inclination $i_0$, a mutual inclination of 39.23° at the eccentricity maximum, no eccentricity growth below 39.2°, and an exactly conserved quantity $\sqrt{1-e^2}\cos i$ (Kozai 1962; Lidov 1962; Naoz 2016). (iii) The nodal precession period of the same set-up at $i$ = 20° from the quadrupole formula of Söderhjelm (1975). The secular backends are tested in the same limits, and in addition on the relativistic apsidal period of an isolated binary ($a$ = 0.05 au, $e$ = 0.3).

**Table 3.** Relativity, ZLK cycles and precession: N-body backend (class R) and secular backends (class own). Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `gr-mercury` | R | 42.98″/century | 42.9803″/century | +0.00 % | 0.5 % | PASS |

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `zlk-emax-65` | R | 0.838047 | 0.84023 | +0.26 % | 1 % | PASS |
| `zlk-emax-80` | R | 0.974548 | 0.976155 | +0.16 % | 1 % | PASS |
| `zlk-imin-65` | R | 39.2315° | 39.3441° | +0.113° | 0.5° | PASS |
| `zlk-below-critical` | R | 0 | 0.00173037 | +0.00173 | 0.01 | PASS |
| `zlk-kozai-constant` | R | 0 | $5.16811{\cdot}10^{-5}$ | $+5.17{\cdot}10^{-5}$ | 0.001 | PASS |
| `nodal-precession` | R | 11351.4 yr | 11339.8 yr | −0.10 % | 1 % | PASS |
| `zlk-period-nbody` | own | 0 | 0.00660493 | +0.0066 | 0.02 | PASS |
| `quad-tp-emax` | own | 0.838047 | 0.838082 | +0.00 % | 0.1 % | PASS |
| `secular-kozai-emax` | own | 0.838047 | 0.838082 | +0.00 % | 0.5 % | PASS |
| `secular-3p1-kozai-emax` | own | 0.838047 | 0.838082 | +0.00 % | 0.5 % | PASS |
| `secular-nodal-precession` | own | 11351.4 yr | 11351.4 yr | −0.00 % | 0.5 % | PASS |
| `secular-gr-apsidal` | own | 7113.77 yr | 7113.77 yr | −0.00 % | 0.5 % | PASS |
| `brouwer-law` | R | 0.5 | 0.59301 | +0.093 | 0.15 | PASS |

**Result.** Table 3 lists the comparisons. The N-body backend gives Mercury's advance to better than $10^{-4}$; with relativity switched off it gives 0″, so the benchmark detects missing physics. The ZLK maxima of the N-body runs lie 0.16–0.26 % above the formula. This small excess is expected and has the right sign: the formula neglects terms of order $a_{\rm in}/a_{\rm out}$ (here 1/20), and the N-body orbit starts at $e = 0.01$ rather than exactly zero. The quantity $\sqrt{1-e^2}\cos i$, averaged over the first and the last quarter of a 20 000-yr run, drifts by $5{\cdot}10^{-5}$ while $e$ swings between 0 and 0.84; its instantaneous value fluctuates by about 0.5 % because the elements are osculating (computed from the instantaneous positions and velocities). Below the critical inclination the eccentricity stays below 0.002. The ZLK period of the N-body run is 0.66 % longer than the quadrupole secular period. The nodal precession period agrees with Söderhjelm's formula to 0.10 %.

The three secular backends agree with the same analytic limits to better than $10^{-4}$ (columns "Difference" of 0.00 %). This is expected and is a strong test of their implementation: they solve exactly the averaged equations of these limits, without the short-period terms that make the N-body values deviate slightly. The test-particle $e_{\rm max}$ is reproduced by the triple code, by the 3+1 branch of the nested code and by the 2+2 code with the perturber replaced by a binary of negligible size.

**Numerical quality of the integrator.** The energy error of a high-order integrator with round-off errors grows like a random walk, $\propto t^{1/2}$ (Brouwer's law; Rein & Spiegel 2015). Six realisations over $3{\cdot}10^4$ orbits give a growth exponent of 0.59 (`brouwer-law`), within the tolerance of 0.15 set by the scatter of six realisations. This is a test of IAS15 as SWINGERsim uses it, not of SWINGERsim's physics.

## 2.2 Stability: hierarchy changes and chaos indicators

A hierarchical configuration is **dynamically stable** if it keeps its hierarchy (who orbits whom) for a very long time. The **Mardling–Aarseth criterion** (MA01; Mardling & Aarseth 2001) gives the smallest ratio of outer periastron to inner semi-major axis for which a triple is stable. **MEGNO** (Mean Exponential Growth factor of Nearby Orbits; Cincotta & Simó 2000) measures how fast neighbouring trajectories diverge: it tends to 2 for regular (quasi-periodic) motion and grows without bound for chaotic motion. SWINGERsim computes MEGNO with REBOUND's variational equations.

**Table 4.** Detection of instability and chaos (class R).

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `stability-unstable` | R | 1 | 1 | +0 | 0 | PASS |
| `stability-stable` | R | 0 | 0 | +0 | 0 | PASS |
| `megno-regular` | R | 2 | 1.99959 | −0.00041 | 0.05 | PASS |
| `megno-chaotic` | R | 1 | 1 | +0 | 0 | PASS |

An equal-mass coplanar 2+2 is placed at half and at 2.5 times the MA01 limit and integrated for 100 outer periods. The first must change its hierarchy (1 = yes) and does, ejecting a star after about two outer orbits; the second must not (0 = no change) and does not. The margin of a factor of two on either side is needed because MA01 was derived for triples, not for quadruples. The MEGNO of a regular hierarchical triple converges to 1.998 (expected 2), and a 2+2 inside the unstable region reaches MEGNO > 4 (1 = yes). The stability tools as a whole are tested against a published neural classifier in Chapter 9.

## 2.3 Reproduction of Naoz et al. (2013)

Naoz et al. (2013) showed that octupole-order terms – the next order beyond the quadrupole in the expansion in $a_{\rm in}/a_{\rm out}$, important when the outer orbit is eccentric – can flip an inner orbit from prograde to retrograde. Two of their figures serve as level-B benchmarks, with the input values taken from the captions and the text.

**Input.** *Fig. 3* (eccentric ZLK flip): a 1 $M_\odot$ star with a Jupiter-mass planet at 6 au and a brown dwarf of 40 Jupiter masses at 100 au with $e$ = 0.6, initial mutual inclination 65°. *Fig. 4* (the PSR B1620−26 set-up): a massive inner binary flipping at quadrupole order. The text gives the range of the mutual inclination as 57.5°–106.7°. The caption states initial inclinations of the two orbits to the total angular momentum of 6.75° and 63.25°; this corresponds to an angular-momentum ratio $G_1/G_2 = \sin 63.25°/\sin 6.75° = 7.60$, whereas the quoted masses and $a_2$ = 50 au give 6.79. An independent quadrupole calculation reproduces the published range with 7.60 (57.53°–106.70°) and not with 6.79 (57.3°–105.0°). The benchmark therefore uses $a_2$ = 39.9 au, which gives the published ratio.

**Table 5.** Naoz et al. (2013), Figs. 3 and 4: N-body backend (class R) and secular backends (own).

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `naoz13-fig4-imin` | R | 57.5° | 57.2391° | −0.261° | 0.5° | PASS |
| `naoz13-fig4-imax` | R | 106.7° | 106.93° | +0.23° | 0.5° | PASS |
| `quad-naoz13-fig4-imin` | own | 57.5° | 57.5376° | +0.0376° | 0.1° | PASS |
| `quad-naoz13-fig4-imax` | own | 106.7° | 106.721° | +0.0207° | 0.1° | PASS |
| `naoz13-fig3-flip-time` | R | 3.8 Myr | 3.9868 Myr | +4.92 % | 10 % | PASS |
| `naoz13-fig3-imax` | R | 142.6° | 143.01° | +0.408° | 2° | PASS |
| `secular-naoz13-fig3-flip-time` | own | 3.8 Myr | 4.04981 Myr | +6.57 % | 10 % | PASS |
| `secular-naoz13-fig3-imax` | own | 142.6° | 142.556° | −0.0435° | 2° | PASS |

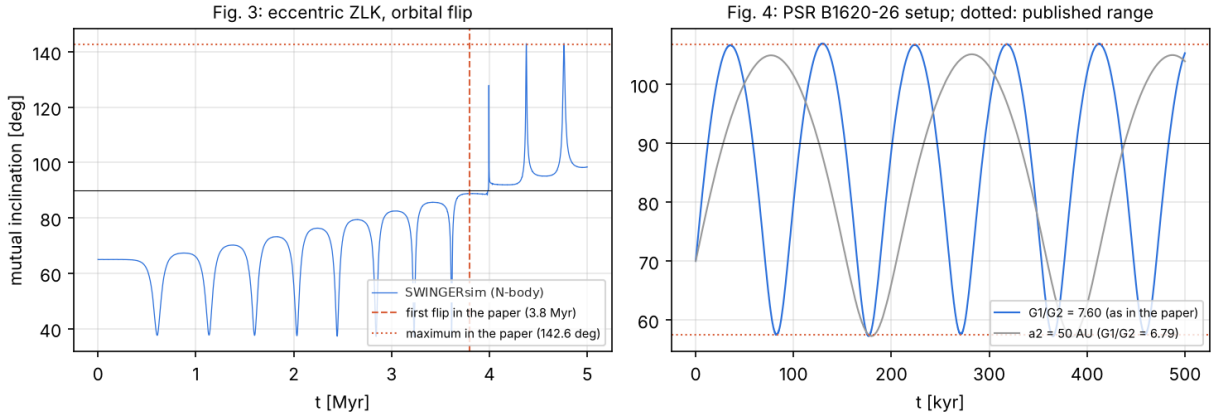



**Figure 2.** SWINGERsim reproducing Naoz et al. (2013). Both panels show the mutual inclination of the inner and outer orbit (vertical axis, degrees) against time; the horizontal black line marks 90°, the boundary between prograde and retrograde. *Left:* their Fig. 3, the eccentric ZLK flip, N-body run over 5 Myr (blue line). The dashed red vertical line is the first flip read from the published figure (3.8 Myr), the dotted red horizontal line the published maximum after the flip (142.6°). *Right:* their Fig. 4, the PSR B1620−26 set-up over 500 kyr; blue with the angular-momentum ratio of the paper ($G_1/G_2$ = 7.60), grey with the quoted $a_2$ = 50 au ($G_1/G_2$ = 6.79); the dotted red lines mark the published range 57.5°–106.7°.

**Result.** Figure 2 shows both set-ups. In the left panel, look at the ZLK minima before the flip: they lie at 38°, and their envelope rises slowly towards 90°, as in the published figure. The orbit first crosses 90° at 3.99 Myr in the N-body run and 4.05 Myr in the secular backend, against 3.8 Myr read from the published figure (+4.9 % and +6.6 %; the tolerance of 10 % reflects the reading of a time from a figure and the sensitivity of the flip time to the phase of the cycles). After the flip the inclination reaches 143.0° (N-body) and 142.6° (secular), against 142.6°. In the right panel, the blue curve oscillates exactly between the two dotted lines: the secular quadrupole code reproduces the published range to 0.04° (57.54°–106.72°), the N-body backend to 0.26° (57.24°–106.93°), the difference being the short-period terms that the averaged equations omit. The grey curve shows that the quoted $a_2$ = 50 au would give a narrower range; the published curve belongs to the ratio in the caption.

## 2.4 Reproduction of Vokrouhlický (2016)

Vokrouhlický (2016) developed the secular theory of a **2+2 quadruple** – two binaries A and B on a common outer orbit – and applied it to a V994 Her-like system. His quantity of interest is $J_{AB}$, the angle between the orbital planes of A and B: it controls whether both binaries can eclipse at the same time. He derived an analytic upper limit of $J_{AB}$ and illustrated four set-ups (his Figs. 2, 3, 4 and 7).

**Input.** The V994 Her-like system of the paper, set up from sky-frame elements: Fig. 2 with circular A and B and nodes $\Omega_A$ = 10°, $\Omega_B$ = 15°; Fig. 3 with the observed $e_A$ = 0.031, $e_B$ = 0.125; Fig. 4 with $\Omega_A$ = 35°, $\Omega_B$ = 40°; Fig. 7 as Fig. 4 plus the tidal quadrupole distortion of the stars. For Fig. 7 the paper tunes one deformability per binary to the observed apsidal periods of about 116 yr (A) and 111 yr (B), with radii $R \propto M^{0.8}$ (2.4, 2.1, 1.6 and 1.7 $R_\odot$). SWINGERsim uses the same radii, tides without rotation and dissipation, and apsidal constants $k_2$ = 0.00754 (A) and 0.00333 (B), for which Sterne's formula gives exactly these apsidal periods for the isolated binaries. The four set-ups were integrated with the N-body backend (25–60 minutes each) and with the 2+2 secular backend (seconds each).

**Table 6.** Vokrouhlický (2016): angles and eclipse probabilities (all class own). Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `vok16-v994-JA` | own | 11.6° | 11.6462° | +0.0462° | 0.1° | PASS |
| `vok16-v994-JB` | own | 15.8° | 15.7932° | −0.00678° | 0.1° | PASS |
| `vok16-jab-max-analytic` | own | 5.2° | 5.20074° | +0.000736° | 0.05° | PASS |
| `vok16-fig2-jab-min` | own | 3.5° | 3.47152° | −0.0285° | 0.15° | PASS |
| `vok16-fig2-jab-max` | own | 5.2° | 5.20069° | +0.000687° | 0.15° | PASS |
| `vok16-fig2-jab-1myr` | own | 5.2° | 5.2006° | +0.000633° | 0.15° | PASS |
| `vok16-fig3-jab-max` | own | 6° | 6.20874° | +0.209° | 0.5° | PASS |
| `vok16-eclipse-prob-A` | own | 0.366 | 0.369265 | +0.89 % | 2 % | PASS |
| `vok16-eclipse-prob-B` | own | 0.397 | 0.399075 | +0.52 % | 2 % | PASS |

The first two rows of Table 6 check SWINGERsim's conversion of sky-frame elements into the tilts $J_A$, $J_B$ of the binaries to the outer orbit – an independent check of the angle conventions. The analytic limit of $J_{AB}$ is 5.201° (paper 5.2°). The secular backend reproduces the ranges of $J_{AB}$ quoted in the text for Figs. 2 and 3 to 0.03° and 0.21°, and over 1 Myr the maximum stays at the analytic limit (the paper finds the same over 100 Myr). SWINGERsim's eclipse criterion applied to isotropically oriented orbits gives the chances that A and B eclipse, 0.369 and 0.399, against the published 0.366 and 0.397.

**Table 7.** Vokrouhlický (2016), Figs. 2, 3, 4 and 7: published behaviour against the N-body backend (class R) and the 2+2 secular backend (class own). Each row is one set-up; the second column gives what the paper shows, the third and fourth the ranges SWINGERsim computes.

| Set-up | Paper | SWINGERsim N-body | SWINGERsim secular |
|---|---|---|---|
| Fig. 2: circular A and B | $J_{AB}$ 3.5°–5.2°, $e_A$, $e_B$ near zero | $J_{AB}$ 3.41°–5.19° over 50 kyr, $e$ < 0.005 | $J_{AB}$ 3.47°–5.20°, $e_A$ < 0.0051 |
| Fig. 3: observed $e_A$, $e_B$ | $J_{AB} \approx$ 1°–6°, small oscillations of $e$ | $J_{AB}$ 0.94°–6.30°, $e_A$ 0.029–0.036, $e_B$ 0.123–0.140 | $J_{AB}$ 0.78°–6.21°, $e_A$ 0.029–0.036, $e_B$ 0.123–0.138 |
| Fig. 4: $\Omega_A$ = 35°, $\Omega_B$ = 40° | $J_A \approx$ 35°, $J_B \approx$ 40°, $e_B$ above 0.6, $J_{AB}$ from 0° to more than 70° | $J_A$ = 35.4°, $J_B$ = 40.3°, $e_B$ up to 0.631, $J_{AB}$ 3.5°–73.4° | $e_B$ up to 0.610, $J_{AB}$ 3.2°–71.1° |
| Fig. 7: Fig. 4 + tidal distortion | order restored: small oscillations of $e$ and $i$ | $e_B$ 0.124–0.136, $e_A$ 0.029–0.032, $J_{AB}$ 0.4°–5.8°, apsidal periods 113 and 108 yr | $e_B$ 0.123–0.134, $J_{AB}$ 0.29°–5.91°, apsidal periods 113.1 and 108.4 yr |

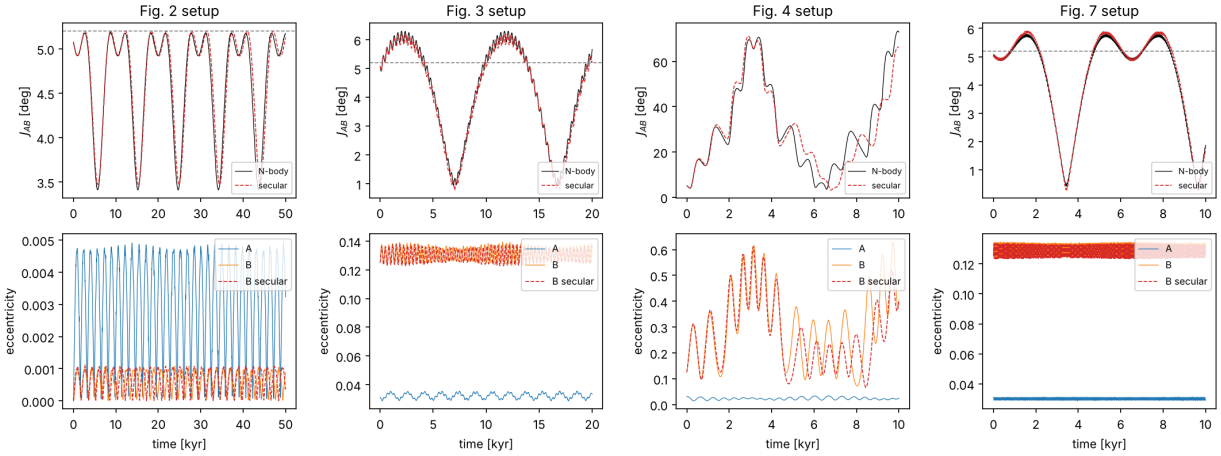



**Figure 3.** SWINGERsim reproducing Vokrouhlický (2016), Figs. 2, 3, 4 and 7 (one column per set-up). *Top row:* the angle $J_{AB}$ between the orbital planes of the two binaries (degrees) against time (kyr); black: N-body backend; red dashed: 2+2 secular backend; grey dashed: the analytic limit 5.2° of the paper. *Bottom row:* the eccentricities of A (blue) and B (orange; red dashed for the secular run) against time.

**Result.** Figure 3 and Table 7 compare the four set-ups. In the first column, $J_{AB}$ oscillates between 3.4° and 5.2° and never crosses the grey analytic limit; the black and red curves overlap. In the second column the observed eccentricities let $J_{AB}$ swing between about 1° and 6°, as in the paper. The third column is the interesting one: with larger tilts, B enters ZLK cycles and reaches $e_B$ = 0.63, $J_{AB}$ wanders up to 73°, and the N-body and secular curves agree for about 4 kyr and then separate. The paper reports exactly this for its own two integrators ("they start to significantly deviate after ≃5 kyr"): the motion is chaotic, and two correct integrations of it diverge. In this set-up the closest periastron of B reaches 1.06 times the sum of the stellar radii, which supports the paper's conclusion that the point-mass model is incomplete here. The fourth column adds the tidal bulges: the fast apsidal precession they cause (113 and 108 yr) detunes the ZLK cycles of B, its eccentricity stays at 0.124–0.136 instead of reaching 0.63, and the planes of A and B stay within 6° of each other – the order that the paper restores in its Fig. 7. Energy is conserved to $2.5{\cdot}10^{-12}$ in the N-body runs. Quantitatively, the published ranges are reproduced to 0.1–0.3° wherever the motion is regular.

## 2.5 Reproduction of Pejcha et al. (2013): an ensemble

Pejcha et al. (2013) integrated thousands of equal-mass 2+2 quadruples with an N-body code and counted how often a binary reaches an extreme eccentricity, $1 - e < 10^{-3}$, within 300 Kozai times of binary A. The inclinations of A and B to the outer orbit were not drawn isotropically but set on a grid in $(\cos i_A, \cos i_B)$, with 100 random set-ups of the other angles at every grid point; SWINGERsim draws them uniformly in $\cos i$ (isotropically), the continuous counterpart of a uniform grid. For their set F6 ($a_A$ = 0.573 au, $e_A$ = 0.3, $a_B$ = 0.351 au, $e_B$ = 0.1, $a_{AB}$ = 6 au, $e_{AB}$ = 0.3, $t_K$ = 64.9 yr) they found 35.8 % for A and 53.0 % for B – 4.8 and 10 times more than for the equivalent triples. This "enhancement" in 2+2 quadruples is the main result of the paper. Vokrouhlický (2016, Sect. 4) repeated the experiment with his secular model ($a_{AB}$ = 5 au, 11 kyr) and found 31.0 % and 48.1 %.

An **ensemble** in SWINGERsim is a set of runs (**members**) whose unknown parameters are drawn from a **prior** (a distribution of plausible values); the result is the fraction of members with a given outcome, with a 68 % Wilson interval as its statistical error.

**Table 8.** Pejcha et al. (2013) and Vokrouhlický (2016): fractions of binaries reaching $1 - e < 10^{-3}$. The N-body benchmarks are manual (class R), the secular ones slow (class own). Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `pejcha13-F6-fA` | R | 0.358 | 0.24 | −0.118 | 0.14 | PASS |
| `pejcha13-F6-fB` | R | 0.530 | 0.41 | −0.12 | 0.15 | PASS |
| `vok16-pejcha-fA` | own | 0.31 | 0.21 | −0.10 | 0.07 | **EXPLAINED** |
| `vok16-pejcha-fB` | own | 0.481 | 0.4025 | −0.0785 | 0.07 | **EXPLAINED** |

**Table 9.** The F6 ensembles in detail. Column 2 is the number of members, columns 3–4 the fractions of A and B reaching $1 - e < 10^{-3}$ with their 1σ statistical errors, column 5 the published values.

| Set-up | Members | Fraction A | Fraction B | Published |
|---|---|---|---|---|
| F6, N-body, 300 $t_K$ | 100 | 0.24 ± 0.04 | 0.41 ± 0.05 | 0.358, 0.530 (N-body) |
| F6, secular, 300 $t_K$ | 400 | 0.215 ± 0.021 | 0.358 ± 0.024 | – |
| F6, secular, 900 $t_K$ | 200 | 0.325 ± 0.033 | 0.505 ± 0.035 | – |
| Vokrouhlický set-up, secular, 11 kyr (≈ 300 $t_K$) | 400 | 0.21 | 0.40 | 0.310, 0.481 (secular) |
| the same, secular, 50 kyr | 400 | 0.36 | 0.57 | 0.310, 0.481 (secular) |
| F6 with B replaced by a point mass (triple), secular | 400 | 0.020 ± 0.007 | – | about 0.07 (N-body), 0.03 analytic |
| the same triple, N-body | 100 | 0.06 ± 0.02 | – | about 0.07 |

**What agrees** (Tables 8 and 9; in Table 9 each row is one ensemble, and the fractions are given with their 1σ errors where the ensemble was large enough to state them). (i) *The triple.* With B replaced by a point mass, the N-body backend gives 6 ± 2 % of members reaching $1 - e < 10^{-3}$, as Pejcha et al. (about 7 %). The secular backend gives 2 % (the analytic quadrupole estimate is about 3 %): in a triple of this mild hierarchy the short-period terms raise the eccentricity peaks (Luo, Katz & Dong 2016). (ii) *The enhancement.* In N-body, binary A of the quadruple reaches the limit 4.0 times more often than in the triple (paper: 4.8), B more often than A, and no member gets there from outside the ZLK window – the main result of both papers. (iii) *The two SWINGERsim backends agree with each other* within the statistical errors: for the same 100 members 0.24 vs 0.24 (A) and 0.41 vs 0.39 (B), although individual chaotic members end differently.

**What differs, and why.** At the nominal 300 Kozai times SWINGERsim finds about two thirds of the published quadruple fractions, with both backends and in both set-ups, while the triple agrees. Four tests narrow the cause down. An independent secular code, SecularMultiple (Hamers & Portegies Zwart 2016), on the same 200 F6 members gives 0.19 (A) and 0.37 (B) at the same multipole orders, and 0.225 and 0.385 with all orders, against SWINGERsim's 0.23 and 0.37 for these members. SWINGERsim's secular equations integrated with a fixed 0.1-yr fourth-order Runge–Kutta step, as in Vokrouhlický (2016), give the same fractions as the adaptive integrator (0.275 and 0.30 for 40 members). And the fractions keep growing with the length of the run (Figure 4).

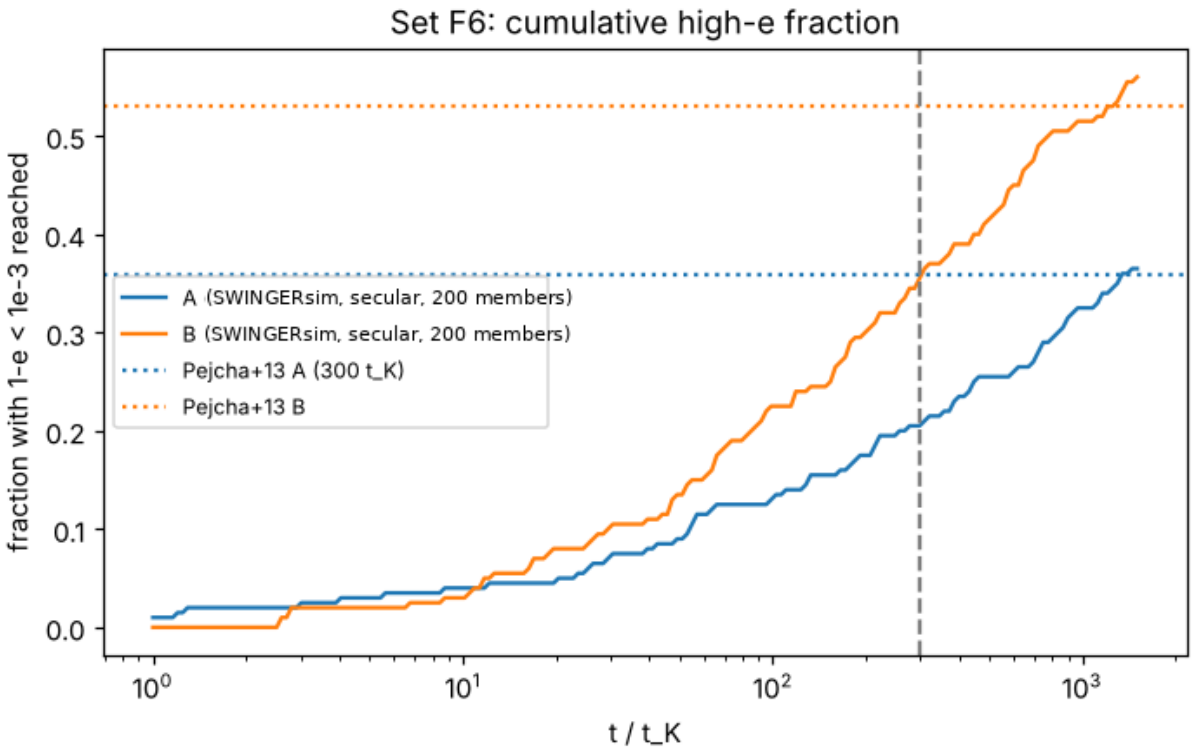



**Figure 4.** Set F6 of Pejcha et al. (2013): the fraction of members that have reached $1 - e < 10^{-3}$ at least once (vertical axis) against the time in Kozai times of binary A (horizontal axis, logarithmic); SWINGERsim secular backend, 200 members. Blue: binary A; orange: binary B. The dotted horizontal lines are the published fractions; the dashed vertical line marks 300 $t_K$, the stated length of the published runs.

Figure 4 shows the cumulative fraction against time. The curves only rise, because a member that has reached the limit stays counted. They do not saturate: they grow roughly logarithmically, by 0.2 (A) and 0.3 (B) per factor of ten in time, and reach the published values at about 1500 $t_K$ (A) and 1200 $t_K$ (B). Fang, Thompson & Hirata (2018) found the same non-saturation in wider 2+2 systems (about 10 % per factor of ten). With such growth, the published numbers depend strongly on the exact length of the runs. Running the set of Vokrouhlický (2016) to 50 kyr gives 0.36 and 0.57: the published 0.31 and 0.48 lie between SWINGERsim at 11 kyr and at 50 kyr (Figure 6, bottom right, in Chapter 4). Three independent integrations (SWINGERsim secular, SWINGERsim N-body with REBOUND, SecularMultiple) agree with each other, and the integration method of the secular paper is ruled out as the cause. The difference therefore lies in the effective length of the published runs (the published fractions correspond to 4–5 times the stated length) and possibly in the sampling of their inclination grid, which Pejcha et al. note "might not be distributed uniformly". The N-body benchmark passes at three standard deviations; the secular benchmark is EXPLAINED by the length of the published runs. The morphology of the high-eccentricity region in the plane of $(\cos J_A, \cos J_B)$ agrees with Fig. 11 of Vokrouhlický (2016): a diagonal band for A and a broad band $|\cos J_B| \lesssim 0.6$ for B.

## 2.6 Comparison with MSE

**Reference.** MSE (Multiple Stellar Evolution; Hamers et al. 2021) is an independent code for hierarchical multiples. It was built from its public repository and run on the same 3+1 and 2+2 quadruples as SWINGERsim's secular backends, with the same Hamiltonian (MSE's binary–triplet, hexadecupole and dotriacontupole terms switched off). A third comparison is a *fate*: a 3+1 whose inner binary (0.3 au) is driven by ZLK cycles into Roche-lobe overflow – MSE's event "RLOF start" against SWINGERsim's first Roche-lobe filling at periastron, with

the same radii.

**Table 10.** SWINGERsim's secular backends against MSE (level C, class own). Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `mse-3p1-inner-emax` | own | 0.9707 | 0.970714 | $+1.37{\cdot}10^{-5}$ | 0.002 | PASS |
| `mse-3p1-inner-t-emax` | own | 49 325 yr | 49 322.8 yr | −0.00 % | 0.2 % | PASS |
| `mse-3p1-mid-emax` | own | 0.2245 | 0.224461 | $-3.86{\cdot}10^{-5}$ | 0.002 | PASS |
| `mse-2p2-A-emax` | own | 0.9494 | 0.949492 | $+9.2{\cdot}10^{-5}$ | 0.002 | PASS |
| `mse-2p2-B-emax` | own | 0.7758 | 0.776227 | +0.000427 | 0.002 | PASS |
| `mse-3p1-rlof-time` | own | 380.03 yr | 380.1 yr | +0.02 % | 0.2 % | PASS |

**Result (Table 10).** The eccentricity maxima agree to $10^{-4}$ or better and the time of the maximum of the 3+1 inner binary to 2 yr in 49 kyr. For the 2+2, the first three ZLK cycles of A and all of B agree, after which the two runs drift apart in phase, as expected in a chaotic regime. MSE's extra multipole terms change $e_{\max}$ by less than 0.001 in these systems. The onset of Roche-lobe overflow agrees to 0.07 yr in 380 yr. Two practical differences are worth knowing: MSE switches to direct N-body integration when a system becomes dynamically unstable, whereas SWINGERsim's secular backends only report the stability diagnostics (the user then switches to the N-body backend); and MSE loses the argument of periastron of an orbit with an inclination of exactly 0°, so the comparison uses 0.5–2°.

## 2.7 Scattering and the statistical escape theory

SWINGERsim can set up encounters of binaries and single stars (scattering experiments) and integrate them with REBOUND until the outcome is clear. Antognini & Thompson (2016, Table 3) give the cross-sections of binary–binary encounters (equal masses, $a_1 = a_2$, circular orbits, velocity at infinity equal to the critical velocity $v_c$ at which the total energy is zero, impact parameters up to $7a$). Stone & Leigh (2019) derived the statistics of the break-up of a resonant ("democratic") three-body system from ergodic arguments; SWINGERsim implements their distributions.

**Table 11.** Scattering experiments (class R) and the statistical escape theory (own). Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `scatter-hyperbola-deflection` | own | 173.548° | 173.548° | $+1.4{\cdot}10^{-12}$° | $10^{-6}$° | PASS |
| `at16-binbin-single-ionization` | R | 2.645 | 2.6873 | +1.60 % | 10 % | PASS |
| `at16-binbin-exchange-ionization` | R | 3.259 | 3.2305 | −0.88 % | 10 % | PASS |
| `at16-binbin-exchange` | R | 0.302 | 0.37165 | +23.06 % | 30 % | PASS |
| `escape-theory-energy-slope-L0` | own | −3.5 | −3.5026 | −0.0026 | 0.05 | PASS |
| `escape-theory-energy-slope-highL` | own | −4 | −4.02479 | −0.0248 | 0.1 | PASS |
| `escape-theory-mean-e-L0` | own | 0.8133 | 0.812817 | −0.000483 | 0.01 | PASS |
| `escape-theory-mean-e-highL` | own | 0.7 | 0.710776 | +0.0108 | 0.02 | PASS |
| `escape-theory-vs-nbody-light-escaper` | own | 0.769 | 0.75369 | −0.0153 | 0.07 | PASS |

**Result (Table 11).** The set-up of a single hyperbolic passage gives the exact two-body deflection angle. The cross-sections (in units of $\pi a^2$) come from 12 000 encounters, against

$10^6$ in the paper. Single ionization and exchange + ionization agree to 1.6 % and 0.9 %. The paper states the normalisation $\pi(a_1^2 + a_2^2)$, but its numbers agree with $\pi a^2$ of one binary; the benchmarks use the latter (a factor-two ambiguity in the paper). Exchanges come out 23 % more frequent than published; with 91 exchange events this is 1.8 standard deviations, and the tolerance of 30 % is the statistical error of so few events. Stable triples are rare in both (0.008 against 0.020 ± 0.002). Two of the 12 000 encounters had no final outcome within the integration limit (0.02 %) and are left out of the cross-sections. The slopes of the energy distribution of the escaping binary and its mean eccentricity reproduce the closed forms of Stone & Leigh (2019) to 0.0026–0.025 (slopes) and 0.0005– 0.011 (mean $e$). In resonant binary–single encounters (1 + 0.6 $M_\odot$ binary, 0.8 $M_\odot$ star) integrated with N-body, the lightest star escapes in 75.4 % of the cases against 76.9 % from the theory. The 76.9 % is not a published number: it is SWINGERsim's own evaluation of the escape distribution of Stone & Leigh (2019) for these masses, so this benchmark compares SWINGERsim's N-body integrations with its own implementation of the theory.

## 2.8 Hierarchies of more than four stars

SWINGERsim describes any hierarchy of nested binaries (2+2+1, 3+1+1, 2+(2+1), 2+2+2, 3+3 or a custom nesting) and is tested up to ten stars. Two kinds of evidence cover the larger hierarchies.

**Secular against N-body for a sextuple (level C).** A 2+2+2 sextuple whose binary A is inclined by 60° to the orbit of the quadruple AB, and whose AB orbit is inclined by 50° to the outer orbit, is integrated with the N-body backend (IAS15 + 1PN, 400 yr, 98 s) and with the nested secular backend (binary pairs to octupole order; Hamers et al. 2015; Hamers & Portegies Zwart 2016; 1 s). A 2+2+1 quintuple tests the ZLK cycles of the orbit of the two binaries driven by a fifth star.

**V994 Her as a sextuple (level D).** V994 Her is a triply eclipsing sextuple (Zasche et al. 2023): two binaries A and B on a mutual orbit and a third binary C on a wide orbit around them. That orbit is unknown: the median of the period distribution that Zasche et al. estimate is close to 3000 yr, while their photodynamical model uses a fixed circular orbit seen face-on with a period of about 30 kyr. SWINGERsim takes 3000 yr ($e$ = 0.3) for it. With the published masses, radii and the other orbits (all coplanar), SWINGERsim's N-body run without relativity and tides gives the apsidal advance that the other stars cause in each binary; with relativity it gives the relativistic share. Both are expressed as fractions of the observed rates (1.86, 3.66 and 1.68 °/yr).

**Table 12.** Larger hierarchies. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `sextuple-secular-emax` | own | 0.75207 | 0.751821 | −0.03 % | 2 % | PASS |
| `sextuple-secular-zlk` | own | 70.2 yr | 71.04 yr | +1.20 % | 5 % | PASS |
| `sextuple-nbody-emax` | R | 0.75207 | 0.75207 | +0.00 % | 0.2 % | PASS |
| `quint-out-zlk-emax` | R | 0.81272 | 0.8152 | +0.31 % | 1 % | PASS |
| `v994s-dyn-A` | R | 0.12 | 0.0999365 | −16.72 % | 25 % | PASS |
| `v994s-dyn-B` | R | 0.05 | 0.0519361 | +3.87 % | 25 % | PASS |
| `v994s-dyn-C` | R | 0 | $1.23{\cdot}10^{-6}$ | $+1.23{\cdot}10^{-6}$ | 0.01 | PASS |
| `v994s-gr-A` | R | 0.10 | 0.0938886 | −6.11 % | 10 % | PASS |
| `v994s-gr-B` | R | 0.08 | 0.0748582 | −6.43 % | 10 % | PASS |
| `v994s-gr-C` | R | 0.06 | 0.0811201 | +35.20 % | 10 % | **EXPLAINED** |

**Result (Table 12).** The nested secular backend gives the largest eccentricity of A to 0.03 % and its ZLK period to 1.2 % of the N-body run, at a hundredth of the cost. The ranges of the other orbits agree as well (AB: $e$ = 0.190–0.327 secular, 0.179–0.324 N-body; mutual

inclination of B 6.6°–23.5° and 6.6°–23.2°). The N-body result itself is reproducible to the last digit (`sextuple-nbody-emax`). For V994 Her the dynamical shares are 10.0 % (A), 5.2 % (B) and 0 (C), against the published 12 %, 5 % and "negligible". The difference for A lies within the effect of the uncertainty of the outer eccentricity ($e_{AB}$ = 0.687 +0.050/−0.037), which changes the rate by up to 25 % – hence the tolerance. The relativistic shares are 9.4 % (A) and 7.5 % (B), against the quoted 10 % and 8 %. For C SWINGERsim gives 8.1 %, the paper 6 %. This difference is EXPLAINED: the 1PN rate computed by hand from the paper's own Table 3 (1.81 + 1.077 $M_\odot$, $P$ = 1.960 d, $e$ = 0.189) is 0.136 °/yr, which is 8.1 % of the observed 1.68 °/yr – SWINGERsim's value. The 6 % is probably a slip in the paper (the primary's mass alone gives 6 %); A and B, computed the same way, match.

The stability of larger hierarchies is judged by the algebraic criteria applied to every nested triple and by the N-body ghost test (Chapter 9); the neural classifiers were trained on quadruples only, and SWINGERsim applies them to every 2+2 and 3+1 sub-quadruple, which is an approximation outside their training set.

## 2.9 The environment: gas, the Galactic tide, passing stars and gravitational waves

Four effects act on a multiple star from outside or over very long times.

- A **parametric gas phase** (for young systems) damps the semi-major axis, the eccentricity and the inclination to the gas with prescribed time scales, $\dot{a} = -a/\tau_a$, $\dot{e} = -e/\tau_e$, and an alignment of the orbit normal with $\tan(i/2) = \tan(i_0/2)\, e^{-t/\tau_i}$ (exponential migration, e.g. Papaloizou & Larwood 2000). The benchmarks check that the integrated orbits follow these definitions.
- The **Galactic tide** of the disc (Heisler & Tremaine 1986) drives a ZLK-like oscillation of very wide orbits; for an orbit inclined by $i$ to the disc and without the radial terms the maximum eccentricity is $e_{\max} = \sqrt{1 - \frac{5}{4}\cos^2 i}$ (Grishin & Perets 2022, eq. 6).
- **Passing stars** kick a binary; for a distant parabolic passage the first-order change of the eccentricity vector is given by Hamers & Samsing (2019, eq. 9).
- **Gravitational waves** shrink and circularise close orbits at the orbit-averaged rates of Peters (1964). The observed orbital decay of the binary pulsar PSR B1913+16 (Weisberg & Huang 2016) is the level-D reference.

**Table 13.** The environment and gravitational waves. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `gas-migration-a` | own | 5.48812 au | 5.48812 au | −0.00 % | $10^{-4}$ % | PASS |
| `gas-damping-e` | own | 0.111565 | 0.111565 | +0.00 % | $10^{-4}$ % | PASS |
| `gas-alignment-i` | own | 5.63999° | 5.63943° | −0.0005610.01° | | PASS |
| `galactic-tide-emax-60` | own | 0.829156 | 0.829181 | $+2.5{\cdot}10^{-5}$ | 0.01 | PASS |
| `galactic-tide-emax-60-nbody` | R | 0.82916 | 0.82908 | −0.01 % | 1 % | PASS |
| `flyby-first-order-kick` | own | 0 | 0.00594917 | +0.00595 | 0.03 | PASS |
| `flyby-nbody-backend` | R | 0 | 0.00738403 | +0.00738 | 0.05 | PASS |
| `gw-hulse-taylor-pdot` | own | $-2.40263{\cdot}10^{-12}$ | $-2.4021{\cdot}10^{-12}$ | +0.02 % | 0.1 % | PASS |
| `gw-secular-circular-decay` | own | 1 | 1 | −0.00 % | 0.01 % | PASS |
| `gw-nbody-decay` | R | 1 | 1.0003 | +0.03 % | 1 % | PASS |

**Result (Table 13).** An orbit of $2{\cdot}10^4$ au inclined by 60° to the disc reaches $e$ = 0.82918 in the secular backend and 0.82908 in the N-body backend, against the analytic 0.82916; the energy of the averaged system is conserved to $10^{-10}$. The first-order kick formula agrees with

a direct IAS15 integration of a parabolic passage at a periastron distance of 20 $a$ to 0.6 % (0.2 % at 40 $a$); the same passage integrated as a particle in the N-body backend agrees with it to 0.7 %. The Peters rates give the orbital decay of PSR B1913+16 (masses 1.438 and 1.390 $M_\odot$, $P$ = 0.323 d, $e$ = 0.617) to 0.02 % of the general-relativity value; the secular backend follows the analytic decay of a circular binary to better than $10^{-4}$, the N-body backend (where SWINGERsim applies the orbit-averaged loss between integration steps) the rate to $3{\cdot}10^{-4}$.

# Chapter 3

# Stars: structure, tides, evolution and mass exchange

The stars of a multiple system are not points. They have a size and an internal structure, they are distorted by tides and rotation, they evolve, lose mass in winds and exchange mass with their companions. This chapter tests every module of SWINGERsim that describes this, from the structure constants of a single star to a binary that evolves into contact and merges.

## 3.1 Structure constants, Roche lobes and rotating stars

**Definitions.** The **apsidal-motion constant** $k_2$ measures how strongly a star is distorted by a tidal or centrifugal force; with the radius $R$ it sets the extra apsidal precession of an eccentric binary, which scales as $k_2 R^5$. The **radius of gyration** $\beta$ (or $r_g^2 = I/MR^2$) sets the moment of inertia $I$. The **Roche lobe** is the region around a star in a binary within which its own gravity dominates; its volume-equivalent radius $R_L$ is given to better than 1 % by the formula of Eggleton (1983). A star rotating rigidly with angular velocity $\Omega$ has, in the Roche model, an equatorial radius that grows with $\Omega$ and reaches 1.5 times the polar radius at the critical rotation.

**References.** The classical apsidal constants of polytropes of index $n$ = 1.5 and 3 (Brooker & Olle 1955); the published tables of $\log k_2$ and $\beta$ along stellar tracks of Claret (2004, up to 125 $M_\odot$) and Claret (2023); the exact Roche potential, integrated directly by the benchmark, for the Eggleton formula and for SWINGERsim's surface mesh; the closed forms of the Roche model of a rotating star; and the rotational correction of $k_2$, $\delta \log k_2 = -0.87\,\lambda_s$ with $\lambda_s = 2V^2/(3gR)$, that is a factor $10^{-0.87 \cdot 2V^2/3gR}$ (Claret 1999).

**Table 14.** Structure constants, Roche lobes and rotating stars. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `polytrope-k2-n1.5` | own | 0.1433 | 0.143279 | −0.01 % | 0.2 % | PASS |
| `polytrope-k2-n3` | own | 0.0144 | 0.014443 | +0.30 % | 0.5 % | PASS |
| `claret2023-k2-2msun` | own/tab | −2.478 | −2.47794 | $+6{\cdot}10^{-5}$ | 0.002 | PASS |
| `claret2023-beta-1msun` | own/tab | 0.2265 | 0.226499 | $-6.3{\cdot}10^{-7}$ | 0.0005 | PASS |
| `claret2004-log-k2` | own/tab | −2.81523 | −2.81569 | −0.000463 | 0.002 | PASS |
| `claret2004-beta` | own/tab | 0.161706 | 0.161699 | $-6.6{\cdot}10^{-6}$ | 0.001 | PASS |
| `k2-rotation` | own | 1 | 1 | +0.00 % | $10^{-7}$ % | PASS |
| `roche-eggleton-q1` | own | 0.379861 | 0.378921 | −0.25 % | 1 % | PASS |
| `roche-eggleton-q0.1` | own | 0.2054 | 0.206773 | +0.67 % | 1 % | PASS |

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `roche-lobe-volume` | own | 1 | 1.00509 | +0.51 % | 1 % | PASS |
| `rotating-star-critical` | own | 1.5 | 1.5 | −0.00 % | $10^{-4}$ % | PASS |
| `rotating-star-equator` | own | 0 | $-2.2\cdot10^{-16}$ | $-2.2\cdot10^{-16}$ | $10^{-9}$ | PASS |

**Result (Table 14).** SWINGERsim integrates the Lane–Emden and Radau equations for its default stellar model; the apsidal constants of the polytropes agree with Brooker & Olle (1955) to 0.01 % and 0.3 % (the automatic tests also check the analytic $n = 1$ values to $10^{-8}$). The interpolation of the Claret tables in $\log g$ along the tracks reproduces the published rows to their last digit ($6\cdot10^{-5}$ in $\log k_2$, $6\cdot10^{-7}$ in $\beta$). The Eggleton formula, which SWINGERsim uses for all Roche-lobe filling factors, differs from the exact volume-equivalent radius by −0.25 % at $q = 1$ and +0.67 % at $q = 0.1$, within the 1 % that Eggleton states for all mass ratios. The surface mesh of a Roche-filling star (used for light curves of distorted stars, Chapter 4) encloses a volume whose equivalent radius is 0.51 % above Eggleton's; the residual is the resolution of the mesh.

## 3.2 Apsidal motion and tidal dissipation

**Definitions.** Tides have a conservative part – the static bulge, which adds apsidal precession – and a dissipative part – the lag of the bulge behind the line joining the stars, which transfers angular momentum between the orbit and the spins and dissipates energy. SWINGERsim's N-body backend uses the REBOUNDx effect `tides_spin` (the constant-time-lag equilibrium tide of Eggleton et al. 1998; Lu et al. 2023) with stellar parameters set by SWINGERsim; the secular backends use SWINGERsim's own orbit-averaged equations (Hut 1981).

**References.** The apsidal period from the tidal and rotational distortion of both stars (Sterne 1939); the orbit-averaged rates $\dot{a}$, $\dot{e}$ and $\dot{\Omega}$ of the constant-time-lag model (Hut 1981; Leconte et al. 2010); the circularisation rate of Hut (1981, eq. 10) for the nested secular backend; and the three contributions to the apsidal motion of the eclipsing pair A of the observed quadruple CzeV343 (tidal, rotational and relativistic) computed by Pejcha et al. (2022, Table 3).

**Table 15.** Apsidal motion and tidal dissipation. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `apsidal-sterne-static` | R | 318.227 yr | 318.201 yr | −0.01 % | 0.5 % | PASS |
| `apsidal-sterne-rotating` | R | 291.213 yr | 291.185 yr | −0.01 % | 0.5 % | PASS |
| `tides-leconte-a` | R | 1 | 0.999285 | −0.07 % | 1 % | PASS |
| `tides-leconte-e` | R | 1 | 0.999356 | −0.06 % | 1 % | PASS |
| `tides-leconte-spin` | R | 1 | 1.00113 | +0.11 % | 1 % | PASS |
| `tides-leconte-e-eccentric` | R | 1 | 0.996923 | −0.31 % | 1 % | PASS |
| `secular-nested-tides-hut81` | own | 1 | 1 | −0.00 % | $10^{-4}$ % | PASS |
| `czev343-apsidal-tidal` | own | $4.14\cdot10^{-4}$ rad/d | $4.22272\cdot10^{-4}$ rad/d | +2.00 % | 3 % | PASS |
| `czev343-apsidal-rotational` | own | $6.4\cdot10^{-5}$ rad/d | $6.32533\cdot10^{-5}$ rad/d | −1.17 % | 5 % | PASS |
| `czev343-apsidal-gr` | own | $1.47\cdot10^{-5}$ rad/d | $1.47874\cdot10^{-5}$ rad/d | +0.59 % | 2 % | PASS |
| `czev343-apsidal-period` | R | 33.4 yr | 34.491 yr | +3.27 % | 6 % | PASS |

**Result (Table 15).** An isolated eccentric binary of extended stars ($R/a \approx 0.14$, $k_2$ = 0.015, $e$ = 0.3), first without rotation and then rotating synchronously, precesses in a 60-yr N-body run with the apsidal period of Sterne's formula to $10^{-4}$. With a time lag, the N-body rates of change of $a$, $e$ and the spin agree with the orbit-averaged theory to 0.06–0.31 %, both at $e$ = 0.3 and at $e$ = 0.6 with a star spinning at 1.2 times the mean motion. The rates are taken from the difference between runs with and without dissipation, which removes the short-period terms of the conservative tide. These benchmarks also fix a convention: REBOUNDx's `tau` is twice SWINGERsim's time lag; without SWINGERsim's factor ½ the rates would be twice as large. The nested secular backend reproduces Hut's circularisation rate exactly (the same equation). For CzeV343 the three contributions agree with Pejcha et al. (2022) to 0.6–2 %, and the full N-body quadruple with tides and rotation gives an apsidal period of pair A of 34.5 yr against the observed 33.4 ± 0.2 yr (Pejcha et al. 2022, Sect. 4.3) (+3.3 %; tolerance 6 %, the uncertainty of the stellar parameters). CzeV343 is studied in full in Part IV.

## 3.3 Stellar tracks of low- and intermediate-mass stars

**Definitions.** SWINGERsim follows stellar evolution with tabulated tracks. The **SSE tracks** are the analytic formulae of Hurley, Pols & Tout (2000), evaluated once with the published SSE Fortran routines on a grid of 320 masses × 5 metallicities and interpolated by SWINGERsim along "equivalent evolutionary points" (fractions of each evolutionary phase); they end at helium ignition. The **MIST tracks** (Choi et al. 2016; Dotter 2016) are MESA models converted from the published EEP archives, from 0.1 to 300 $M_\odot$, with and without rotation, up to the end of the asymptotic giant branch (AGB). The envelope quantities (mass and radius of the convective envelope) follow BSE (Hurley et al. 2002).

**References.** The ZAMS radii of Tout et al. (1996); the present Sun; the helioseismic base of the solar convection zone, 0.713 $R_\odot$ (Christensen-Dalsgaard et al. 1991); the published MIST EEP ages; MIST against SSE (two families of models from different codes); the Reimers wind on the red giant branch; and the radius of the fully convective M dwarf CM Dra A, which standard models underpredict (Feiden & Chaboyer 2014).

**Table 16.** Stellar tracks of low- and intermediate-mass stars (class own/tab: SWINGERsim's interpolation of published tracks). Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `sse-zams-radius-1` | own/tab | 0.88825 $R_\odot$ | 0.888312 $R_\odot$ | +0.01 % | 0.1 % | PASS |
| `sse-zams-radius-2` | own/tab | 1.79434 $R_\odot$ | 1.79435 $R_\odot$ | +0.00 % | 0.1 % | PASS |
| `sse-sun-radius` | own/tab | 1 $R_\odot$ | 0.984843 $R_\odot$ | −1.52 % | 5 % | PASS |
| `sse-sun-luminosity` | own/tab | 1 $L_\odot$ | 0.95786 $L_\odot$ | −4.21 % | 5 % | PASS |
| `sse-sun-convection-zone` | own/tab | 0.713 | 0.733903 | +0.0209 | 0.03 | PASS |
| `mist-grid-tams-age-1` | own/tab | 9919.42 Myr | 9919.42 Myr | +0.00 % | 0.001 % | PASS |
| `mist-interpolation` | own/tab | 1 | 1.01068 | +1.07 % | 2 % | PASS |
| `mist-ms-lifetime-1` | own/tab | 9878 Myr | 11 003.5 Myr | +11.39 % | 15 % | PASS |
| `mist-ms-lifetime-2` | own/tab | 1054 Myr | 1164.17 Myr | +10.45 % | 15 % | PASS |
| `mist-tams-radius-2` | own/tab | 3.913 $R_\odot$ | 3.56583 $R_\odot$ | −8.87 % | 15 % | PASS |
| `sse-wind-rgb-1msun` | own/tab | 0.04158 $M_\odot$ | 0.0421789 $M_\odot$ | +1.44 % | 10 % | PASS |

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `cmdra-a-radius-excess-solar` | own/tab | 0.035 | 0.0335989 | −0.0014 | 0.015 | PASS |
| `cmdra-a-radius-excess-lowz` | own/tab | 0.06 | 0.0641756 | +0.00418 | 0.015 | PASS |

**Result (Table 16).** The SSE grid reproduces the ZAMS radii of Tout et al. (1996) to $10^{-4}$, also for 2.5 $M_\odot$, which lies between grid masses. A 1 $M_\odot$, $Z = 0.02$ star at 4.57 Gyr has $R = 0.985$ $R_\odot$ and $L = 0.958$ $L_\odot$, within the 5 % accuracy that Hurley et al. (2000) quote for their formulae; the base of its convective envelope lies at 0.734 $R$, 0.021 above the helioseismic value. The MIST grid reproduces the published EEP ages exactly, and a track interpolated from neighbours at twice the grid spacing (1.14 $M_\odot$ from 1.12 and 1.16 $M_\odot$) has a radius at the end of the main sequence within 1.1 % of the tabulated one. The main-sequence lifetimes of 1 and 2 $M_\odot$ are 11 % and 10 % longer in MIST than in SSE, and the radius of 2 $M_\odot$ at the end of the main sequence is 9 % smaller. These are the known differences between the two families (convective overshooting, solar abundance scale $Z = 0.02$ against 0.0142); they set the systematic uncertainty of ages and of the onset of Roche-lobe overflow in SWINGERsim, and the tolerance of 15 %. The SSE Reimers wind with $\eta = 0.1$ removes 0.0422 $M_\odot$ from a 1 $M_\odot$ star up to helium ignition, against 0.0416 $M_\odot$ in the MIST model with the same law. CM Dra A is 3.4 % larger than the MIST radius of its mass at solar metallicity and 6.4 % at [Fe/H] = −0.3, as Feiden & Chaboyer (2014) find for standard models (3–4 % and 5–7 %): SWINGERsim reproduces the known radius inflation problem of M dwarfs rather than hiding it.

## 3.4 Massive stars and giants to the end of the AGB

**References.** For massive stars: leave-one-out interpolation of the MIST grid (a track computed from its neighbours and compared with the tabulated one); the wind recipe of Vink et al. (2001, eq. 24); the hydrogen-burning lifetimes of the Geneva models (Ekström et al. 2012), an independent code with and without rotation. For giants: the white-dwarf initial–final mass relation of Cummings et al. (2018), measured for 80 white dwarfs in 13 clusters with MIST progenitor ages (scatter 0.06 $M_\odot$). The mass of the helium–carbon core at the end of the MIST track is almost the mass of the future white dwarf.

**Table 17.** Massive stars and the end of the AGB. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `mist-interpolation-massive-r` | own/tab | 1 | 0.999431 | −0.06 % | 1 % | PASS |
| `mist-interpolation-massive-m` | own/tab | 1 | 1.00107 | +0.11 % | 3 % | PASS |
| `mist-vink2001` | own/tab | 1 | 1.05511 | +5.51 % | 10 % | PASS |
| `mist-geneva-tauh-15-0` | own/tab | 11.015 Myr | 12.3538 Myr | +12.15 % | 15 % | PASS |
| `mist-geneva-tauh-15-4` | own/tab | 13.447 Myr | 12.6825 Myr | −5.69 % | 15 % | PASS |
| `mist-geneva-tauh-60-0` | own/tab | 3.53 Myr | 3.71791 Myr | +5.32 % | 15 % | PASS |
| `mist-geneva-tauh-60-4` | own/tab | 4.465 Myr | 3.94657 Myr | −11.61 % | 15 % | PASS |
| `mist-geneva-tauh-120-0` | own/tab | 2.672 Myr | 2.78283 Myr | +4.15 % | 15 % | PASS |
| `mist-geneva-tauh-120-4` | own/tab | 3.178 Myr | 3.33128 Myr | +4.82 % | 15 % | PASS |
| `agb-final-mass-1p5` | own/tab | 0.609 $M_\odot$ | 0.565426 $M_\odot$ | −0.0436 $M_\odot$ | 0.1 $M_\odot$ | PASS |

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `agb-final-mass-2p5` | own/tab | 0.689 $M_\odot$ | 0.610396 $M_\odot$ | −0.0786 $M_\odot$ | 0.1 $M_\odot$ | PASS |
| `agb-final-mass-3p0` | own/tab | 0.745 $M_\odot$ | 0.653775 $M_\odot$ | −0.0912 $M_\odot$ | 0.1 $M_\odot$ | PASS |
| `agb-final-mass-4p0` | own/tab | 0.899 $M_\odot$ | 0.835342 $M_\odot$ | −0.0637 $M_\odot$ | 0.1 $M_\odot$ | PASS |

**Result (Table 17).** The radius of a 60 $M_\odot$ star in the middle of its main sequence, interpolated from 55 and 65 $M_\odot$, is within 0.06 % of the tabulated one, and the present mass of 100 $M_\odot$ at the end of the main sequence within 0.11 %. The limits of the interpolation lie later in the evolution: from the late main sequence on, the inflated envelopes of 40–70 $M_\odot$ stars are not resolved by the grid, with radius errors up to 20–40 % and mass errors up to 7 % (for example −24 % in radius for 60 $M_\odot$ at the end of the main sequence without rotation). The MIST mass-loss rate of a 40 $M_\odot$ star follows Vink et al. (2001) with an offset of +5.5 %, which is the metallicity passed to the recipe. The hydrogen-burning lifetimes of 15, 60 and 120 $M_\odot$ agree with the Geneva models to 4–12 %, with and without rotation. The masses at the end of the main sequence differ more (60 $M_\odot$ without rotation: 46.6 against 36.3 $M_\odot$; 120 $M_\odot$ with rotation: 25.3 against 34.6 $M_\odot$): this spread between model families, not the interpolation, is the uncertainty of the mass history of very massive stars.

The white-dwarf masses of MIST stars of 1.5–4 $M_\odot$ lie 0.04–0.09 $M_\odot$ below the relation of Cummings et al. (2018), the offset the authors themselves note for these models; the tolerance of 0.1 $M_\odot$ is that offset plus the scatter of the relation. The benchmark compares the core, because the MIST tracks stop with an envelope of up to 0.15–0.2 $M_\odot$ that the post-AGB phase would remove. Derived quantities are checked by the automatic tests: the Sun's convective envelope has 0.034 $M_\odot$ and 0.29 $R$, with a turnover time of 21 d.

## 3.5 Inertial waves and tidal circularisation

**Definitions.** In convective envelopes the tidal flow is damped by turbulent viscosity (the **equilibrium tide**). In rotating stars the tide also excites **inertial waves**, waves restored by the Coriolis force, which can dissipate much more energy (Ogilvie 2013; Barker 2020, 2022). The efficiency is expressed as the modified tidal quality factor $Q'$ (larger $Q'$ = weaker dissipation). The **circularisation period** of a population of binaries of the same age is the period below which the orbits are circular; observers define it from the final orbits (Meibom & Mathieu 2005).

**References.** The closed formula for the inertial-wave dissipation of a two-layer star (Ogilvie 2013, in the form of Barker 2020, eq. 39); $Q' \approx 10^7$ for 0.3 and 1 $M_\odot$ main-sequence stars rotating in 10 d (Barker 2022, from MESA profiles); circularisation on the pre-main sequence, 7.2–8.5 d (Zahn & Bouchet 1989); the circularisation period of pre-main-sequence binaries, 7.1 ± 1.2 d, and that of the 4-Gyr open cluster M67, 12.1 +1.0/−1.5 d (both Meibom & Mathieu 2005, Table 3; quoted by Mazeh 2008).

**Table 18.** Inertial waves and tidal circularisation. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `iw-two-layer` | own | 1 | 1.00031 | +0.03 % | 0.5 % | PASS |
| `iw-q-solar` | own | 7 | 7.0122 | +0.0122 | 0.3 | PASS |
| `iw-q-m-dwarf` | own | 7 | 6.94697 | −0.053 | 0.3 | PASS |
| `tides-circularization-pms` | own | 7.85 d | 7.9307 d | +1.03 % | 10 % | PASS |

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `tides-circularization-pms-population` | own | 7.1 d | 6.5831 d | −0.517 d | 1.2 d | PASS |
| `tides-circularization-m67` | own | 12.1 d | 3.7847 d | −68.72 % | 30 % | **EXPLAINED** |
| `tides-circularization-m67-iw` | own | 12.1 d | 7.4572 d | −38.37 % | 30 % | **EXPLAINED** |
| `tides-circularization-m67-final` | own | 12.1 d | 7.5337 d | −4.57 d | 1.2 d | **EXPLAINED** |

**Result (Table 18).** SWINGERsim's full-profile solver of the inertial-wave dissipation reproduces the closed two-layer formula to 0.03 % for a smoothed two-layer star; this checks the solver and its normalisation. For 0.3 and 1 $M_\odot$ main-sequence stars rotating in 10 d it gives $\log Q' = 6.95$ and 7.01, as Barker (2022). Twin 1 $M_\odot$ binaries followed on MIST pre-main-sequence tracks from 1 Myr with the equilibrium tide and inertial waves are circular up to 7.9 d when they reach the main sequence (Zahn & Bouchet: 7.2–8.5 d), and a population of such binaries at 3 Myr has a circularisation period of 6.6 d in the observers' definition (final period, $e < 0.01$), against the observed 7.1 ± 1.2 d of pre-main-sequence binaries.

The three M67 benchmarks are EXPLAINED as an open point of SWINGERsim's tidal model. With the equilibrium tide of Hurley et al. (2002) alone, twin solar-type binaries started on the main sequence with $e = 0.3$ are circular after 4 Gyr only below 3.8 d; with inertial waves below 7.5 d; and with the whole history from the pre-main sequence 7.5 d in the final period. The pre-main-sequence phase is reproduced, but the main sequence adds only about 1 d, whereas M67 requires growth to 12 d. Barker (2022) finds that inertial waves can explain the circularisation periods of solar-type binaries, with a circularisation period that grows with main-sequence age as observed. SWINGERsim's estimate of the inertial-wave dissipation is also frequency-averaged, but it uses a two-layer approximation (Ogilvie 2013) of the SSE and MIST structures and follows the spins of the stars, whereas Barker computes the dissipation from full MESA profiles and assumes synchronised spins. The difference therefore lies in the stellar structure and the spin history; which of the two causes it is not resolved here. (Resonance locking, which SWINGERsim does not model, acts mainly on the pre-main sequence, with a negligible main-sequence contribution; Zanazzi & Wu 2021.) The tolerance is not widened; the difference is stated as a limit of SWINGERsim's tidal model.

## 3.6 Mass loss by winds

**References.** A binary whose star loses mass slowly (compared with the orbital period) keeps the adiabatic invariant $aM$ (Jeans 1924; Hadjidemetriou 1963). Within one orbit the eccentricity follows $e = e_0 + \Psi(1-e_0^2)^{3/2} \sin f/(1+e_0 \cos f)$, obtained by integrating eq. 4 of Veras et al. (2011) over the true anomaly $f$, where $\Psi$ is the mass lost per orbit in units of the total mass; an initially circular orbit reaches $e = 2\Psi$. An instantaneous loss of the fraction $\delta$ of the mass gives $a'/a = (1-\delta)/(1-2\delta)$ and $e' = \delta/(1-\delta)$ (Blaauw 1961; Hills 1983). In a triple whose inner primary loses mass, the expanding orbits can cross the stability limit (the "triple evolution dynamical instability" of Perets & Kratter 2012).

In the N-body backend the masses change through the REBOUNDx operator `modify_mass` on SWINGERsim's mass-loss schedule (class R); in the secular and evolving backends the orbits are rescaled by SWINGERsim's own equations (class own).

**Table 19.** Mass loss. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `ml-adiabatic-invariant` | R | 0 | $3.57{\cdot}10^{-9}$ | $+3.57{\cdot}10^{-9}$ | $10^{-6}$ | PASS |

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `ml-osculating-e-03` | R | 0 | 0.0126471 | +0.0126 | 0.05 | PASS |
| `ml-osculating-e-07` | R | 0 | 0.0123538 | +0.0124 | 0.05 | PASS |
| `ml-circular-emax` | R | 2 | 2.00265 | +0.13 % | 2 % | PASS |
| `ml-impulsive-a` | R | 1.21429 | 1.21491 | +0.05 % | 1 % | PASS |
| `ml-impulsive-e` | R | 0.176471 | 0.176892 | +0.24 % | 1 % | PASS |
| `ml-zlk-nbody-secular` | own | 1 | 1.0021 | +0.21 % | 1 % | PASS |
| `ml-tedi-onset` | R | 1 | 1.06475 | +6.47 % | 15 % | PASS |
| `wind-transfer-jeans` | own | 0 | $1.69{\cdot}10^{-11}$ | $+1.69{\cdot}10^{-11}$ | $10^{-8}$ | PASS |

**Result (Table 19).** A binary whose star loses 0.3 $M_\odot$ slowly keeps $aM$ to $4{\cdot}10^{-9}$ in the N-body backend; the secular backends keep it exactly by construction. Within two orbits the eccentricity follows the formula above to 1.3 % of $\Psi$ for $e_0$ = 0.3 and 0.7, and a circular orbit reaches $e/\Psi$ = 2.003. Veras et al. print the denominator of their eq. 17 as $1 - e_0 \cos f$; that form does not follow from their eq. 4 and disagrees with the N-body runs by about $\Psi$, so we take it for a misprint. Losing 15 % of the mass within $10^{-3}$ of an orbit gives the instantaneous results to 0.05 % and 0.24 %; the residual is the finite duration of the loss. While a ZLK perturber loses half its mass, the N-body and the secular backend agree on the time of the fourth ZLK maximum to 0.2 %. For the masses of Perets & Kratter (2012) (5 + 2 $M_\odot$ inner binary, 2.25 $M_\odot$ tertiary) and the inner primary losing 3 $M_\odot$ in 8000 yr, the inner eccentricity starts to grow 6.5 % after the adiabatically expanding orbits cross the MA01 limit; the system had not broken up after a further 2000 outer orbits, in line with the long survival on the edge found by Toonen et al. (2022). With winds and mass transfer together, a 25 + 18 $M_\odot$ binary keeps $a(M_1 + M_2)$ to $2{\cdot}10^{-11}$ before its transfer begins.

## 3.7 Mass transfer and rejuvenation

**Definitions.** A star that fills its Roche lobe transfers mass to its companion (**Roche-lobe overflow**). How the orbit responds depends on how much of the transferred mass the accretor keeps and how much angular momentum the lost mass carries away. SWINGERsim's mass-transfer module is its own code: it follows the donor on its track, computes the rate, the response of the orbit and of the Roche lobe, and the stability of the transfer. A main-sequence accretor that gains mass is **rejuvenated**: its fresh hydrogen makes it look younger (Hurley et al. 2000).

**References.** The closed forms of the orbit response with constant fractions of the mass lost from the donor and from the accretor (Soberman, Phinney & van den Heuvel 1997, eqs. 20–28), and for conservative transfer and isotropic re-emission (Postnov & Yungelson 2014, eqs. 37 and 43); the response $\zeta_L = d\ln R_L / d\ln M_d$ of the Roche lobe, approximated by $2.13q - 1.67$ (Tout et al. 1997, quoted by Hurley et al. 2002); the **BSE Algol example** of Hurley et al. (2002, Sect. 3.1) – 2.9 + 0.9 $M_\odot$, $Z$ = 0.02, started at $P$ = 8 d and $e$ = 0.7, which tides have circularised to $P$ = 2.7 d when the primary fills its Roche lobe; the benchmarks start from this state – with BSE run independently in COSMIC 4.2.1 (Breivik et al. 2020); and the age correction of a rejuvenated accretor (Hurley et al. 2000).

**Table 20.** Mass transfer and rejuvenation (all class own). Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `mt-soberman-general` | own | 1.29573 $a/a_0$ | 1.29573 $a/a_0$ | −0.00 % | $10^{-5}$ % | PASS |
| `mt-conservative` | own | 2.25 $a/a_0$ | 2.25 $a/a_0$ | −0.00 % | $10^{-5}$ % | PASS |

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `mt-isotropic-reemission` | own | 2.03003 $a/a_0$ | 2.03003 $a/a_0$ | −0.00 % | $10^{-5}$ % | PASS |
| `mt-zeta-lobe-q2` | own | 2.59 | 2.60707 | +0.0171 | 0.05 | PASS |
| `mt-bse-algol-donor` | own | 0.42 $M_\odot$ | 0.404336 $M_\odot$ | −0.0157 $M_\odot$ | 0.05 $M_\odot$ | PASS |
| `mt-cosmic-algol-donor` | own | 0.401 $M_\odot$ | 0.404336 $M_\odot$ | +0.00334 $M_\odot$ | 0.03 $M_\odot$ | PASS |
| `mt-bse-algol-accretor` | own | 2.85 $M_\odot$ | 2.51105 $M_\odot$ | −0.339 $M_\odot$ | 0.15 $M_\odot$ | **EXPLAINED** |
| `mt-bse-algol-period` | own | 30 d | 21.9301 d | −26.90 % | 15 % | **EXPLAINED** |
| `mt-bse-algol-giant-phase` | own | 30 d | 30.3561 d | +1.19 % | 3 % | PASS |
| `rejuvenation-hurley` | own | 1 | 0.999174 | −0.08 % | 0.2 % | PASS |

**Result (Table 20).** The numerically integrated orbit response agrees with the closed forms to $3{\cdot}10^{-13}$ (general fractions: 30 % of the transferred mass lost from the donor, 40 % re-emitted from the accretor, 30 % accreted) and $2{\cdot}10^{-12}$ (conservative, isotropic re-emission). $\zeta_L$ at $q = 2$ is 2.607 against the fit's 2.59; the fit is accurate to 0.03 for $q = 0.5$–3. In the BSE Algol example SWINGERsim, re-modelling the stars as BSE does, ends the first transfer with a donor of 0.404 $M_\odot$ (paper 0.42 $M_\odot$, BSE in COSMIC 0.401 $M_\odot$; both PASS), an accretor of 2.51 $M_\odot$ and $P = 21.9$ d. The paper gives 2.85 $M_\odot$ and 30 d. These two benchmarks are EXPLAINED because the published history is not self-consistent: from 2.7 d to 17 d at the end of the Hertzsprung gap the orbital angular momentum would have to *grow* by at least 8 % (an upper bound without any loss gives 13.5 d), whereas the following giant phase, from 17 d to 30 d, is exactly conservative. Started from the published state at the end of the Hertzsprung gap (0.53 + 2.74 $M_\odot$, 17 d), SWINGERsim reproduces this giant phase (30.4 d against 30 d). The BSE code itself, run in COSMIC 4.2.1 with the same input and the BSE accretion limit, ends at 3.2 d with an accretor of 1.07 $M_\odot$: how much mass a thermal-time-scale accretor keeps depends on its thermal time scale (COSMIC uses $3.13{\cdot}10^7$ instead of $10^7$ yr in the corresponding equation) and on the implementation of BSE. It is the least certain quantity of Algol evolution, and SWINGERsim lets the user set it (`accretion: fractions`). The age of a 2 $M_\odot$ accretor after gaining 1 % of its mass follows the rejuvenation formula of Hurley et al. (2000) to 0.08 %.

## 3.8 Common envelope, magnetic braking, the Applegate mechanism and tidal capture

**Definitions.** If mass transfer becomes dynamically unstable, the envelope of the donor engulfs both stars (**common envelope**); in the energy formalism the orbital energy released by the spiral-in unbinds the envelope with an efficiency $\alpha_{\rm CE}$. **Magnetic braking** is the loss of spin angular momentum of a cool star through its magnetised wind. The **Applegate mechanism** explains cyclic period changes of close binaries by a magnetic cycle that changes the quadrupole moment of one star; the question is whether the star has enough energy for it. **Tidal capture**: two stars that pass close enough lose enough orbital energy to tides to become bound.

**References.** The 35-h orbit after the common envelope of a cataclysmic-variable progenitor of Tout et al. (1997) – 7 + 1 $M_\odot$, $P = 1200$ d, whose primary fills its Roche lobe on the AGB at 5.36 $M_\odot$ with a core of 1.014 $M_\odot$ (this is not the cataclysmic-variable example of Hurley et al. 2002, Sect. 3.2, which starts from 6.0 + 1.3 $M_\odot$ in a circular 630-d orbit) – and the 0.04-d orbit after the second common envelope of the BSE Algol example; the impulsive response of an outer orbit to the sudden loss of the envelope (Hadjidemetriou 1963; Veras et al. 2011);

the wind torque of Matt et al. (2015) and the Skumanich law $\Omega \propto t^{-1/2}$ (Skumanich 1972); the thin-shell energy ratios of HU Aqr, NN Ser and V471 Tau (Völschow et al. 2016, Table 4, column of the thin-shell model of Tian et al. 2009); and the capture radii of Ray et al. (1987), quoted by Portegies Zwart & Meinen (1993), and of McMillan et al. (1987).

**Table 21.** Common envelope, magnetism and tidal capture. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `ce-tout97-period` | own | 35 h | 34.84 h | −0.46 % | 10 % | PASS |
| `ce-bse-algol-second-period` | own | 0.04 d | 0.0365638 d | −8.59 % | 15 % | PASS |
| `ce-impulsive-outer-a` | R | 1 | 1.01274 | +1.27 % | 3 % | PASS |
| `ce-impulsive-outer-e` | R | 0 | 0.00442155 | +0.00442 | 0.02 | PASS |
| `matt15-sun-period` | own | 27.97 d | 27.7635 d | −0.74 % | 2 % | PASS |
| `matt15-unsaturated-powerlaw` | own | 0.5 | 0.491949 | −0.00805 | 0.02 | PASS |
| `applegate-huaqr` | own | 0.1 | 0.105659 | +5.66 % | 10 % | PASS |
| `applegate-nnser` | own | 3.3 | 3.30338 | +0.10 % | 10 % | PASS |
| `applegate-v471tau` | own | 0.014 | 0.0143688 | +2.63 % | 10 % | PASS |
| `tidal-capture-radius-n15-mcmillan` | own | 3.4 | 3.33454 | −1.93 % | 5 % | PASS |
| `tidal-capture-radius-n3` | own | 2.72 | 2.36599 | −13.01 % | 5 % | **EXPLAINED** |
| `tidal-capture-radius-n15` | own | 3.84 | 3.33454 | −13.16 % | 5 % | **EXPLAINED** |

**Result (Table 21).** The energy formalism reproduces the 35-h orbit (34.8 h; the donor fills its lobe in the orbit widened by its wind) and the 0.04-d orbit (0.037 d with $\alpha_{\rm CE}$ = 3). A 1.8 + 0.5 $M_\odot$ binary whose giant goes through a common envelope inside a circular 10-yr outer orbit (N-body) widens the outer orbit 1.3 % more than the impulsive prediction and makes it eccentric to within 0.004 of the prediction; the residual is the finite duration of the ejection. A solar model started at $P$ = 1 d spins at 27.76 d at the solar age (the solar value $2\pi/\Omega_\odot$ is 27.97 d), and the spin follows the Skumanich law (ratio 0.492 against 0.5 between 1.14 and 4.55 Gyr). The Applegate energy ratios agree with Völschow et al. to 0.1–5.7 %.

The two tidal-capture benchmarks are EXPLAINED as a difference of definition. SWINGERsim uses the fits of Portegies Zwart & Meinen (1993) (as implemented in NBODY6); for two 0.6 $M_\odot$ stars meeting at 10 km/s they give capture radii of 2.37 (polytrope $n$ = 3) and 3.33 (n = 1.5) stellar radii, 13 % less than the values of Ray et al. (1987) quoted in that paper. At the quoted radii, however, the tidal energy is 5 and 13 times smaller than the orbital energy that must be removed, so the quoted values cannot belong to the same velocity and stellar structure; they belong to other conventions of Ray et al. The same fits agree with McMillan et al. (1987), about 3.4 $R$ for $n$ = 1.5, to 1.9 % (PASS).

## 3.9 Contact binaries and the dynamical onset of mass transfer

**Definitions.** A **contact binary** is a pair whose stars both overfill their Roche lobes and share a common envelope (W UMa stars). It loses angular momentum (Gazeas & Stępień 2008) and merges when it reaches the Darwin instability or loses mass through the outer Lagrange point L2 (Rasio 1995). A **dynamical onset** of mass transfer happens when ZLK cycles drive the eccentricity of an inner binary so high that a star touches its Roche lobe at periastron; if tides then circularise the orbit while conserving angular momentum, the

circular separation is $a(1-e_c^2)$, and the donor fills $(1+e_c)$ times less of its lobe than at the contact (Hut 1981).

**Table 22.** Contact binaries and the dynamical onset. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `contact-jdot` | own | 2.18893 Gyr | 2.18898 Gyr | +0.00 % | 0.1 % | PASS |
| `dynamic-onset-circularisation` | own | 1 | 1 | −0.00 % | 0.01 % | PASS |
| `dynamic-onset-nbody-vs-secular` | own | 1 | 1.0173 | +1.73 % | 3 % | PASS |

**Result (Table 22).** The angular-momentum loss time of a 1.0 + 0.5 $M_\odot$ contact binary with $P$ = 0.35 d, computed in SWINGERsim's units, equals the value computed in cgs units from Gazeas & Stępień (2008) to $2{\cdot}10^{-5}$. The filling of the donor after circularisation at $e_c$ = 0.89 times $(1+e_c)$ is 1 to $2{\cdot}10^{-9}$. The time of the contact driven by ZLK cycles (about 61 yr) differs by 1.7 % between the N-body and the secular backend; the difference is the short-period terms of the outer orbit ($a_{\rm out}/a_{\rm in}$ = 15), which the double-averaged equations omit. Both backends place the onset at the eccentricity at which $a(1-e_c) = R/r_L$, with $r_L$ the Eggleton lobe radius in units of $a$. The full evolution of a binary into contact and its merger is shown in Chapter 8.

# Chapter 4

# Observables: light curves, timings and pulsations

Astronomers do not observe orbits; they observe light. This chapter tests how SWINGERsim turns its orbits and stars into what a telescope records: light curves of eclipsing stars (spherical, distorted and in contact), the dimming by dust discs, radial velocities, the times of eclipse minima and their deviations from a strict period, and the timing of pulsations. All modules in this chapter are SWINGERsim's own code, except the PHOEBE hybrid model, which calls the external code PHOEBE 2.5 (Prša et al. 2016) for the light of Roche-distorted stars.

## 4.1 Light curves of spherical stars

**Definitions.** SWINGERsim's basic photometric model (called **tier A** in the software) treats stars as spheres with a radially symmetric limb darkening (the four-coefficient law of Claret 2000) and computes the flux blocked in an eclipse by Gauss–Legendre integration over the visible disk. The limb darkening comes from the tables of Claret & Bloemen (2011, ATLAS atmospheres) and, for stars hotter than 15 000 K, of Reeve & Howarth (2016, non-LTE TLUSTY atmospheres). Ellipsoidal variation – the light change of a tidally elongated star – is added as the leading term of Morris (1985).

**References.** The analytic occultation of a limb-darkened star by a dark disk (Mandel & Agol 2002) as implemented in batman (Kreidberg 2015); the eclipse depths of a detached binary of spheres in ellc (Maxted 2016); the ellipsoidal variation of ellc's Roche model; PHOEBE 2.5 with the same physics (spheres, black bodies, Claret limb darkening); the published TLUSTY coefficients and the overlap of TLUSTY and ATLAS at 15 000–30 000 K.

**Table 23.** Light curves of spherical stars and limb darkening. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `lc-occultation-small` | own | 0.987891 | 0.987891 | $-1.9\cdot10^{-12}$ | $10^{-7}$ | PASS |
| `lc-occultation-limb` | own | 0.945236 | 0.945236 | $+3.7\cdot10^{-9}$ | $10^{-7}$ | PASS |
| `lc-occultation-large` | own | 0.128979 | 0.128979 | $-5.5\cdot10^{-9}$ | $10^{-7}$ | PASS |
| `lc-detached-primary` | own | 0.380821 | 0.380868 | $+4.7\cdot10^{-5}$ | $10^{-4}$ | PASS |
| `lc-detached-secondary` | own | 0.144658 | 0.144654 | $-3.9\cdot10^{-6}$ | $10^{-4}$ | PASS |
| `lc-ellipsoidal` | own | 0.0027949 | 0.00260619 | −6.75 % | 10 % | PASS |
| `phoebe-sphere-vs-tier-a` | own | 0 | 0.00037 | +0.00037 | 0.001 | PASS |
| `ld-tlusty-row` | own/tab | 1.18262 | 1.18262 | +0 | $10^{-4}$ | PASS |
| `ld-tlusty-vs-atlas` | own/tab | 1 | 0.997587 | −0.24 % | 2 % | PASS |

**Result (Table 23).** For small (radius ratio 0.1), grazing and large (1.2) occulters the flux agrees with the analytic model to better than $10^{-8}$: the integration is exact for spheres in practice. The eclipse depths of a detached binary agree with ellc to $5{\cdot}10^{-5}$, which is the accuracy of ellc's integration grid. Against PHOEBE 2.5 with the same physics, the largest difference of the normalised light curves over a whole orbit ($e$ = 0.1) is $3.7{\cdot}10^{-4}$ of the flux. The small-distortion ellipsoidal term is 6.8 % below ellc's Roche model at $R/a$ = 0.15; at $R/a$ = 0.10, 0.20 and 0.25 the deficit is 3, 11 and 17 %. This benchmark marks the limit of the spherical model: closer pairs need the surface meshes of the next section. The TLUSTY coefficients are reproduced at the grid points, and the disk-averaged intensities of TLUSTY and ATLAS agree to 0.2–0.7 % where both apply, so the switch between them at 15 000 K is smooth.

## 4.2 Distorted and contact stars on surface meshes

**Definitions.** For stars that are far from spherical SWINGERsim builds **surface meshes**: the surface of a rigidly rotating oblate star or of a star filling (or overfilling) its Roche lobe is divided into elements of 64 × 128 directions, each with its own temperature (gravity darkening), limb darkening and projected area; the eclipses are computed element by element. The same meshes give the **centre of light** of a star: in a contact binary the radial velocity measured from spectral lines is the flux-weighted velocity of the visible surface, which differs from the velocity of the centre of mass (and during an eclipse shows the Rossiter–McLaughlin effect; Rossiter 1924; McLaughlin 1924).

**Table 24.** Surface meshes, contact binaries and centre-of-light velocities. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `mesh-sphere-eclipse` | own | 0 | 0.00026 | +0.00026 | 0.001 | PASS |
| `contact-vs-phoebe` | own | 0 mag | 0.0030 mag | +0.0030 mag | 0.03 mag | PASS |
| `rv-light-vs-phoebe` | own | 0 | 0.022 | +0.022 | 0.05 | PASS |

**Result (Table 24).** Meshes of spheres reproduce the exact sphere eclipses of the previous section to $2.6{\cdot}10^{-4}$ of the flux; the edges of the stars in front are anti-aliased by the projected size of each element. A contact binary ($q$ = 0.42, fill-out factor 0.3, $i$ = 80°) has the same V amplitude as in PHOEBE 2.5 with black bodies and the same limb darkening to 0.003 mag. The centre-of-light velocity of its secondary differs from PHOEBE's flux-weighted velocity by at most 2.2 % of the 52-km/s amplitude; the largest differences occur inside the eclipses, where the two codes' meshes resolve the occulted edge differently. Automatic tests add that out of eclipse the offset of a spherical star is exactly zero, that the Rossiter–McLaughlin signal is antisymmetric about mid-eclipse with the expected size $v_{\mathrm{rot}}(R_{\mathrm{occ}}/R)^2$, and that velocities of a contact binary computed with the centre of light but fitted with the centre of mass give a $\chi^2$ more than twice as large and masses several per cent too large, whereas fitting with `fit.rv.centre: light` returns the masses within 3σ.

## 4.3 Dust discs as occulters

**Definitions.** Some eclipses are caused not by a star but by a dark dust disc around a companion, or by a ring around a whole binary. SWINGERsim represents a disc as a **slab** – a cylinder of outer radius $r_{\mathrm{out}}$, an optional inner hole $r_{\mathrm{in}}$, thickness $H$ and optical depth $\tau$. A star behind it is dimmed by $e^{-\tau L/H}$, where $L$ is the length of the line of sight inside the slab. The edge can be sharp or soft, the disc can emit and scatter light, and it can be fixed, precess at a prescribed rate or precess as a rigid body under the torque of its binary or of a companion. A disc around an eccentric binary can settle **polar**, perpendicular to the binary

orbit; above a critical inclination it librates around the polar orientation (Farago & Laskar 2010; Martin & Lubow 2017).

**References.** Exact geometry for the transmission; the test-particle precession rate for a narrow rigid ring (Zanazzi & Lai 2018, eq. 2); $\cos I_{\rm crit} = \sqrt{5e^2/(1+4e^2)}$ for the polar libration; and three observed systems: ε Aur (the F supergiant eclipsed every 27.1 yr for about two years by the disc around its B-star companion; orbit of Stefanik et al. 2010, low-mass model of Hoard et al. 2010, disc of Kloppenborg et al. 2015), HD 98800 B (a polar circumbinary disc, inclined 26° to the sky; Kennedy et al. 2019; orbit of Zúñiga-Fernández et al. 2026) and EE Cep (a B star eclipsed every 5.6 yr with depths that change with a period of 11–12 orbits, 62–67 yr; Gałan et al. 2012; Pieńkowski et al. 2020).

**Table 25.** Dust discs. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `disc-strip-uniform` | own | 0.623838 | 0.62475 | +0.000912 | 0.005 | PASS |
| `disc-face-on-tau` | own | 0.496585 | 0.496585 | +0 | $10^{-9}$ | PASS |
| `disc-inclined-tau` | own | 0.246597 | 0.246597 | $-4{\cdot}10^{-16}$ | $10^{-9}$ | PASS |
| `disc-ring-precession` | own | 1 | 1.00018 | +0.02 % | 0.1 % | PASS |
| `disc-polar-critical` | own | 37.7612° | 37.7612° | $+6{\cdot}10^{-9}$° | 0.2° | PASS |
| `epsaur-mid-eclipse` | own | JD 2455429.5 | JD 2455414.5 | −15 d | 30 d | PASS |
| `epsaur-duration` | own | 740 d | 638 d | −13.78 % | 15 % | PASS |
| `epsaur-depth` | own | 0.75 mag | 0.83134 mag | +0.0813 mag | 0.15 mag | PASS |
| `hd98800-polar-disc` | own | 26° | 28.4735° | +2.47° | 3° | PASS |
| `eecep-disc-precession` | own | 65 yr | 183.163 yr | +181.79 % | 15 % | **EXPLAINED** |

**Result: geometry and dynamics (Table 25).** An opaque edge-on slab of half-thickness 0.3 $R$ across a uniform star transmits 0.6248 of the light against the exact 0.6238: the stellar disk is sampled on a sunflower lattice of 4000 points, which resolves a sharp edge to about $10^{-3}$ of the flux. Face-on and inclined discs of optical depth 0.7 transmit exactly $e^{-0.7}$ and $e^{-1.4}$. A narrow rigid ring at four binary separations precesses at the test-particle rate to 0.02 %, and the rigid-disc equation reproduces the critical inclination of polar libration for $e$ = 0.5 (37.76°) exactly.

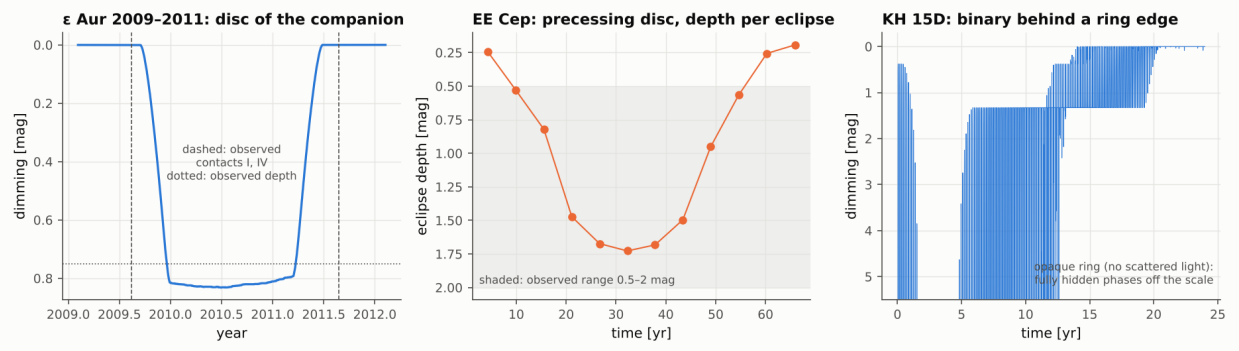



**Figure 5.** Dust discs as occulters, with a sharp-edged opaque disc. In all panels the vertical axis is the dimming in magnitudes, growing downwards as astronomers plot it. *Left* (ε Aur): the computed light curve of the 2009–2011 eclipse against Julian date; dashed vertical lines: the observed first and last contacts; dotted horizontal line: the observed depth. *Middle* (EE Cep, illustrative disc): the depth of each eclipse (points) over 70 yr as the disc precesses in a prescribed 65 yr; the grey band is the observed range of depths. *Right* (KH 15D, illustrative ring): 25 years of the light curve of a young binary disappearing behind the edge of a precessing circumbinary ring and re-emerging.

**Result: observed systems.** Figure 5 shows the three discs with a sharp edge. For ε Aur (left), the time of mid-eclipse follows from the spectroscopic orbit alone and falls 15 d before the observed mid-point of the contacts (2009 August 16 to 2011 August 26). The eclipse is 638 d long against the observed 740 d (−14 %) and 0.83 mag deep against 0.75 mag. The shorter duration is the sharp edge: the observed ingress takes about 145 d (Stencel 2013), so the real edge is soft. The depth is 0.08 mag too large because the model omits the light of the B star inside the disc and the light scattered by the disc. For HD 98800 B, a disc put perpendicular to the orbit of B and along its eccentricity vector – the polar alignment predicted for discs around eccentric binaries – is inclined 28.5° to the sky, computed from the published orbit without any fit; the observed disc is inclined 26°. In the middle panel, an illustrative EE Cep disc (1.6 au across, $\tau = 1$, tilted 15°) whose precession is *prescribed* at 65 yr produces eclipse depths of 0.19–1.73 mag, close to the observed 0.5–2 mag. In the right panel, an illustrative KH 15D ring (opaque, 3–5 au, tilted 20°, precessing in 600 yr) reproduces the observed sequence: first one star of the binary peeks out near apastron (spikes), then both are hidden for years, then the binary re-emerges.

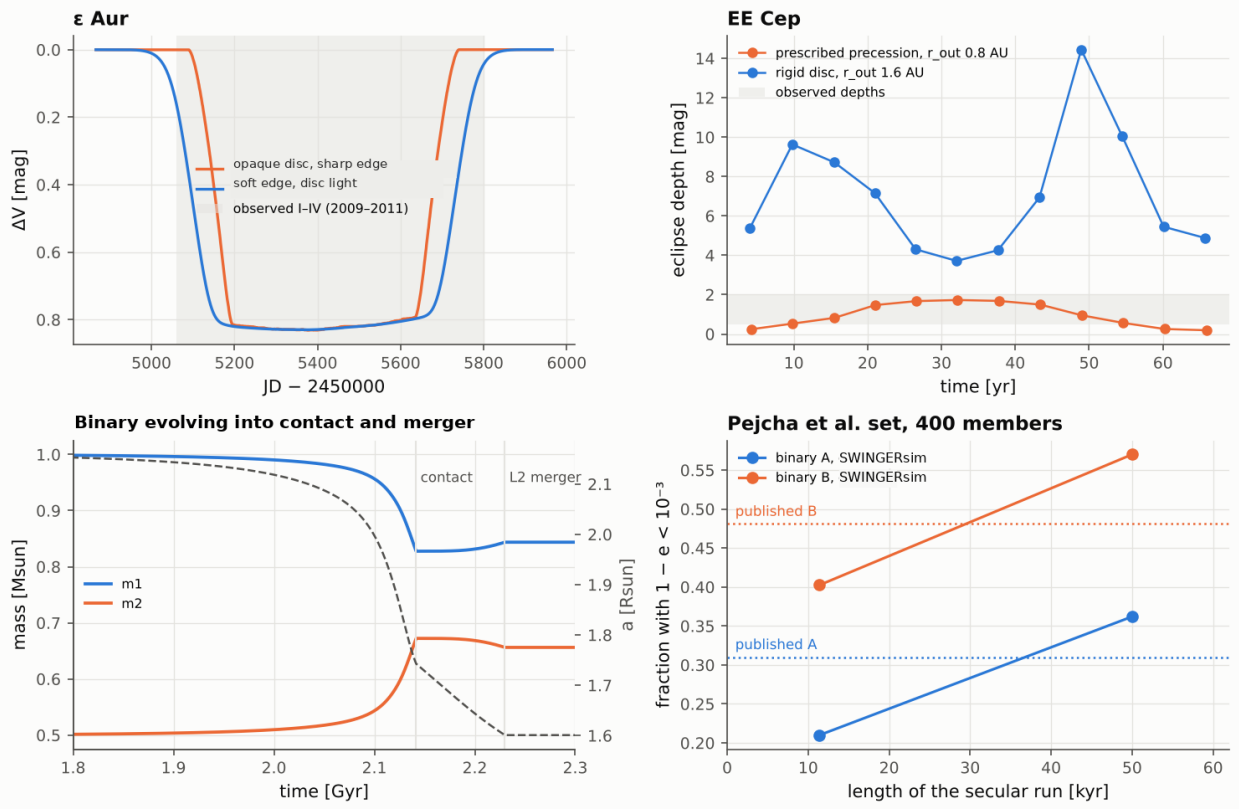



**Figure 6.** Discs with soft edges and rigid dynamics, contact evolution and the Pejcha fractions. *Top left* (ε Aur): dimming against Julian date for the opaque sharp-edged disc (orange) and for a disc with a soft edge and its own light (blue); grey area: the observed first to last contact. *Top right* (EE Cep): the depth of each eclipse (magnitudes, points connected by lines) over 65 years, for the disc of outer radius 0.8 au whose 65-yr precession is prescribed (orange) and for the rigid disc of outer radius 1.6 au whose torque-driven precession has 65 yr (blue); grey band: observed depths. *Bottom left:* masses of the two stars (solid, left axis) and their separation (dashed, right axis) of a 1.0 + 0.5 $M_\odot$ binary evolving into contact and merging (see Chapter 8). *Bottom right:* the fraction of the Pejcha et al. (2013) set (Vokrouhlický set-up, 400 members) whose binary A (blue) or B (orange) has reached $1 - e < 10^{-3}$ after secular runs of 11 kyr and 50 kyr; dotted: the published fractions.

**Result: soft edges and torques.** Figure 6 shows what happens when the discs get a soft edge, their own light, and a precession computed from the torque instead of prescribed. For ε Aur (top left) the soft edge makes the ingress take 141 d, as observed (about 145 d; the sharp edge gives 89 d), but the eclipse becomes 829 d long, 12 % longer than observed; the width of the soft edge is assumed, not fitted, and the disc adds only 0.1 % of light. For EE Cep the illustrative disc of outer radius 0.8 au, made rigid, precesses in 183 yr under the torque

of the B star – not in 65 yr. The benchmark is EXPLAINED as a constraint on the disc rather than an error: the rate scales as $r_{\rm out}^{3/2}$, and 65 yr requires $r_{\rm out} \approx 1.6$ au (or a heavier B star). With that size and the illustrative thickness and optical depth, however, the eclipses become 3.7–14 mag deep and 36–158 d long (top right, blue), far outside the observed 0.5–2 mag. A disc that is rigid and precesses with the observed period must therefore also be thinner or less opaque than the illustrative one. The KH 15D ring of 3–5 au, made rigid, turns by only 2.6° in 60 yr (period about 6500 yr) and hides the binary 98 % of the time; the change observed within decades needs 1000–3000 yr, so the ring must be closer to the binary. These are constraints on the discs, which SWINGERsim derives because its disc dynamics passes the analytic benchmarks above. The bottom panels of Figure 6 belong to Chapter 8 and to the Pejcha ensembles of Chapter 2: the published fractions (dotted) lie between the values after 11 kyr and after 50 kyr.

## 4.4 Eclipse timings and O−C diagrams

**Definitions.** An observer predicts the times of minima of an eclipsing binary with a constant period, $T_n = T_0 + nP$, and plots the **O−C diagram** (observed minus computed time against cycle number $n$). Any structure in it is a signal. The **light-travel-time effect** (LTTE) of a third body moves the minima by up to $a_{12} \sin i/c$, where $a_{12}$ is the semi-major axis of the binary's orbit around the common centre of mass (Irwin 1952). The **dynamical** eclipse-timing variation (ETV) is the change of the binary's period by the pull of the third body. **Apsidal motion** moves the primary and secondary minima in opposite directions. Inside the binary itself, the secondary eclipse is delayed by the **Rømer delay** $(P/\pi c)(K_2 - K_1)$, because the two stars are at different distances from us at the two conjunctions (Kaplan 2010). A steady period change $\dot{P}$ bends the O−C into a parabola, $\frac{1}{2}\dot{P}Pn^2$ (Kwee 1958; Hilditch 2001).

SWINGERsim offers three ways to compute minima: exact conjunctions of the N-body integration, including the light travel time (`swingersim timings`); minima measured from a synthetic light curve as an observer would; and the PHOEBE hybrid, which computes the light of Roche-distorted stars with PHOEBE 2.5 in chunks of the N-body orbit. For observed minima it provides two fits: **stage 1**, the analytic O−C model used by the SQUADRA group in OCFit (Gajdoš & Parimucha 2019) – LTTE, apsidal motion in the series of Giménez & Bastero (1995) and a quadratic term for every pair – and **stage 2**, the joint N-body model of all eclipsing pairs (tested in Parts II–IV).

**Table 26.** Eclipse timings and the stage-1 fit. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `timing-ltte-amplitude` | own | 41.5837 min | 41.5863 min | +0.01 % | 0.1 % | PASS |
| `phenom-romer-delay` | own | 27.4517 s | 27.3578 s | −0.34 % | 1.5 % | PASS |
| `phoebe-romer-delay` | ext | 27.452 s | 27.287 s | −0.60 % | 2 % | PASS |
| `pdot-parabola` | R | 1 | 1.00005 | +0.00 % | 0.1 % | PASS |
| `ocfit-gimenez-bastero` | own | 0 s | 0.00063 s | +0.00063 s | 0.01 s | PASS |
| `ocfit-lite-amplitude` | own | 1 | 1 | −0.00 % | $10^{-4}$ % | PASS |
| `ocfit-nbody-asini-A` | own | 1 | 0.99864 | −0.14 % | 1 % | PASS |
| `ocfit-nbody-asini-B` | own | 1 | 1.0024 | +0.24 % | 1 % | PASS |
| `ocfit-nbody-P3` | own | 1 | 0.99996 | −0.004 % | 0.5 % | PASS |

**Result (Table 26).** The synthetic O−C of a binary on a circular edge-on outer orbit of 10 au (equal masses) has the LTTE semi-amplitude of Irwin (1952) to $10^{-4}$; the dynamical delays of this wide system are below 0.01 min. In a 2 + 1 $M_\odot$ ZAMS binary with $P$ = 5 d and $i$ = 89°, the secondary minima measured from the light curve come 27.36 s after half the period,

against the Rømer delay of 27.45 s (−0.34 %); the PHOEBE hybrid gives 27.29 s (−0.60 %). A binary whose period changes by $4{\cdot}10^{-6}$ d/yr over 60 yr, integrated with N-body and period kicks, gives the quadratic coefficient of its minima to $5{\cdot}10^{-5}$; a stage-2 fit of 160 such minima recovers $\dot{P}$ within its error (below 5 % of the value). In the stage-1 fit, SWINGERsim's apsidal term computed from exact conjunctions agrees with the nine-term series of Giménez & Bastero (1995) used by OCFit to 0.6 ms at $e$ = 0.3 (at $e$ = 0.6 the truncated series is off by 1.6 s), and its LTTE term equals OCFit's. Finally, the eclipse timings of an N-body 2+2 with a 1500-d outer orbit ($e$ = 0.3, $i$ = 60°), fitted in stage 1 with one shared outer orbit, return the Keplerian $a \sin i$ of both pairs to 0.14 % and 0.24 % and the outer period to $4{\cdot}10^{-5}$. The remaining difference is the dynamical delay, of order $(3/8\pi)P_{\rm in}^2/P_{\rm out}$, which the analytic stage-1 model does not contain.

## 4.5 Pulsation timing

**Definitions.** A pulsating star is a clock. In a binary its pulsations arrive early and late by the light travel time across its orbit, exactly as eclipses do. SWINGERsim includes the pulsations in its light curves, with periods from published relations (period–luminosity of δ Scuti stars, pulsation constant $Q$, Cepheid models), and measures the delays as an observer does: it fits the pulsation phase in short segments of the light curve (Murphy et al. 2014).

**References.** For a circular orbit the amplitude of the time delays is $a_1 \sin i/c$ (Murphy et al. 2014, Sect. 2.7, eq. 10); for an eccentric orbit the ratio of the second to the first harmonic of the delays depends only on the eccentricity; KIC 9651065, a δ Scuti star in a 272.7-d orbit (Murphy et al. 2014, Table A1); the period–luminosity relation of 2375 Galactic δ Scuti stars (Poro et al. 2024, eq. 10) with the fundamental pulsation constant $Q$ = 0.0329 d (Breger 1979); and the period–radius relation of rotating Geneva Cepheid models (Anderson et al. 2016, Table 5) with Cepheid periods from Bono et al. (2000).

**Table 27.** Pulsation timing and pulsation periods. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `pulsation-time-delay-circular` | own | 53.859 s | 53.859 s | +0.00 % | 0.01 % | PASS |
| `murphy2014-kic9651065-a1` | own | 165.57 s | 167.767 s | +1.33 % | 6 % | PASS |
| `murphy2014-kic9651065-a2-a1` | own | 0.231986 | 0.219651 | −0.0123 | 0.025 | PASS |
| `delta-scuti-pl-1p8` | own/tab | 0 mag | −0.0252 mag | −0.0252 mag | 0.25 mag | PASS |
| `delta-scuti-pl-2p0` | own/tab | 0 mag | −0.115 mag | −0.115 mag | 0.25 mag | PASS |
| `cepheid-period-radius-6` | own/tab | 0 dex | −0.0077 dex | −0.0077 dex | 0.05 dex | PASS |
| `cepheid-period-radius-7` | own/tab | 0 dex | +0.0039 dex | +0.0039 dex | 0.05 dex | PASS |

**Result (Table 27).** A δ Scuti-like pulsator (1.7 $M_\odot$, pulsation period 0.1 d) with a 0.5 $M_\odot$ companion ($P$ = 100 d, $i$ = 60°), sampled like Kepler long-cadence data and analysed in 2-d segments, gives the delay amplitude $a_1 \sin i/c$ = 53.859 s to $3{\cdot}10^{-7}$ after the segment averaging is corrected by its sinc factor. For KIC 9651065, with the published orbit ($a_1 \sin i$ = 0.37 au, $e$ = 0.468, $\varpi$ = 2.01 rad), the first harmonic of the delays is 167.8 s against the measured 165.57 ± 2.03 s (+1.3 %; the tolerance of 6 % is the uncertainty of $a_1 \sin i$), and the ratio of the harmonics 0.220 against 0.232 ± 0.013, which checks the eccentric-orbit structure of the delays. MIST stars of 1.8 and 2.0 $M_\odot$, where they cross $T_{\rm eff}$ = 7500 K on the main sequence, lie 0.03 and 0.12 mag from the δ Scuti period–luminosity relation (the

tolerance of 0.25 mag reflects the uncertainty of $Q$, up to 18 %). MIST Cepheids of 6 and 7 $M_{\odot}$ on their blue loops have radii within 0.008 and 0.004 dex of the Geneva period–radius relation; the tolerance of 0.05 dex is the difference that Anderson et al. (2016) find between predicted and empirical radii, and the two chains of models share no code. The light curves and delays of real pulsators in multiple systems are shown in Chapter 7 and Chapter 8.

# Chapter 5

# QZ Car against Brož et al. (2022)

QZ Car (HD 93206) in the Carina nebula is a 2+2 quadruple of O stars, among the most massive stars known: the eclipsing binary Ac (Ac1 25.5 + Ac2 33.2 $M_\odot$, $P$ = 6.0 d), the eccentric binary Aa (Aa1 69.8 + Aa2 8.85 $M_\odot$, $P$ = 20.7 d) and an outer orbit of 14 728 d (about 40 yr). Brož et al. (2022) fitted 40 years of eclipses, spectra and interferometry with their N-body code Xitau and published a complete model. It is the most massive system in our tests and a strict test of SWINGERsim's geometry and photometry, because the orbit sizes and the light shares of the four stars are known precisely. Their Table 2 is the input of SWINGERsim's QZ Car configurations; radii and effective temperatures are taken from their Tables 2 and 3. In the configuration the four stars are m1 = Ac1, m2 = Ac2, m3 = Aa1 and m4 = Aa2.

## 5.1 Geometry, light and the evolutionary masses

**Table 28.** QZ Car benchmarks. The first three rows test SWINGERsim's hierarchical initialisation (semi-major axes from periods and masses), the next four its photometry (V-band light shares from radii and temperatures), the last three the MIST tracks (the mass of a single-star model with the observed radius and temperature). Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `qzcar-broz-a1` | own | 54.032 $R_\odot$ | 53.9946 $R_\odot$ | −0.07 % | 0.2 % | PASS |
| `qzcar-broz-a2` | own | 136.132 $R_\odot$ | 136.081 $R_\odot$ | −0.04 % | 0.2 % | PASS |
| `qzcar-broz-a3` | own | 13 053 $R_\odot$ | 13 046 $R_\odot$ | −0.05 % | 0.2 % | PASS |
| `qzcar-broz-lv-ac1` | own | 0.22 | 0.220775 | +0.000775 | 0.005 | PASS |
| `qzcar-broz-lv-ac2` | own | 0.133 | 0.133856 | +0.000856 | 0.005 | PASS |
| `qzcar-broz-lv-aa1` | own | 0.635 | 0.635024 | $+2.4\cdot10^{-5}$ | 0.005 | PASS |
| `qzcar-broz-lv-aa2` | own | 0.01 | 0.0103458 | +0.000346 | 0.005 | PASS |
| `qzcar-evolutionary-mass-ac1` | own/tab | 25.5 $M_\odot$ | 27.0258 $M_\odot$ | +5.98 % | 15 % | PASS |
| `qzcar-evolutionary-mass-ac2` | own/tab | 33.2 $M_\odot$ | 26.0243 $M_\odot$ | −21.61 % | 15 % | **EXPLAINED** |
| `qzcar-evolutionary-mass-aa1` | own/tab | 69.8 $M_\odot$ | 42.5172 $M_\odot$ | −39.09 % | 15 % | **EXPLAINED** |

Table 28 lists the ten benchmarks; the paragraphs below read it group by group.

**Geometry (level B).** The three semi-major axes follow from the published periods and masses to 0.04–0.07 %; the tolerance of 0.2 % covers the rounding of the masses in Table 2 to three digits.

**Light (level C).** Black bodies with the Claret/TLUSTY limb darkening and the published radii and temperatures give the shares of the four stars in the V-band light – 0.221, 0.134, 0.635 and 0.010 – to 0.001 of the values that Brož et al. derived from synthetic spectra. The tolerance of 0.005 is the difference between a black body and a model atmosphere at these temperatures.

**Evolutionary masses (level D).** The **evolutionary mass** of a star is the mass of the MIST track that reaches its observed radius and temperature; the **dynamical mass** comes from the orbits. For a normal single star they agree. Ac1 is a 27.0 $M_\odot$ star of 5.6 Myr on the MIST tracks, within 6 % of its dynamical mass. Ac2 (26.0 $M_\odot$, 5.1 Myr) and above all Aa1 (42.5 $M_\odot$, 3.9 Myr) do not fit their dynamical masses of 33.2 and 69.8 $M_\odot$. These two benchmarks are EXPLAINED: both stars are likely mass gainers. The nominal model of Brož et al. (2022, Table 2) fits the temperature of Ac2 (32 979 K) and leaves its rotation free; the fit gives 410 km/s, the signature of a star that has accreted angular momentum in the Ac1 → Ac2 transfer that Brož et al. describe. For Aa1, Brož et al. note that it is extremely massive compared with Aa2 and suggest past transfer Aa2 → Aa1. A gainer is rejuvenated (fresh hydrogen mixed into its core) and looks younger and less massive than a single star of the same mass; single-star tracks cannot describe it. Placed at their dynamical masses, the MIST tracks reach the observed radii at 6.0 Myr (Ac1), 3.5 Myr (Ac2) and 2.5 Myr (Aa1), and the rotating tracks never give Aa1 its observed 28 $R_\odot$ at 70 $M_\odot$: single-star tracks cannot make the system coeval. SWINGERsim therefore lets a star start with an effective age (`stars.<name>.age_offset_myr`) so that every component begins with its observed radius, and it rejuvenates gainers during a run (Chapter 3).

## 5.2 Synthetic minima

**Input and method.** The minima of the eclipsing pair Ac were computed over one outer period (40.5 yr) with the spherical model from a V-band light curve with 10-min cadence and yearly observing seasons (4933 minima), and over one year with the PHOEBE hybrid, which deforms the stars into their Roche shapes (4000 triangles per star).

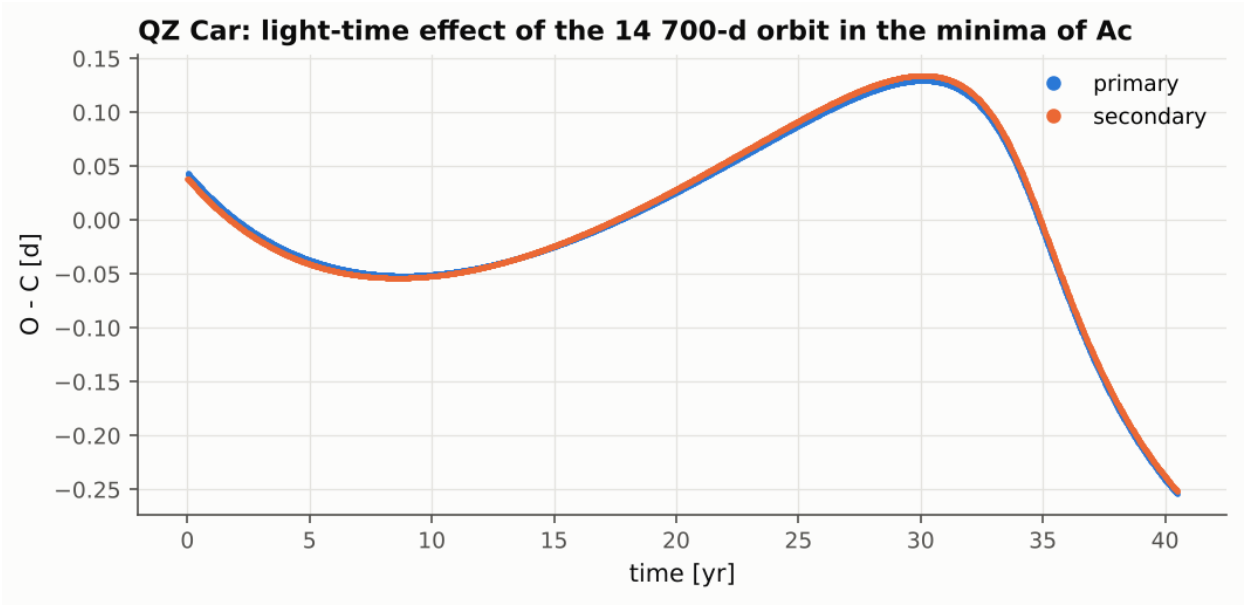



**Figure 7.** O−C diagram of the synthetic minima of the eclipsing pair Ac of QZ Car over one outer period of about 40 years. Horizontal axis: time; vertical axis: O−C in days, the time of each minimum minus the time predicted with a constant period. Blue: primary minima; orange: secondary minima. The wave is the light-travel-time effect of the outer orbit.

**Result.** Figure 7 shows the O−C of the 4933 minima. If Ac were alone, all points would lie on a horizontal line. Instead they follow a wave of about ±0.2 d (an O−C range of 0.38 d): while Ac moves away from us on the outer orbit its minima come late, while it approaches they come early. The wave is not a sinusoid because the outer orbit is eccentric. Its amplitude equals $a_{12} \sin i/c$ of the published outer orbit. Primary and secondary minima lie on the same curve, so both are timed equally well. Measured against the barycentric conjunctions, the primary minima come 16 s late and the secondary minima 10 s early: the Rømer delay inside Ac is ±16 s (Kaplan 2010), and the slightly eccentric orbit of Ac adds the rest. The PHOEBE minima over one year agree with the spherical-model minima to 1 s for the secondary and lie within +6 s and −5 s of the conjunctions over three weeks. The timing of a massive, tidally distorted eclipsing binary is thus reproduced to a few seconds by two independent light-curve models.

## 5.3 Present-day dynamics

**Input.** An N-body ensemble of 2000 members with masses and orbital elements drawn within the uncertainties of Brož et al. (2022) and the unknown nodes of the binaries drawn isotropically, integrated 100 yr forward and 100 yr backward, with the condition that Ac has eclipsed since 1955.

**Result.** All 2000 members stay intact in both directions, and all are stable by the Mardling–Aarseth and Vynatheya et al. (2022) criteria. Over a century the orbits hardly change: the inclination of Ac drifts by a median of $9{\cdot}10^{-4}$ °/yr (16–84 % range −0.0011 to +0.0010 °/yr), about 0.04° in 40 yr, which Brož et al. could not have measured. Ac therefore eclipses for centuries; Aa eclipses in 24 % of the isotropic orientations of its node. QZ Car is a stable laboratory for timing studies.

## 5.4 Evolution to the first Roche-lobe overflow

**Question.** The stars of QZ Car are young (2–6 Myr) but massive stars evolve fast. Which star fills its Roche lobe first, and when?

**Input.** Three ensembles of 1000 members each with the evolving backend and MIST tracks, masses drawn within the published errors: (a) all stars coeval at 2.5 Myr on non-rotating tracks; (b) the same with rotating tracks ($v/v_{\mathrm{crit}}$ = 0.4), which have stronger winds; (c) effective ages chosen so that every star starts with its observed radius (the stars are then not coeval).

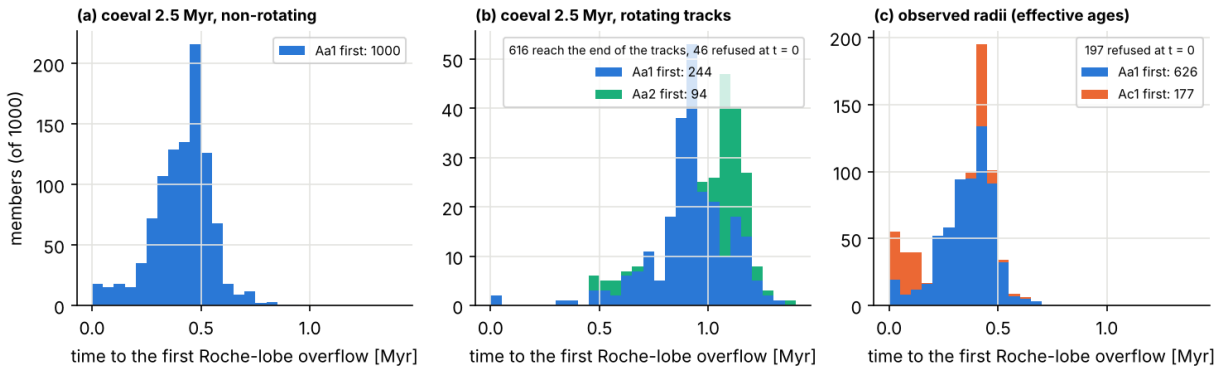



**Figure 8.** QZ Car: time from today to the first Roche-lobe overflow (horizontal axis, Myr) in three ensembles of 1000 members (vertical axis: number of members per 0.05-Myr bin). Colours show which star overflows first: blue Aa1 (the 69.8 $M_\odot$ supergiant), green Aa2, orange Ac1. The legend gives the number of members per star and, at its top, the members that end otherwise. (a) Coeval stars at 2.5 Myr, non-rotating MIST tracks. (b) The same with rotating tracks. (c) Effective ages that give every star its observed radius.

**Result.** Figure 8 shows the three ensembles. (a) With coeval non-rotating stars, all 1000 members end with a Roche-lobe overflow of Aa1, after a median of 0.44 Myr (16–84 %: 0.30–0.53 Myr); the outer orbit widens slightly by the winds (median from 60.5 to 62.1 au). (b) With rotating tracks the stronger winds widen the orbits: only 338 members overflow (Aa1 first in 244, median 0.92 Myr; Aa2 first in 94, median 1.10 Myr), while 616 reach the end of their tracks without overflow, with Aa widened to about 2.3 au and the outer orbit to about 105 au; 46 members are refused, because an Aa1 heavier than about 73.3 $M_\odot$ does not exist at 2.5 Myr on the rotating grid. (c) With the observed radii, 803 members end in an overflow – Aa1 first in 626 (median 0.38 Myr) and Ac1 first in 177 (median 0.12 Myr) – and 197 are refused at the start, because an Ac1 heavier than 26.4 $M_\odot$ at its effective age already overfills its Roche lobe in the observed 6-d orbit. This last number is a result in itself: the observed radius of Ac1 and the fact that Ac is detached bound the mass of Ac1 from above, a constraint the dynamical fit does not contain. In panel (c) the two binaries race: Ac1 wins only when the overflow comes very soon, in members in which Ac1 is drawn close to that upper bound.

There is no published simulation of the future of QZ Car to compare with; the evolution is a prediction. Its ingredients are validated separately: the MIST tracks of massive stars against Geneva models and leave-one-out interpolation, the winds against Vink et al. (2001) (Chapter 3), the widening of orbits by winds against Jeans' law (Chapters 3 and 7). The spread between the three ensembles – a factor of two in the time to the overflow – is the uncertainty of the stellar models of very massive stars, not of the dynamics.

# Chapter 6

# The SQUADRA systems

SQUADRA is the survey of 2+2 quadruple candidates of J. Kolář, P. Zasche and collaborators: systems in which TESS sees two eclipsing pairs in one star image (Kolář et al. 2025: 781 candidates). The question for SWINGERsim is whether the observed behaviour – eclipse depths that change, eclipse times that deviate from a constant period – can be explained by a dynamical model, and which hierarchy is the right one. This chapter does not reproduce a published model, because none exists for these systems; it tests the ensemble machinery of SWINGERsim on real targets: conditioning on observations, the evidence of competing models, the agreement of the secular and N-body backends, and the symmetry of the dynamics in time. The detailed analyses of the SQUADRA systems with real minima and light curves are the subject of Parts II–IV.

**Definitions.** In an ensemble, each member gets a **weight** – the likelihood of the observations given its parameters, for example $\exp(-\chi^2/2)$ for a measured rate of change with its error. The mean weight over all members drawn from the prior is the **evidence** $Z$ of the model; the ratio of the evidences of two models (the Bayes factor) says which explains the data better. The **effective number of members** $n_{\mathrm{eff}} = (\sum w)^2 / \sum w^2$ (Kish 1965) counts how many members actually carry the weight; an $n_{\mathrm{eff}}$ near 1 means that one member carries everything and the evidence is not reliable. The **fate** of a member says how its run ended: intact, collision, ejection, reconfiguration (a change of hierarchy), exchange or breakup.

## 6.1 CzeV4315: 2+2 or 2+(2+1)?

**The system.** CzeV4315 shows two eclipsing pairs: A (6.74 d, eccentric, $e$ = 0.64), whose eclipses have been stable for years, and B (0.92 d), whose inclination changes by −1.4 ± 0.2 °/yr – its eclipses are getting shallower quickly. An eclipse-timing variation with a period of about 119 d is seen. Is it a plain 2+2 quadruple, or a 2+(2+1) quintuple in which B has its own close companion on a 119-d orbit?

**Input.** A 2+2 ensemble of 20 000 members with the secular backend (±300 yr; outer period log-uniform between 40 and 3000 d, isotropic orientations, estimated masses), and a 2+(2+1) ensemble of 10 000 members with the N-body backend and relativity (±60 yr; companion of B on an orbit of 119 ± 3 d). The observations used as conditions: the tilt rate of B, no Roche-lobe overflow, and optionally A eclipsing for at least 5 yr in the past and at least 0.25 yr ahead.

**Table 29.** Evidence of the two models of CzeV4315. Column 3 lists the observations that condition the ensemble; columns 4–5 the evidence and the effective number of members.

| Model | Members | Conditions | Evidence $Z$ | $n_{\mathrm{eff}}$ |
|---|---|---|---|---|
| 2+2, broad prior of the outer period | 20 000 | tilt rate of B | 0.0063 | 177 |

| Model | Members | Conditions | Evidence $Z$ | $n_{\mathrm{eff}}$ |
|---|---|---|---|---|
| 2+2, broad prior of the outer period | 20 000 | + A eclipsing for years | 0.00013 | 4.8 |
| 2+(2+1) | 10 000 | tilt of B + A eclipsing + detached | 0.0104 | 165 |
| 2+(2+1) | 10 000 | the same + intact in both directions of time | 0.0095 | 153 |

**Result.** Table 29 shows that the 2+(2+1) quintuple is favoured over the 2+2 by a factor of about 70 under the same conditions. The observation that discriminates most is that A has eclipsed for years: it removes almost all 2+2 members. The 2+2 members that fit lie at the edge of stability (weighted median MA01 ratio 1.43, outer period about 116 d). Here the secular backend, which assumes a stable hierarchy, cannot be trusted alone. An N-body check of the 40 best 2+2 members over 100 yr with MEGNO finds 4 collisions, 14 chaotic members (MEGNO 4–35, Lyapunov times about 1.5–25 yr) and 22 regular ones: about half of the 2+2 solutions cannot describe a system that exists today, so the 2+2 evidence is an overestimate and the preference for 2+(2+1) is conservative.

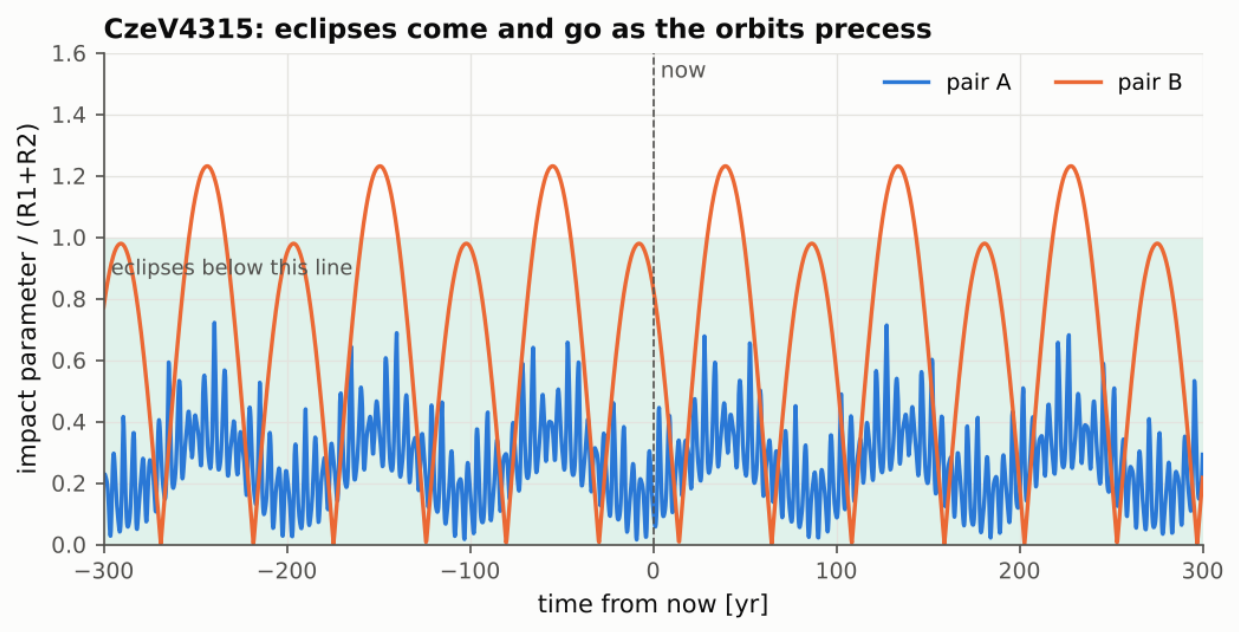



**Figure 9.** The best 2+2 configuration of CzeV4315 in the secular ensemble: the impact parameter of both pairs over 600 years. Horizontal axis: time from 300 yr in the past to 300 yr ahead (dashed line: today). Vertical axis: the smallest projected separation of the two stars of a pair at conjunction, in units of the sum of their radii; below 1 (shaded band) the pair eclipses. Blue: pair A; orange: pair B.

Figure 9 shows why a 2+2 has difficulty. Pair A (blue) stays far below 1 – it eclipses all the time. Pair B (orange) swings up and down as its orbital plane precesses; it crosses 1 every few decades, so its eclipses vanish for years and return (B eclipses 80 % of the time, in windows of about 20 yr). Today B is on its way down – the observed fast change of its eclipse depth. A 2+2 reproduces this tilt rate only with outer periods of 50–170 d (median 116 d), close to the stability limit; in the 2+(2+1) model the 119-d companion of B drives the precession of B directly.

**Time symmetry.** Conservative dynamics is symmetric in time: a configuration that is unstable will show it in the past as often as in the future. In 20 000 runs of the 2+(2+1) ensemble (10 000 members in both directions), 5.99 % of the members fail forward and 5.97 % backward. 3.93 % fail in both directions within the window, where independent failures

would give 0.36 %: instability is a property of the configuration, and the starting phase only decides when it shows. The cumulative failure fraction grows without saturating (0.4 % at 20 yr, 2.4 % at 40 yr, 6.0 % at 60 yr), as expected near the stability boundary, where the disruption time grows roughly exponentially with the hierarchy (Hayashi et al. 2022; Mushkin & Katz 2020). The 4.2 % of runs that end in a collision do so by three routes: ZLK cycles of B driven by its companion (mutual inclinations 60°–120°; B reaches $e \approx 0.45$, where its stars touch), an eccentricity of the companion pumped to about 0.9 by pair A on the outer orbit, so that its periastron comes within 1–3 semi-major axes of B and the inner triple becomes unstable, and – rarely – a growth of the eccentricity of A from 0.64 to about 0.86.

## 6.2 CzeV4315 with tides and rotation

**Question.** The 2+(2+1) ensemble used point masses with relativity. Real stars are distorted by tides and rotation; the distortion adds a fast apsidal precession and dissipates energy. With the apsidal constants of these stars (Claret 2023) the tidal and rotational precession of B has a period of about 9 yr, shorter than the ZLK periods of 14–118 yr. Does this change which members survive?

**Input.** The same prior and random seed as before, so that members 0–4999 are exactly the same draws, plus tides and rotation; N-body, 100 yr forward and backward (10 000 runs).

**Table 30.** CzeV4315 with tides and rotation: fates of the 5000 members after 100 yr in each direction.

| Fate (100 yr) | Forward | Backward |
|---|---|---|
| intact | 94.68 % | 94.34 % |
| collision | 2.50 % | 2.52 % |
| ejection | 1.26 % | 1.38 % |
| reconfiguration | 0.92 % | 1.12 % |
| exchange | 0.36 % | 0.38 % |
| breakup | 0.28 % | 0.26 % |
| not intact in both directions | 4.20 % (0.30 % if independent) | |

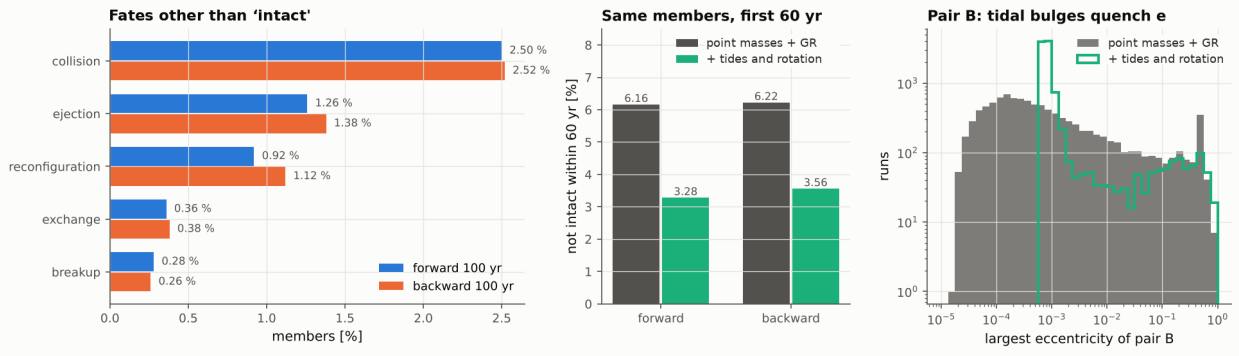



**Figure 10.** CzeV4315 with tides and rotation. *Left:* for every fate other than intact, the percentage of the 5000 members that ended so within 100 yr; blue bars: runs into the future, orange bars: runs into the past. *Middle:* the percentage of the same members that did not stay intact within the first 60 yr, with point masses and relativity (dark grey) and with tides and rotation added (green), forward and backward. *Right:* histogram (both axes logarithmic) of the largest eccentricity that the 0.92-d pair B reached in each run, with point masses (grey filled) and with tides and rotation (green outline).

**Result.** Figure 10 and Table 30 show three things. In the left panel the blue and orange bars have almost the same length for every fate (collisions 2.50 % forward, 2.52 % backward): the

time symmetry holds with tides as well. In the middle panel the failure rate within the first 60 yr drops from 6.2 % to 3.3–3.6 % when tides and rotation are added; 223 runs that collided with point masses stay intact. The right panel shows why: with point masses the eccentricity of B is pumped to anything up to 1 (median 0.45 in the runs that collided), whereas with tides almost all runs stay near $10^{-3}$ (median 0.0015) – the tidal and rotational bulges make the orbit of B precess so fast that its companion can no longer pump its eccentricity. This suppression of ZLK cycles by short-range forces is known (Fabrycky & Tremaine 2007; Liu, Muñoz & Lai 2015); here it appears in a real candidate. Rerun with tides over 100 yr, examples of all three collision routes survive, while a deep plunge of the companion (periastron at 1.6 $a_B$) still collides: short-range forces remove the secular collisions and part of the dynamical ones. The practical conclusion: point masses overestimate the failures of compact 2+(2+1) systems by a factor of two, and the ensembles of such systems need tides and rotation. CzeV4315 is analysed with its real minima in Part IV.

## 6.3 ASASSN-V J075203

**The system.** In this 2+2 candidate, B tilts at 1.03 ± 0.2 °/yr, A keeps a constant inclination (0 ± 0.3 °/yr) and A shows a fast apsidal motion (apsidal period $U \approx 3.5 \pm 1.5$ yr). Can a 2+2 reproduce all three at once?

**Table 31.** ASASSN-V J075203: the four steps of the analysis. Column 3 gives what each step found.

| Step | Members | Result |
|---|---|---|
| isotropic prior, secular | 20 000 | no member fits all three observations (best $\chi^2$ 6.2; $n_{\mathrm{eff}}$ = 1.0) |
| outer orbit near the plane of A ($i \sim N(80.7°, 3°)$, $P_{\mathrm{out}}$ 40–600 d), secular ±100 yr | 40 000 | $n_{\mathrm{eff}}$ = 53; 32 members with $\chi^2$ < 3; $P_{\mathrm{out}}$ 50–80 d, A almost coplanar with the outer orbit |
| N-body ±30 yr of the 32 | 32 | 2 collide; 7 fit all three ($\chi^2$ < 3.5, $P_{\mathrm{out}}$ 55–83 d) |
| N-body with MEGNO ±1000 yr of the 7 | 7 | 4 regular, 1 chaotic (Lyapunov time about 13.5 yr), 2 collide within 50–150 yr |

**Table 32.** The seven members that fit all three observations in N-body. Columns: member number, outer period, MA01 stability ratio (above 1 = stable by the criterion), MEGNO (2 = regular) and verdict.

| Member | $P_{\mathrm{out}}$ [d] | MA01 ratio | MEGNO | Verdict |
|---|---|---|---|---|
| 642 | 55.3 | 1.06 | 2 | regular |
| 12744 | 62.3 | 1.16 | 2 | regular |
| 39269 | 66.3 | 1.22 | 2 | regular |
| 38234 | 77.5 | 1.21 | 2 | regular |
| 1043 | – | 1.08 | 40–55 | chaotic, intact for 1000 yr |
| 2654, 30610 | – | 0.87–0.89 | – | collide, excluded |

**Result.** Tables 31 and 32 trace the search. With an isotropic prior one member carries the whole weight ($n_{\mathrm{eff}}$ = 1.0): the evidence of such an ensemble is meaningless and the prior must be focused, which the second step does. The members that fit lie at the stability limit (MA01 0.9–1.2, period ratio $P_{\mathrm{out}}/P_A \approx$ 6–9). There, the orbit averaging of the secular backend fails: the N-body apsidal period of A is about half the secular one, as expected when the period

ratio is small (Luo, Katz & Dong 2016). Only 4 regular members survive all tests, with outer periods of 55–78 d. A 2+2 is therefore possible only in this narrow regular island at the stability limit, and it makes two predictions that existing minima can test: the eccentricity of A oscillates by 0.13–0.17 within years, and the minima of A show an eclipse-timing variation with the 55–78-d outer period. If these are absent, the 2+2 is excluded. The lesson for the software is general: near the stability limit the secular result is a pre-selection, and the N-body backend gives the answer.

## 6.4 Eclipse statistics of 2+2 populations

**Question.** How often do both pairs of a 2+2 eclipse, and how fast do eclipses appear and disappear? This calibrates what SQUADRA can expect to see.

**Input.** 50 000 synthetic 2+2 systems with the secular backend over 3000 yr. Assumed prior: masses uniform in 0.8–3 $M_\odot$, inner periods log-uniform in 0.3–30 d, inner $e$ uniform in 0–0.2, outer periods log-uniform from 100 d to 3000 yr, outer $e$ uniform in 0–0.8, isotropic orientations, stable by the criterion of Vynatheya et al. (2022).

**Table 33.** Eclipse statistics of the synthetic 2+2 population. Rows: fraction of pairs eclipsing today, and the fractions whose eclipses stop (among those eclipsing today) or start (among those not eclipsing today) within 10, 100 and 1000 yr. Columns: pair A and pair B.

| Quantity | A | B |
|---|---|---|
| eclipsing now | 31.7 % | 32.3 % |
| eclipses stop within 10 / 100 / 1000 yr | 3.4 / 10.2 / 19.0 % | 3.0 / 10.2 / 19.4 % |
| eclipses start within 10 / 100 / 1000 yr | 2.6 / 12.9 / 24.7 % | 2.6 / 13.0 / 24.3 % |

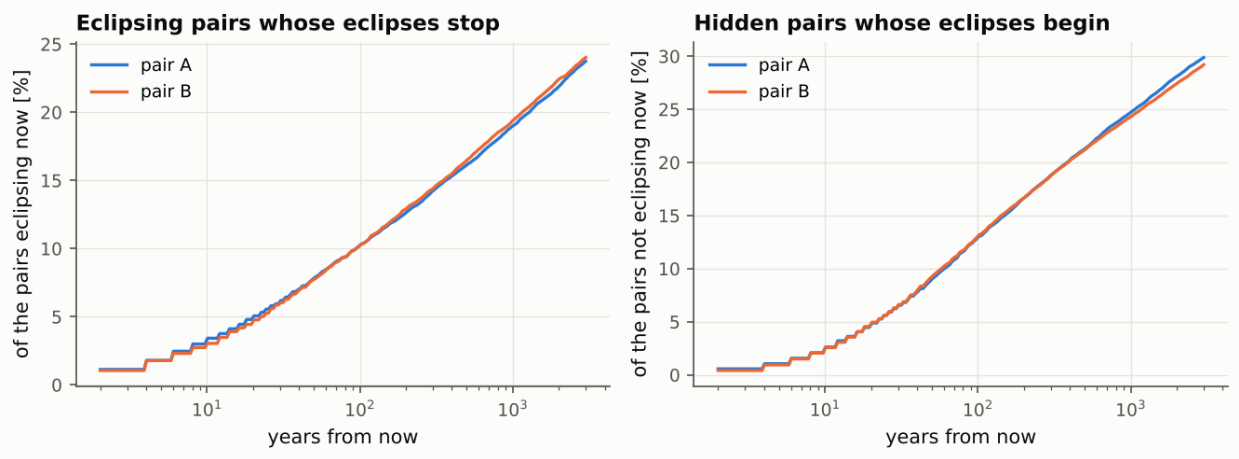



**Figure 11.** Synthetic 2+2 population (50 000 systems): the fraction of eclipsing pairs whose eclipses stop (left) and of non-eclipsing pairs whose eclipses begin (right) within a given time from today (horizontal axis, logarithmic, in years). Blue: pair A; orange: pair B.

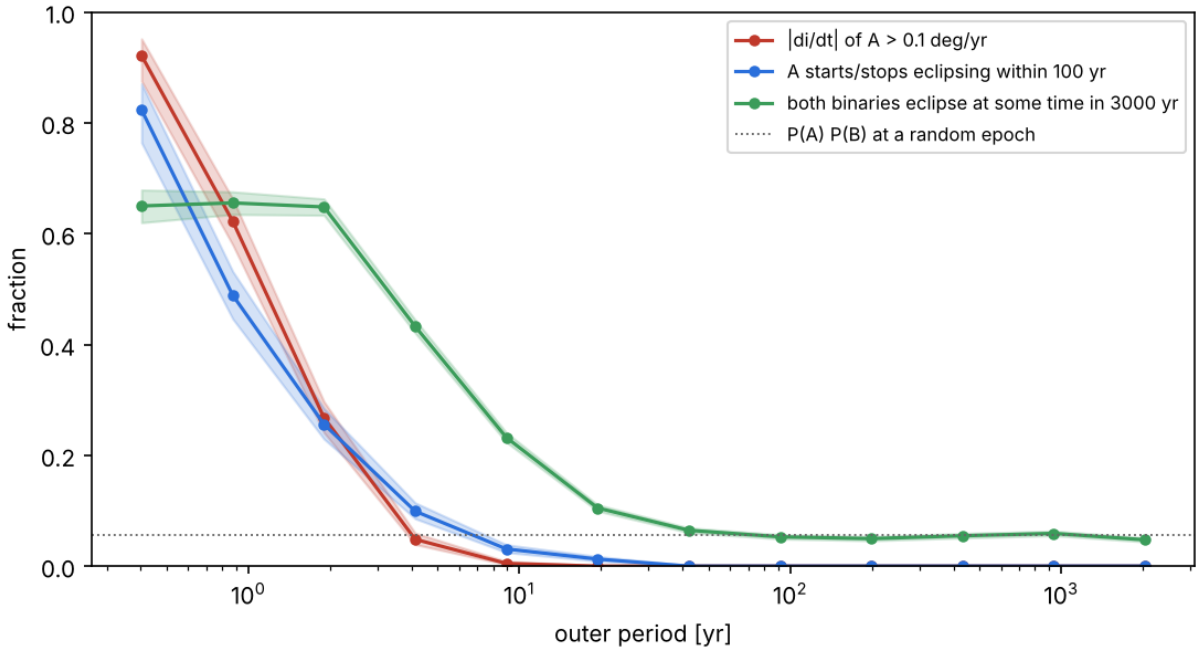



**Figure 12.** The same population sorted by the outer period (horizontal axis, logarithmic). Each point is the fraction of systems in that period bin; the shaded band is its statistical uncertainty. Red: pair A tilts faster than 0.1°/yr. Blue: pair A starts or stops eclipsing within 100 yr. Green: both pairs eclipse at some time within 3000 yr; the dotted line is the fraction expected at a random moment for independent pairs.

**Result (Table 33).** In Figure 11, read for example at 100 years: about 10 % of the eclipsing pairs stop eclipsing and about 13 % of the hidden ones start. The curves of A and B lie on top of each other, as they must for pairs drawn from the same prior, and P(B eclipses | A eclipses) equals P(B eclipses) – a check that the eclipse statistics treat isotropic orbits correctly (cf. Vokrouhlický 2016, Sect. 3.3). Both pairs eclipse together 10 % of the time on average. Figure 12 shows that fast changes need short outer periods: among eclipsing pairs a tilt faster than 0.1 °/yr occurs in 92 % of the systems with outer periods below 0.6 yr, in 27 % at about 2 yr, 5 % at 4 yr and never beyond 10 yr. Rates above 1 °/yr – the rates measured for CzeV4315 and ASASSN-V J075203 – need outer periods of 105–460 d, which agrees with the observed fast-tilting systems (outer periods of 0.3–3 yr; Kolář et al. 2025; V994 Her; CzeV4315). The probability that a pair eclipses, 32 % for this prior, agrees with the geometric estimate $(R_1 + R_2)/a$ averaged over the prior. The population is assumed, not fitted; a comparison with the 781 candidates needs the selection function of TESS, which is outside the scope of SWINGERsim.

## 6.5 CzeV343

The fourth SQUADRA system of Part I is CzeV343, a 2+2 quadruple in a period resonance studied dynamically by Pejcha et al. (2022). Its apsidal-motion contributions are four of SWINGERsim's benchmarks (Chapter 3): the tidal, rotational and relativistic rates of pair A agree with the paper to 0.6–2 %, and the full N-body quadruple gives the observed apsidal period to 3.3 %. CzeV343 is analysed in full in Part IV.

# Chapter 7

# A catalogue of published multiple stars

The benchmarks of Chapters 2–4 test one effect at a time. A real multiple star tests them together, with stars from every part of the Hertzsprung–Russell diagram. The SWINGERsim distribution contains 29 configurations of 23 published systems (Table 34): Wolf–Rayet and O stars, β Cep, δ Sct and Cepheid pulsators, giants, young T Tauri stars, white dwarfs, a millisecond pulsar, and stars hidden by dust discs. Each configuration names its sources (paper and table), every conversion (for example, the relative orbit uses $\omega$ = published $\omega$ of the primary + 180°; the mean anomaly follows from the time of periastron) and every assumption, marked ASSUMED or UNKNOWN; where a quantity is unknown, an ensemble draws it from a prior. Every quantity that SWINGERsim computes and that has also been published is compared below. A difference is called *agrees* if it lies within the published uncertainty or about 10 %, *explained* if it has been traced to the input, a definition or the published numbers, and *prediction* if nothing has been published to compare with.

**Table 34.** The catalogue. Column 2 gives the hierarchy in SWINGERsim's notation (2+1 = a binary with a third star; 1+2 = a single star orbited by a binary), column 3 the stars, column 4 the main source of the configuration.

| System | Hierarchy | Stars | Source |
|---|---|---|---|
| Algol | 2+1 | B8 V + K0 IV (fills its lobe) + Am | Baron et al. 2012 |
| BU CMi | 2+2 | four A stars | Pribulla et al. 2023 |
| VW LMi | 2+2 | contact W UMa pair + G pair | Pribulla et al. 2020 |
| ξ Tau | 3+1 | B9 V + B9 V, B5 V, F5 V | Nemravová et al. 2016 |
| TIC 120362137 | 3+1 | F + F + F + G | Borkovits et al. 2026 |
| TIC 454140642 | 2+2 | four F stars | Kostov et al. 2021 |
| TIC 470710327 | 2+1 | B + B + O9.5 V | Eisner et al. 2022 |
| δ Ori A | 2+1 | O9.5 II + B2 V + B0 IV | Oplištilová et al. 2023 |
| σ Ori | 2+1 | O9.5 V + B0.5 V + B | Schaefer et al. 2016 |
| λ Sco | 2+1 | β Cep + pre-main-sequence star + B2 V | Tango et al. 2006 |
| Polaris | 2+1 | first-overtone Cepheid + F6 V + F3 V | Evans et al. 2024 |
| HD 181068 | 1+2 | red giant + G + K | Borkovits et al. 2013 |

| System | Hierarchy | Stars | Source |
|---|---|---|---|
| PSR J0337+1715 | 2+1 | millisecond pulsar + two white dwarfs | Ransom et al. 2014 |
| KIC 4150611 | 2+(2+1) | δ Sct/γ Dor + M + M, G + G | Hełminiak et al. 2017 |
| HD 98800 | 2+2 + disc | four T Tauri stars, 10 Myr, polar disc | Zúñiga-Fernández et al. 2026; Kennedy et al. 2019 |
| HD 5980 | 2+2 | WR/LBV + WN4, O supergiant + unseen | Koenigsberger et al. 2014 |
| HD 150136 | 2+1 | O3 V + O5.5 V + O6.5 V | Mahy et al. 2018 |
| α Cen | 2+1 | G2 V + K1 V + M5.5 V | Akeson et al. 2021; Kervella et al. 2017 |
| GG Tau | 2+(2+1) + ring | five T Tauri stars / brown dwarf | Köhler 2011; Keppler et al. 2020 |
| TIC 168789840 | 2+2+2 | three eclipsing F + K/M pairs | Powell et al. 2021 |
| ε Aur | 1+1 + disc | F supergiant + B star in a dust disc | Stefanik et al. 2010; Hoard et al. 2010 |
| EE Cep | 1+1 + disc | B5 III + dark disc (illustrative) | Pieńkowski et al. 2020 |
| KH 15D | 2 + ring | T Tauri binary behind a ring (illustrative) | Winn et al. 2006 |

The dust-disc systems are discussed in Chapter 4; the others follow, grouped by the physics they test. Classes in the sense of Table 1: precession and timing runs use the N-body backend (R) or the secular backends (own), the evolving runs and pulsation timing use SWINGERsim's own code.

## 7.1 Apsidal and nodal precession

**Method.** The turning of the ellipses (apsidal) and of the orbital planes (nodal) is measured directly from the integrated orbital elements – with all stars, relativity, tides and rotation – and compared with the published rates, which the authors derived from eclipse timing, photodynamical models or analytic formulae. Runs cover 30–1000 yr.

**Table 35.** Apsidal and nodal precession in the catalogue. Column 4 is SWINGERsim's value, column 5 the published one, column 6 their difference.

| System | Quantity | SWINGERsim | Published | Difference | Status |
|---|---|---|---|---|---|
| BU CMi | apsidal period of A | 27.8 yr | 28.7 yr (Pribulla et al. 2023) | −3 % | agrees |
| BU CMi | apsidal period of B | 24.8 yr | 25.1 yr | −1 % | agrees |
| ξ Tau | apsidal rate of the 145-d orbit | 2.14 °/yr | 2.02 ± 0.31 °/yr | +6 % (0.4σ) | agrees |
| TIC 120362137 | apsidal period of the 51-d orbit | 32.4 yr | 34.79 yr (Borkovits et al. 2026) | −7 % | agrees |
| TIC 120362137 | nodal period of the outer planes | 282 yr | 265.8 yr | +6 % | agrees |
| TIC 454140642 | apsidal periods of A, B | 69.7, 90.9 yr | 70, 97 yr (Kostov et al. 2021) | −0.4 %, −6 % | agrees |

| System | Quantity | SWINGERsim | Published | Difference | Status |
|---|---|---|---|---|---|
| HD 181068 | nodal period | 12.8 yr | 13–14 yr (Borkovits et al. 2013) | −5 % | agrees |
| VW LMi | apsidal period of the detached pair | 95.1 yr | 95.5 yr (the paper's own N-body model) | −0.4 % | agrees |
| δ Ori A | apsidal rate of the 5.73-d orbit | 2.42 °/yr | 1.45 °/yr | +67 % | explained: input |
| TIC 470710327 | nodal period of the eclipsing pair | 14.1 yr | – | – | prediction |

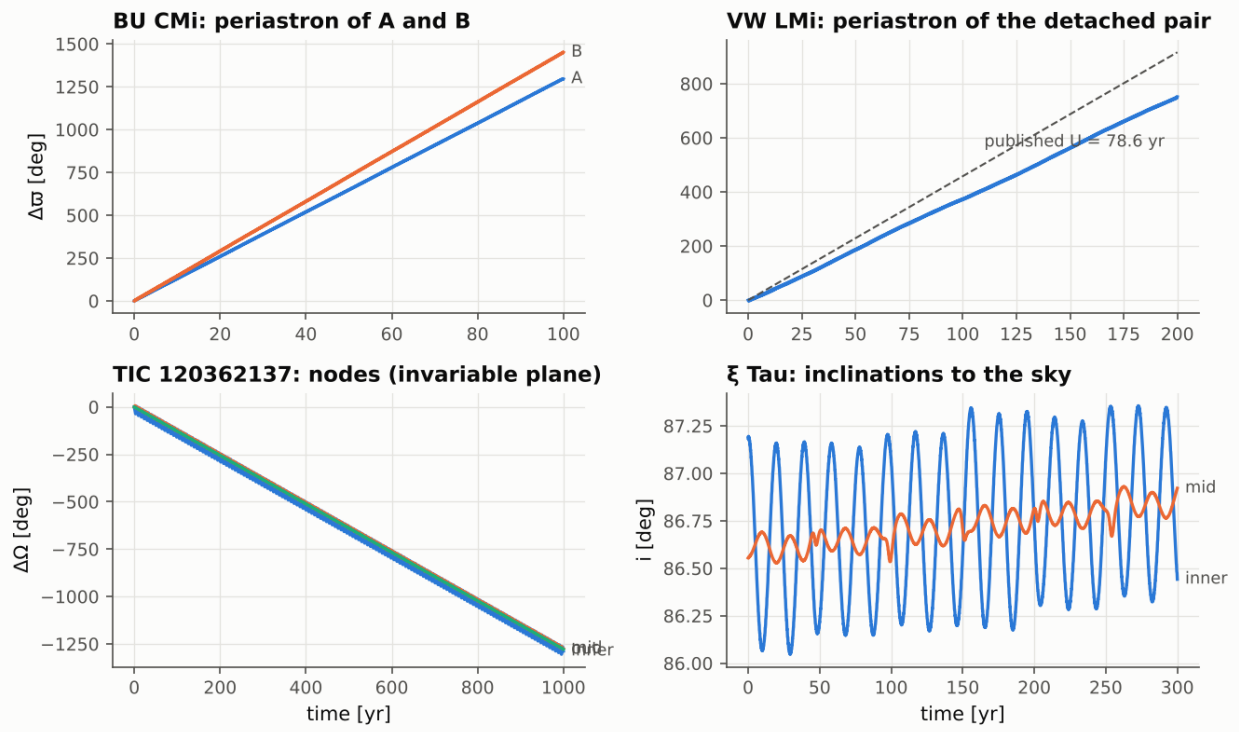



**Figure 13.** Precession measured in four catalogue runs; in every panel an angle grows steadily with time, the slope is the precession rate and 360° divided by the rate is the period. *Top left:* longitude of periastron of the two pairs of BU CMi (A and B). *Top right:* the same for the detached pair of VW LMi (blue) against the dashed line of the 78.6-yr period fitted by Pribulla et al. (2020) to three years of data. *Bottom left:* nodes of the three orbits of TIC 120362137 in the invariable plane (the plane perpendicular to the total angular momentum) over 1000 yr. *Bottom right:* inclinations of the two inner orbits of ξ Tau to the sky over 300 yr.

**Result (Table 35).** Figure 13 shows four of the runs. In BU CMi, 360° are reached after about 28 yr (A) and 25 yr (B), 3 % and 1 % faster than published. In VW LMi the detached pair turns in 95 yr, slower than the dashed 78.6-yr line: the 78.6 yr of Pribulla et al. (2020) is a fit to only three years of radial velocities, whereas the paper's own analytic and N-body models give 96.2 and 95.5 yr (and DDO data of 1998–2008 give 91 ± 11 yr); SWINGERsim agrees with the paper's dynamical model to 0.4 %. The nodes of TIC 120362137 all turn together, backwards, by 360° in about 280 yr: the planes precess as one block. For this system, Table 1 of Borkovits et al. (2026) gives 34.79 yr for the eccentric 51-d orbit, against SWINGERsim's 32.4 yr (−7 %). The nearly circular 3.3-d pair ($e$ = 0.007) is pushed by the tertiary to $e$ between 0.0001 and 0.016; its osculating $\omega$ jumps, and a measured apsidal period is not defined (the paper gives only a nominal theoretical value for it). The inclination of the eclipsing inner pair of ξ Tau wobbles by ±0.6° with a period of about 20 yr, so its eclipse

depth should vary with that period – a prediction.

The one large difference, δ Ori A, comes from the input. Its apsidal motion is dominated by tides and rotation (the binary alone gives an apsidal period of 155 yr), and the tidal part scales as $k_2 R^5$ of the primary. With the radius of Opličtilová et al. (2023) and $k_2$ from Claret (2004), SWINGERsim is 1.7 times faster than the rate of Pablo et al. (2015); the published 1.45 °/yr requires $R_1 \approx 11.5$ R$_\odot$ (11 % smaller) or $\log k_2 \approx -2.85$. The published rate is therefore a measurement of $k_2 R^5$ of Aa1 rather than a test of the dynamics. For TIC 470710327, with a mutual inclination of 17° the 1.1-d pair precesses in 14 yr, and its eclipses should change within years (a prediction).

## 7.2 Eclipse-timing variations

**Method.** `swingersim timings` computes the exact times of conjunction from the N-body integration, including the light travel time, for TIC 470710327 (5 yr), TIC 454140642 (5 yr), BU CMi (30 yr) and ξ Tau (10 yr). The resulting O−C diagrams are what an observer would obtain.

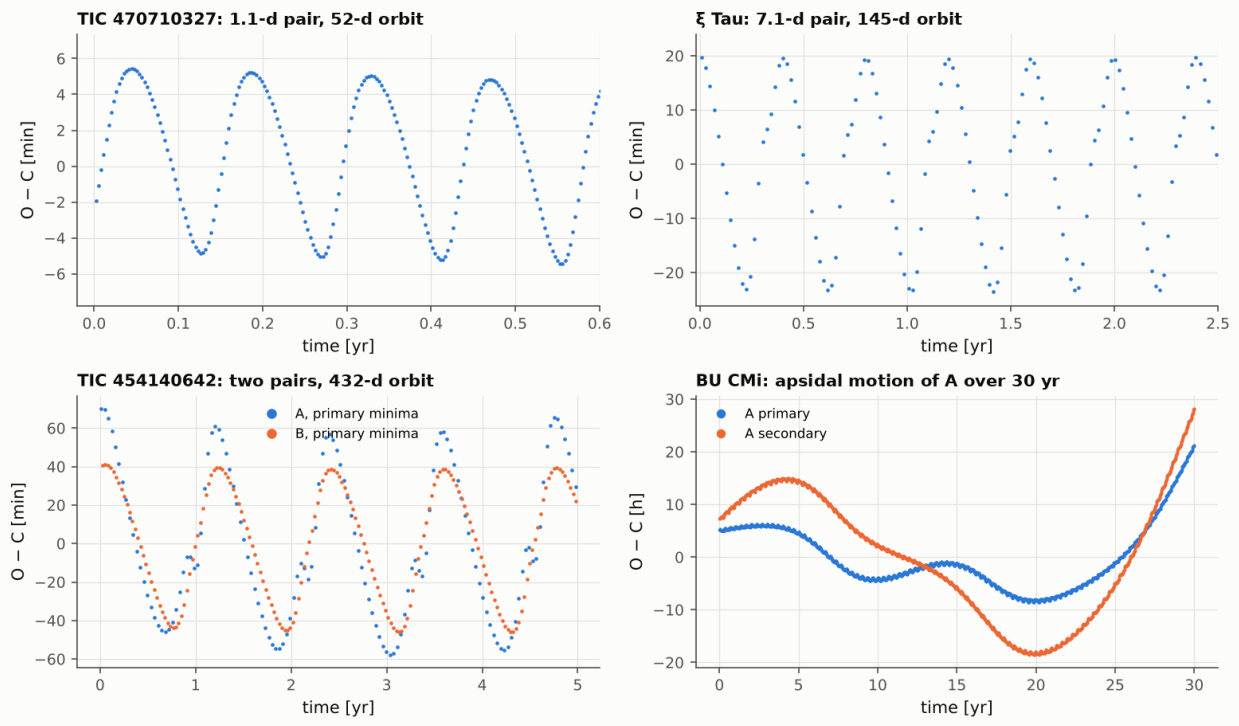



**Figure 14.** Synthetic O−C diagrams of four catalogue systems; horizontal axes: time; vertical axes: how much earlier or later each eclipse comes than a strictly periodic clock predicts. *Top left* (TIC 470710327, minutes): the 1.1-d pair with the 52-d orbit of the third star. *Top right* (ξ Tau, minutes): the 7.1-d pair with the 145-d orbit. *Bottom left* (TIC 454140642, minutes): the primary minima of both pairs, A (blue) and B (orange), with the 432-d outer orbit. *Bottom right* (BU CMi, hours): the primary (blue) and secondary (orange) minima of pair A over 30 yr.

**Table 36.** Eclipse-timing variations. "Semi-amplitude" is half the peak-to-peak O−C of a fitted periodic curve.

| System | Quantity | SWINGERsim | Published | Difference | Status |
|---|---|---|---|---|---|
| TIC 470710327 | ETV semi-amplitude of the 1.1-d pair (52-d orbit) | 4.97 min | ≈ 6 min (read from a figure) | −17 % | agrees |

| System | Quantity | SWINGERsim | Published | Difference | Status |
|---|---|---|---|---|---|
| ξ Tau | ETV semi-amplitude of the 7.1-d pair (145-d orbit) | 19.7 min | – | – | prediction |
| TIC 454140642 | ETV semi-amplitudes of A and B (432-d orbit) | 51.4 and 40.2 min, in phase | dynamical ETV dominates (Kostov et al. 2021) | – | agrees |
| BU CMi | apsidal period of A from the O−C crossings | 27.78 yr | 28.7 yr | −3 % | agrees |
| BU CMi | apsidal period of B from the O−C crossings | 24.79 yr | 25.1 yr | −1 % | agrees |

**Result (Table 36).** In Figure 14 the eclipses of the 1.1-d pair of TIC 470710327 run about ±5 min early and late with the 52-d period of the third star; the published value, read from a figure, is about 6 min. ξ Tau shows ±20 min with the 145-d orbit; the pure light-time effect of that orbit would be at most 8.6 min, so the dynamical ETV dominates – the tertiary is close enough to change the period of the pair within each outer orbit. The bottom-left panel is the instructive one. If only the light travel time mattered, the two pairs of a 2+2 would move in antiphase (A late when B is early), here with about ±8 min (7.6 and 7.9 min). The computed ETVs are five to seven times larger and *in phase*: in this compact quadruple ($P_{\rm out}/P_{\rm in} \approx 32$–42) the gravitational pull of each pair on the other dominates, as Kostov et al. (2021) describe. Antiphase is therefore not a general signature of a 2+2, and a comparison with the minima of a real compact 2+2 must use the full dynamical model, as `swingersim timings` does. In BU CMi the primary and secondary minima of A drift apart and cross again as the ellipse turns; the time between two crossings is half the apsidal period, which gives 27.78 and 24.79 yr – the same as measured from $\omega$ (27.76 and 24.82 yr) by an independent route.

## 7.3 ZLK cycles and long-term survival

**Method.** Three systems with strongly inclined orbits: σ Ori (secular, 3 Myr), HD 98800 (secular, 10 Myr) and α Cen with Proxima (secular with the Galactic tide, 1 Gyr).

**Table 37.** ZLK cycles and survival. Column 3 gives the result of the run, column 4 the published observation.

| System | Run | SWINGERsim | Published | Status |
|---|---|---|---|---|
| σ Ori | secular, 3 Myr | $e$ of Aa–Ab 0.69–0.90, $i_{\rm mut}$ 116°–134°, ZLK period 39.5 kyr; closest periastron 3.4 $(R_1 + R_2)$ | observed $e$ = 0.78 (Schaefer et al. 2016) | agrees; survives |
| HD 98800 | secular, 10 Myr | $e_A$ 0.27–0.63, $e_B$ 0.42–0.80, ZLK periods 46 and 29 kyr | observed 0.48 and 0.78 (Zúñiga-Fernández et al. 2026) | agrees; survives |
| α Cen | secular + Galactic tide, 1 Gyr | Proxima stays bound; $e_{\rm out}$ 0.50 → 0.92, periastron down to 724 au; $i_{\rm mut}$ 64°–83° | bound (Kervella et al. 2017) | agrees; the growth of $e$ is a prediction |

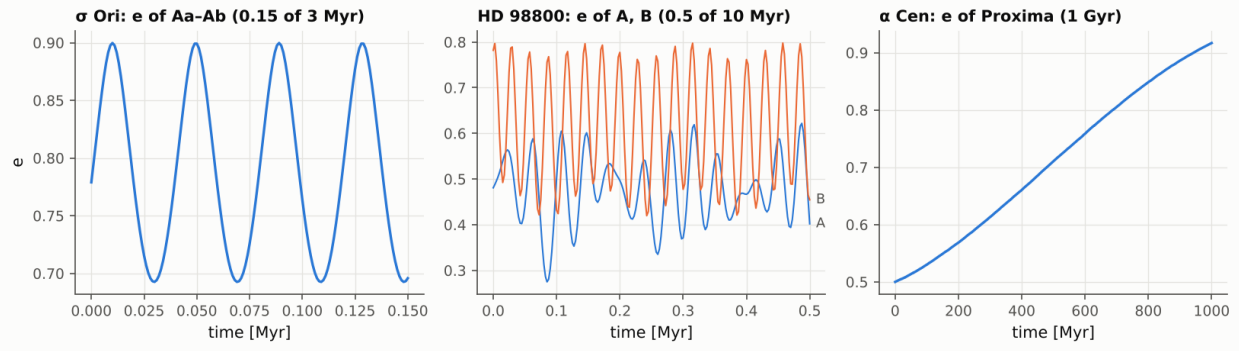



**Figure 15.** ZLK cycles and the Galactic tide: eccentricity (vertical axes) against time. *Left:* σ Ori Aa–Ab, the first 0.15 of the 3-Myr run. *Middle:* the two pairs of HD 98800, A (blue) and B (orange), the first 0.5 of the 10-Myr run. *Right:* the orbit of Proxima around α Cen AB over 1 Gyr.

**Result (Table 37).** In Figure 15 (left) the eccentricity of σ Ori Aa–Ab swings regularly between 0.69 and 0.90 every 40 kyr, and the observed 0.78 lies in the middle of the swing; the closest approach stays 3.4 times the sum of the radii, so the system survives its age. In HD 98800 both pairs oscillate (0.27–0.63 and 0.42–0.80), and the observed 0.48 and 0.78 lie within these ranges: the large observed eccentricities are naturally explained by ZLK cycles, as Zúñiga-Fernández et al. (2026) suggest. In α Cen the Galactic tide raises the eccentricity of Proxima's orbit slowly from 0.50 to 0.92 over a billion years without unbinding it; the ensemble of Ensembles of published systems confirms this with passing stars. The tide itself is validated against the analytic maximum eccentricity of a wide binary (Chapter 2).

## 7.4 Growing stars and winds

**Method.** Systems in which the stars change during the run: HD 181068 (N-body over 30 yr; evolving backend with MIST from 362.2 Myr), HD 150136 (evolving, MIST), HD 5980 (secular with winds, 100 kyr) and PSR J0337+1715 (N-body over 30 yr) as a check of exact coplanarity.

**Table 38.** Growing stars and winds.

| System | Run | SWINGERsim | Published | Status |
|---|---|---|---|---|
| HD 181068 | N-body, 30 yr | coplanar ($i_{\rm mut}$ 0.8°), nodal period 12.8 yr | 0.8 ± 1.4°; 13–14 yr (Borkovits et al. 2013) | agrees |
| HD 181068 | evolving, MIST | the giant grows from 12.4 to 38.3 $R_\odot$ and fills its Roche lobe after 2.1 Myr; tertiary mass transfer ends in a triple common envelope at 2.1055 Myr (age 364.3 Myr) | – | prediction |
| HD 150136 | evolving, MIST | the O3 V primary fills its lobe after 2.45 Myr (age 3.45 Myr) | – | prediction |
| HD 5980 | secular with winds, 100 kyr | $a_{\rm out}$ 41.3 → 42.9 au = $a_0 M_0/M$ exactly; masses 61 → 56.5 and 66 → 64 $M_\odot$ | Jeans (1924) | agrees |
| PSR J0337+1715 | N-body, 30 yr | $i_{\rm mut}$ stays 0.00° | coplanar to 0.01° (Ransom et al. 2014) | agrees |

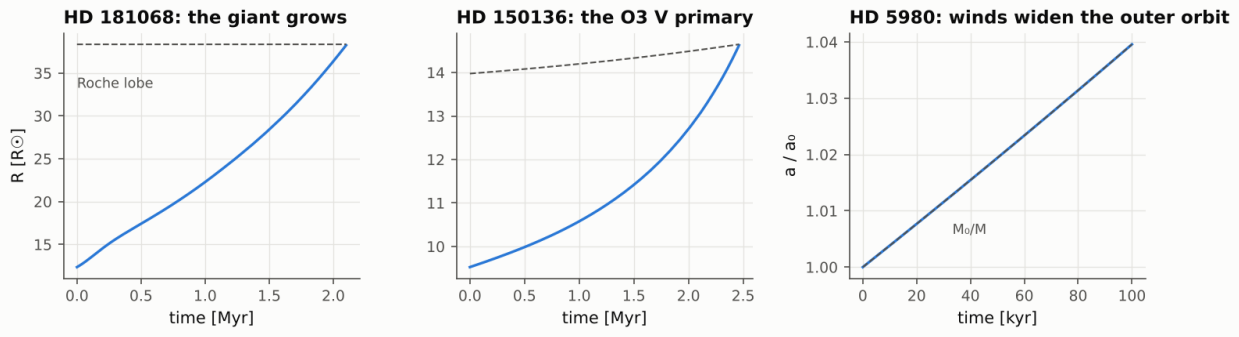



**Figure 16.** Growing stars. *Left and middle:* the radius of a growing star (blue) and the size of its Roche lobe (dashed) against time, for the giant of HD 181068 (whose lobe is around the inner pair) and for the O3 V primary of HD 150136; where the blue line reaches the dashed one the star starts to transfer mass. *Right:* the semi-major axis of the outer orbit of HD 5980 divided by its initial value (blue) and the ratio of the initial to the present total mass (dashed) against time.

**Result (Table 38).** In Figure 16 (left) the giant of HD 181068 grows from 12 to 38 $R_\odot$ in 2.1 Myr until it meets its Roche lobe of 38.3 $R_\odot$. Because the giant (3.0 $M_\odot$) is heavier than the inner pair (0.92 + 0.87 $M_\odot$), the transfer onto the pair is unstable (Hamers et al. 2022), and SWINGERsim ends the run in a triple common envelope at the moment of contact. HD 181068 is the textbook case for mass transfer from a tertiary onto a binary (de Vries et al. 2014; Tokovinin & Moe 2020). The O3 V primary of HD 150136 grows from 9.5 to 14.6 $R_\odot$ in about 2.5 Myr while its lobe grows a little, because its wind widens the orbit; it will become a very young, massive Algol-like system. In the right panel the two lines for HD 5980 coincide: the outer orbit widens exactly as fast as the system loses mass, $a \propto 1/M$, which is Jeans' law for slow mass loss. PSR J0337+1715, a millisecond pulsar with two white dwarfs, stays exactly coplanar, as observed.

## 7.5 Pulsation timing

**Method.** SWINGERsim's light curves contain the pulsations, and the delay is measured as an observer does it: a sinusoid at the pulsation frequency is fitted in short windows and its phase is followed (Chapter 4). Three systems: λ Sco (β Cep pulsator, 1100 d, V band), Polaris (first-overtone Cepheid with the observed period of 3.97 d, 30 yr, V) and KIC 4150611 (δ Sct, 200 d, TESS band).

**Table 39.** Pulsation timing. Column 3 is the semi-amplitude of the delays measured from the synthetic light curve, column 4 the light-time semi-amplitude of the orbit used in the run (Irwin 1952), column 5 the published value.

| System | Pulsation frequency | Measured | Orbit of the run | Published | Difference | Status |
|---|---|---|---|---|---|---|
| λ Sco | 4.679 c/d | 1067 s | 1069 s | 1193 s | −11 % | explained: input |
| Polaris | 0.252 c/d | 1309 s | 1336 s | 1335 s | −2 % | agrees |
| KIC 4150611 | 20.243 c/d | 97.1 s | 97.9 s | 98 ± 9.5 s | −1 % | agrees |

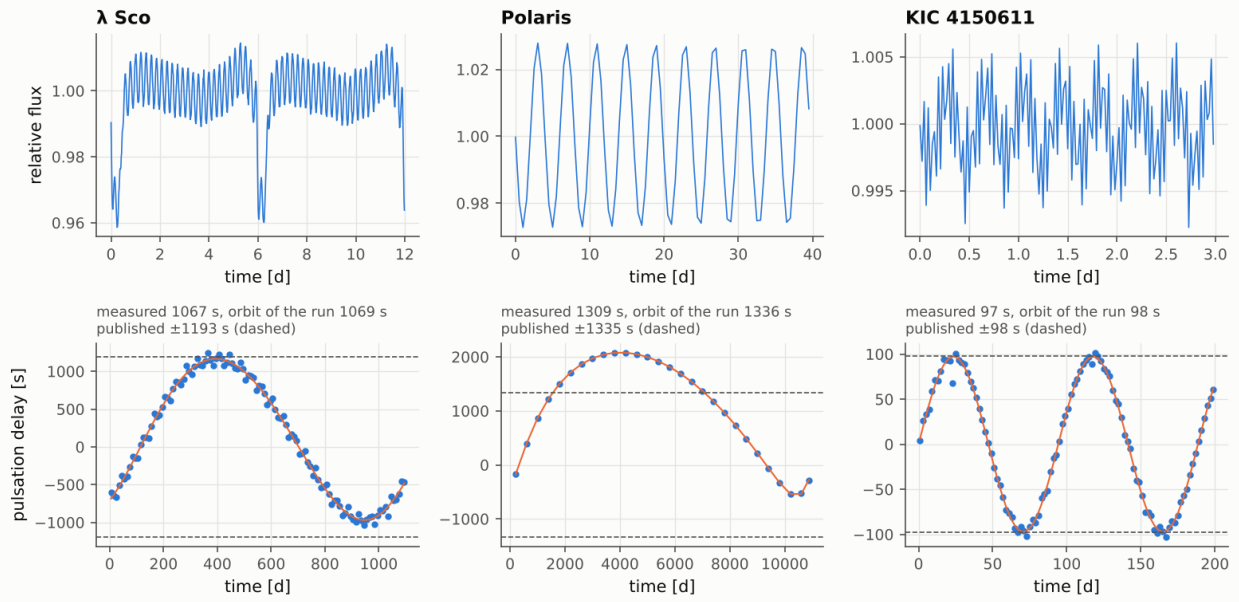



**Figure 17.** Pulsation timing in three catalogue systems. *Top:* a few days of each synthetic light curve (relative flux against time); the regular waves are the pulsations, for λ Sco also two eclipses. *Bottom:* the delay of the pulsation phase in seconds (points), measured in short windows along the whole light curve, with the light-time curve of the orbit that carries the pulsating star (line); the dashed lines mark ± the published semi-amplitude. The text above each panel gives the measured, model and published values.

**Result (Table 39).** In Figure 17 the measured delays (points) follow the light-time curve of the run's own orbit (line) to within 2 % in all three systems: the measurement is accurate. The remaining differences to the published numbers come from the input. For λ Sco the configuration uses the interferometric orbit and masses of Tango et al. (2006), which give $a_1 \sin i = 2.15$ au; the published 1193 s belong to the older orbit of Uytterhoeven et al. (2004). For Polaris the 1430 s often quoted is $a_1 \sin i/c$; the semi-amplitude of the light-time curve is smaller by $\sqrt{1 - e^2 \cos^2 \omega}$, 1335 s (Evans et al. 2024), which SWINGERsim reproduces to 2 % (only one 30-yr orbit is computed, so the curve is not centred on zero; its height from lowest to highest point is twice the semi-amplitude). The first-overtone period that the stellar model of Polaris predicts from the period–luminosity–mass relation is 3.62 d against the observed 3.97 d (−9 %), within the scatter of such relations; the run uses the observed period. For KIC 4150611 the configuration takes the companion mass $M_{\mathrm{Ab}} = 0.86\ \mathrm{M}_\odot$, consistent with the size and period ratios of Kemp et al. (2024); their dynamical mass ratio of 3.61 contradicts these ratios (which imply 1.70) and would give only 54 s. With consistent masses the orbit gives the published 98 s (Hełminiak et al. 2017 from radial velocities; Shibahashi & Kurtz 2012 from frequency modulation). Pulsation timing is thus a sensitive test of the masses in a configuration.

## 7.6 Ensembles of published systems

**Method.** Where a parameter of a real system is unknown (an outer orbit, a node, a mutual inclination), an ensemble draws it from a prior and asks what the observed system implies.

**Table 40.** Ensembles of published systems. Column 3 is the number of members, column 4 the question, column 5 the answer.

| System | Backend | Members | Question | Answer |
|---|---|---|---|---|
| HD 150136 | N-body | 500 | does the unknown mutual inclination matter within a century? | $i_{\mathrm{mut}}$ changes by < 0.04°; the 2.7-d pair stays eclipsing; its inclination drifts by 0–0.57° per century depending on $i_{\mathrm{mut}}$ |
| HD 5980 | N-body | 1000 | is the 2+2 bound and safe? | 29.7 % collide (median largest $e_B$ 0.91), mostly for outer periods below 20 yr |
| GG Tau | N-body | 500 | fate of the young quintuple in 20 kyr | intact 80.8 %, reconfiguration 9.2 %, collision 7.6 %, ejection 2.4 % |
| TIC 168789840 | secular | 500 | how often does each pair eclipse? | A 31 %, B 91 %, C 35 % of the time; all three together 11 % |
| KIC 4150611 | secular | 500 | which outer orbits keep the Ab pair circular, and is the system stable? | with the consistent masses all members stay intact and stable by MA01 and Vynatheya et al. (2022); for unknown outer orbits 42.6 % keep $e$ < 0.05 over 1 Myr, the others are pumped up to 0.7 |
| α Cen | secular + Galactic tide + passing stars | 300 | does Proxima stay bound for 1 Gyr? | 100 % bound; median largest $e$ 0.90; median closest periastron 849 au, minimum 70 au |

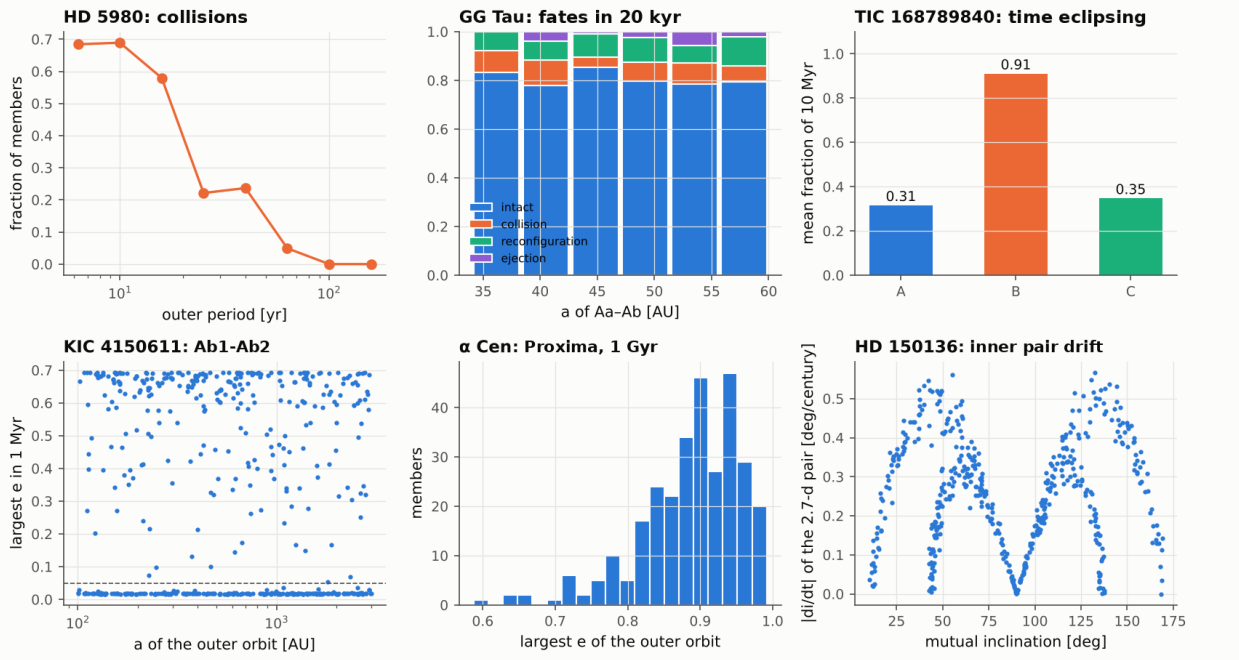



**Figure 18.** Ensembles of published systems, one panel each. *HD 5980:* fraction of members that end in a collision against the outer period. *GG Tau:* fates (blue intact, orange collision, green reconfiguration, violet ejection) per range of the Aa–Ab semi-major axis. *TIC 168789840:* mean fraction of time each of the three pairs eclipses. *KIC 4150611:* each point is a member, the largest eccentricity of the Ab pair against the size of the outer orbit; dashed: $e$ = 0.05. *α Cen:* histogram of the largest eccentricity of Proxima's orbit over 1 Gyr. *HD 150136:* each point is a member, the rate of change of the inclination of the eclipsing pair against the unknown mutual inclination.

**Result (Table 40).** Figure 18 shows the six ensembles. For HD 5980 most members with outer periods below 20 yr collide, long ones are safe: if the two binaries form a bound 2+2, the outer period is likely above 20 yr – a dynamical constraint for the spectroscopists. GG

Tau stays intact in about 80 % of the members everywhere; 74 % of the members that MA01 calls unstable stay intact for 20 kyr, which only shows that 20 kyr is short compared with the time scale of MA01 (MA01 calls 34.8 % stable). In TIC 168789840 pair B eclipses almost always, A and C a third of the time: the triple eclipses seen by TESS are a 1-in-9 coincidence. The KIC 4150611 members either keep the Ab pair circular or pump it to 0.6–0.7 at any outer size. Proxima stays bound in all 300 members, mostly with largest eccentricities of 0.85–0.97 (Kervella et al. 2017 find it bound). For HD 150136 the drift of the inclination vanishes at mutual inclinations of 0°, 90° and 180°, so a measured change of the eclipse depth would determine the mutual inclination; on observable time scales the effects are negligible, as Mahy et al. (2018) state.

## 7.7 Summary of the catalogue

All 23 systems ran without failure. Every quantity that can be compared with a publication – apsidal and nodal periods, eclipse-timing amplitudes, ZLK ranges, coplanarities, light-time amplitudes, Roche-lobe sizes and disc geometries – agrees within the published uncertainty or about 10 %, or the difference has been traced to the input of the configuration (δ Ori A, λ Sco) or to a different definition of the published quantity (VW LMi, TIC 120362137, Polaris). The figure of all comparisons on one scale is in Chapter 11.

# Chapter 8

# Life-cycle examples

The examples of this chapter are not real systems. They follow a multiple star through a whole phase of its life and test the coupling of dynamics and stellar evolution – the part of SWINGERsim that is entirely its own code: the evolving backend, the mass-transfer and common-envelope modules, the contact-binary model and the light curves with pulsations. Each example is a configuration in the SWINGERsim distribution; the numbers below are those of its run with the final code. Where a published result exists for the same physics, it is named; where none exists, the test is that the run obeys the conservation laws and the published behaviour of each ingredient.

**Table 41.** The life-cycle examples. Column 2 gives the system, column 3 the backend and stellar model, column 4 how the run ends, column 5 the reference for the physics.

| Example | System | Run | End | Reference |
|---|---|---|---|---|
| giant tertiary | 1.3 + 1.1 $M_\odot$ pair, 2.0 $M_\odot$ tertiary on a 7.6-au orbit, from 1.2 Gyr | evolving, MIST, winds | Roche-lobe overflow of the giant at 141.83 Myr | Jeans (1924); Cummings et al. (2018) |
| tertiary mass transfer | the same stars, outer orbit 3.62 au | evolving, MIST, tertiary transfer | end of the donor's track at 141.87 Myr | de Vries et al. (2014); Hamers et al. (2022) |
| Algol triple | 2.9 + 0.9 $M_\odot$ in 2.7 d (circular, the state at the onset of overflow), 1.6 $M_\odot$ tertiary in 680 d, from 390 Myr | evolving, MIST, mass transfer, tides | still transferring at 10 Myr | Hurley et al. (2002) |
| contact binary | 1.0 + 0.5 $M_\odot$, $a$ = 0.010 au | evolving, MIST, contact model | merger through L2 at 2.23 Gyr | Gazeas & Stępień (2008); Rasio (1995) |
| common envelope in a triple | 1.8 $M_\odot$ giant + 0.5 $M_\odot$ in a 10-yr outer orbit | N-body, common envelope | post-common-envelope binary | Hadjidemetriou (1963) |
| pulsating triples | δ Sct star in an eclipsing pair with a long-period-variable tertiary; Cepheid tertiary | light curves and minima | – | Irwin (1952) |

Table 41 is a map of the chapter: each row is one example, described in the section of the same name.

## 8.1 A giant tertiary on the thermally pulsing AGB

**Question.** A 2.0 $M_\odot$ tertiary around a 1.3 + 1.1 $M_\odot$ pair on a 7.6-au orbit ($e$ = 0.3) climbs the giant branch and the asymptotic giant branch, loses mass in winds and widens the outer orbit. Does it reach its Roche lobe – that is, does it start to pour mass onto the inner binary?

**Result.** The giant reaches the **thermally pulsing AGB** (TP-AGB), on which the helium shell ignites in flashes every $10^4$–$10^5$ yr and the radius jumps by a factor of 2–4 at each pulse, for a few hundred years only. The run ends with a Roche-lobe overflow of the giant at outer periastron at 141.83 Myr (age 1341.8 Myr), on the peak of a thermal pulse, with $R$ = 452 $R_\odot$. At that moment the giant has 1.06 $M_\odot$ left, and the outer orbit has widened from 7.60 to 9.64 au. The widening follows the adiabatic law $a \propto 1/M$ of Jeans (1924) to $4{\cdot}10^{-15}$ – exactly, because the evolving backend applies it as an invariant. Without the overflow the star would leave a white dwarf whose mass is checked against the initial–final mass relation (Chapter 3). Panel (a) of Figure 19 shows the last 6 Myr: the radius of the giant (blue, logarithmic axis) rises in steps towards its Roche lobe at outer periastron (dashed orange, about 420–450 $R_\odot$), which it touches at the star. Resolving the pulses requires care, because they are short: SWINGERsim places the boundaries of its evolution segments at the nodes of the MIST track and refines them until no radius changes by more than 1 % between two boundaries (Chapter 10).

## 8.2 Tertiary mass transfer

**Definitions.** When a tertiary fills its Roche lobe around an inner binary, SWINGERsim transfers mass at the BSE rate at outer periastron, decides whether the stream forms a circumbinary disc, lets the inner binary accrete with the fractions of Hamers et al. (2022), and responds the inner orbit with an energy or angular-momentum prescription. If the transfer is dynamically unstable (a donor much heavier than the binary), the run ends in a **triple common envelope**. The stream is a prescription, not a hydrodynamical calculation.

**Result.** With the same stars on a tighter outer orbit (3.62 au), the giant starts to transfer at 140.58 Myr and continues in episodes on the thermal pulses until its MIST track ends at 141.87 Myr. The donor goes from 2.0 to 0.70 $M_\odot$, the inner pair gains only 0.002 $M_\odot$ (most of the mass leaves the system), and the outer orbit widens from 3.62 to 5.15 au (panel b of Figure 19). The transfer is stable because the donor is lighter than the inner pair (2.4 $M_\odot$). The opposite case is HD 181068 (Chapter 7): its 3.0 $M_\odot$ giant is heavier than its 1.8 $M_\odot$ inner pair, and SWINGERsim ends the run in a triple common envelope at the first contact, as expected for an unstable transfer.

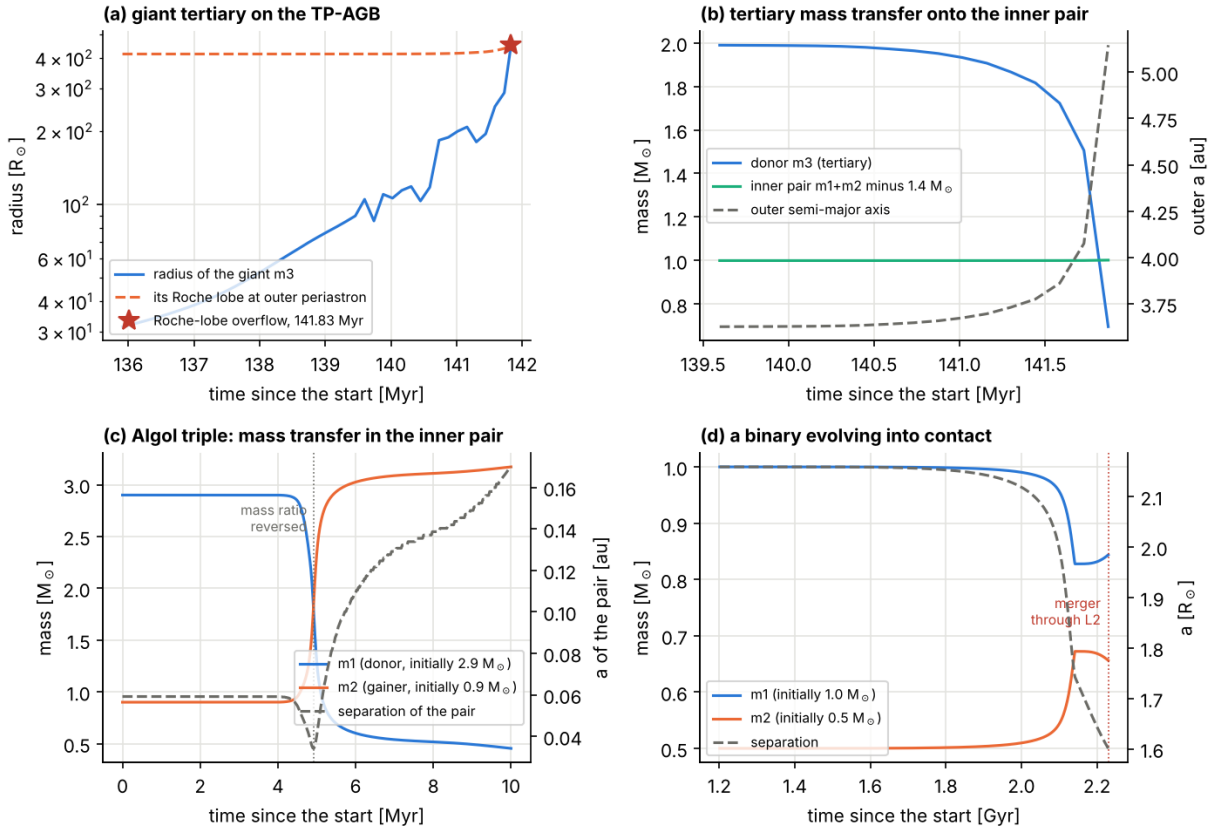



**Figure 19.** Life-cycle examples from the final runs. (a) Giant tertiary: radius of the 2.0 $M_\odot$ giant (blue) and its Roche lobe at outer periastron (dashed orange) over the last 6 Myr before the overflow (red star, 141.83 Myr); logarithmic radius axis in $R_\odot$, time since the start of the run in Myr. (b) Tertiary mass transfer: mass of the donor (blue, left axis), mass of the inner pair minus 1.4 $M_\odot$ (green, left axis, shifted to share the axis) and the outer semi-major axis (dashed grey, right axis, au) during the transfer. (c) Algol triple: masses of the donor m1 (blue) and the gainer m2 (orange) (left axis, $M_\odot$) and the separation of the pair (dashed grey, right axis, au) over 10 Myr; dotted vertical line: the mass ratio reverses. (d) A binary evolving into contact: masses of the 1.0 $M_\odot$ (blue) and 0.5 $M_\odot$ (orange) stars (left axis) and their separation (dashed grey, right axis, $R_\odot$) from 1.2 to 2.23 Gyr; dotted red line: merger through L2.

## 8.3 An Algol triple with mass transfer

**Question.** The BSE Algol example (Hurley et al. 2002, Sect. 3.1) starts as 2.9 + 0.9 $M_\odot$ in an orbit of $P$ = 8 d and $e$ = 0.7; when the primary fills its Roche lobe, tides have made the orbit circular with $P$ = 2.7 d. The run starts from this state, here with a 1.6 $M_\odot$ tertiary in 680 d. The more massive star fills its lobe after the main sequence, transfers mass to its companion, the mass ratio reverses and the orbit widens – the history of Algol itself.

**Result.** Panel (c) of Figure 19 follows the run with MIST tracks and tides. The donor fills its lobe at 3.71 Myr. During the fast, thermal-time-scale phase the separation first shrinks (to 0.034 au, while the donor is still the heavier star) and then grows; the mass ratio reverses at 4.91 Myr. After 10 Myr the donor has 0.454 $M_\odot$ and the gainer 3.172 $M_\odot$, and the separation has grown from 0.059 to 0.170 au; the donor is still transferring on its nuclear time scale. Tides keep the orbit circular ($e < 4{\cdot}10^{-4}$); the role of tides here is discussed in Chapter 10. The quantitative test of this physics is the BSE benchmark of Chapter 3: with the SSE tracks that BSE uses, SWINGERsim strips the donor to 0.404 $M_\odot$ (BSE 0.42 $M_\odot$, BSE in COSMIC 0.401 $M_\odot$), and it reproduces the published giant phase to 1.2 %. With MIST tracks the slow phase is slower than with SSE, because a MIST track is labelled by the initial mass, and a donor that has lost 2 $M_\odot$ keeps the large core of its initial mass. Which tracks to use is a choice of the configuration; the difference is a property of the stellar models, not of the dynamics.

## 8.4 A binary evolving into contact

**Result.** A 1.0 + 0.5 $M_\odot$ binary with $a$ = 0.010 au (panel d of Figure 19) starts to transfer mass from the primary at 1.31 Gyr. The pair comes into contact at about 2.14 Gyr (the kink in the masses); after that the lighter star feeds the primary while the orbit shrinks by angular-momentum loss (Gazeas & Stępień 2008). At 2.23 Gyr, with masses 0.844 and 0.656 $M_\odot$ ($q$ = 0.78) and a separation of 0.0074 au, the pair merges through the outer Lagrange point L2. That contact binaries lose angular momentum and merge within gigayears is the published expectation; the merger here happens at L2 rather than at the Darwin limit ($q \approx 0.09$; Rasio 1995) because the mass ratio is far from extreme. The quantitative benchmark of the angular-momentum loss is `contact-jdot` (Chapter 3), and the light curve of a contact pair agrees with PHOEBE (Chapter 4).

## 8.5 A common envelope inside a triple

**Result.** A 1.8 $M_\odot$ giant (93 $R_\odot$) with a 0.5 $M_\odot$ companion fills its Roche lobe inside a circular 10-yr outer orbit and goes through a common envelope (N-body backend). The 1.37 $M_\odot$ envelope leaves the system, and the 0.43 $M_\odot$ core and the companion spiral in from 0.88 to 0.0093 au. The sudden loss of mass widens the outer orbit and makes it eccentric: by 3.53 times instead of the impulsive 3.48, with $e$ = 0.717 instead of 0.713 (Hadjidemetriou 1963); these are the benchmarks `ce-impulsive-outer-a` and `-e`.

## 8.6 Pulsating triples

**Question.** Light curves and minima of triples with pulsating stars: a δ Sct star (pulsations of about 1 h) in an eclipsing pair with a long-period variable (LPV) red-giant tertiary, and an eclipsing pair with a Cepheid tertiary. Do the pulsations bias the measured minima?

**Input.** A 60-d light curve in the TESS band and minima over 3 yr for the δ Sct triple; a 120-d V-band light curve and minima over 10.5 yr for the Cepheid triple. The minima are measured from the light curves as an observer would.

**Table 42.** Minima of the pulsating triples. Columns: number of minima, scatter of the O−C around the model, mean offset, total range of the O−C.

| System | Minima | Scatter of O−C | Mean offset | O−C range |
|---|---|---|---|---|
| δ Sct + LPV triple | 730 | 136 s | +8.6 s | 17 s (no outer signal in 3 yr) |
| Cepheid triple | 3835 | 137 s | 0.0 s | 5891 s (light time of the outer orbit) |

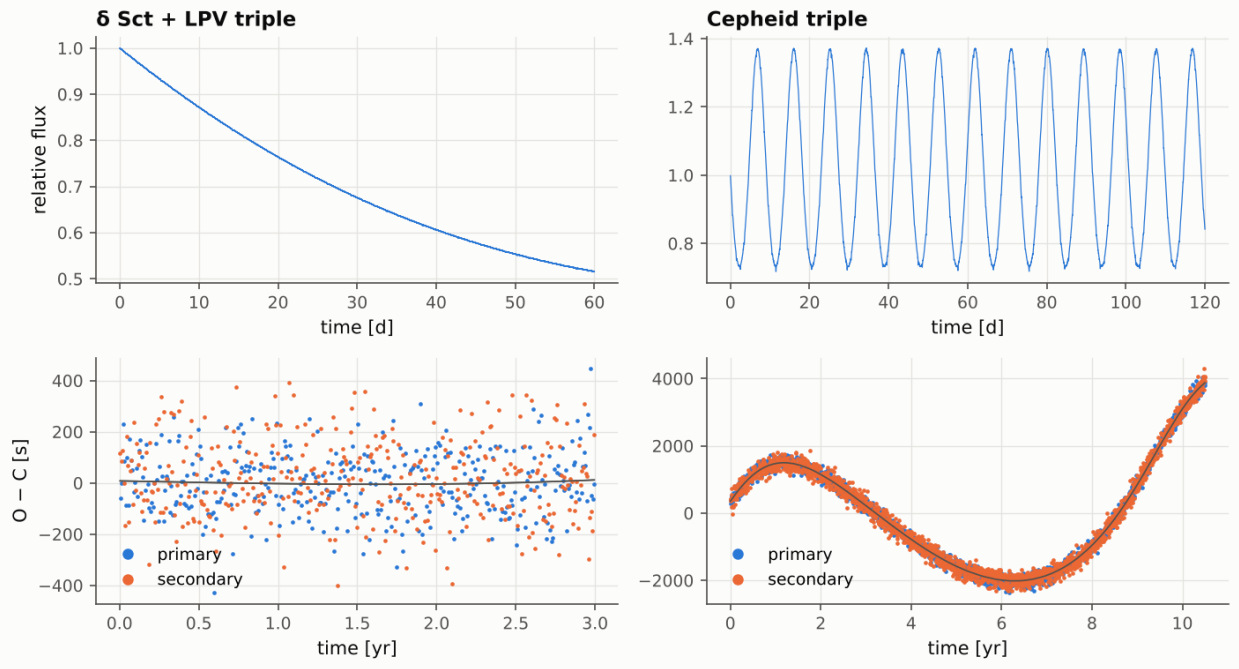



**Figure 20.** Pulsating triples. *Top left:* 60 days of the light curve of the δ Sct triple (relative flux); the slow fall by half is the long-period variable giant, the eclipses and the fast δ Sct pulsations are too small to be seen at this scale. *Top right:* 120 days of the Cepheid triple; the large waves are the 9-d Cepheid pulsations, the eclipses of the companion pair are the small notches. *Bottom:* the O−C of every synthetic minimum (blue primary, orange secondary) in seconds; on the right the black line is the light-time curve of the 10-yr outer orbit.

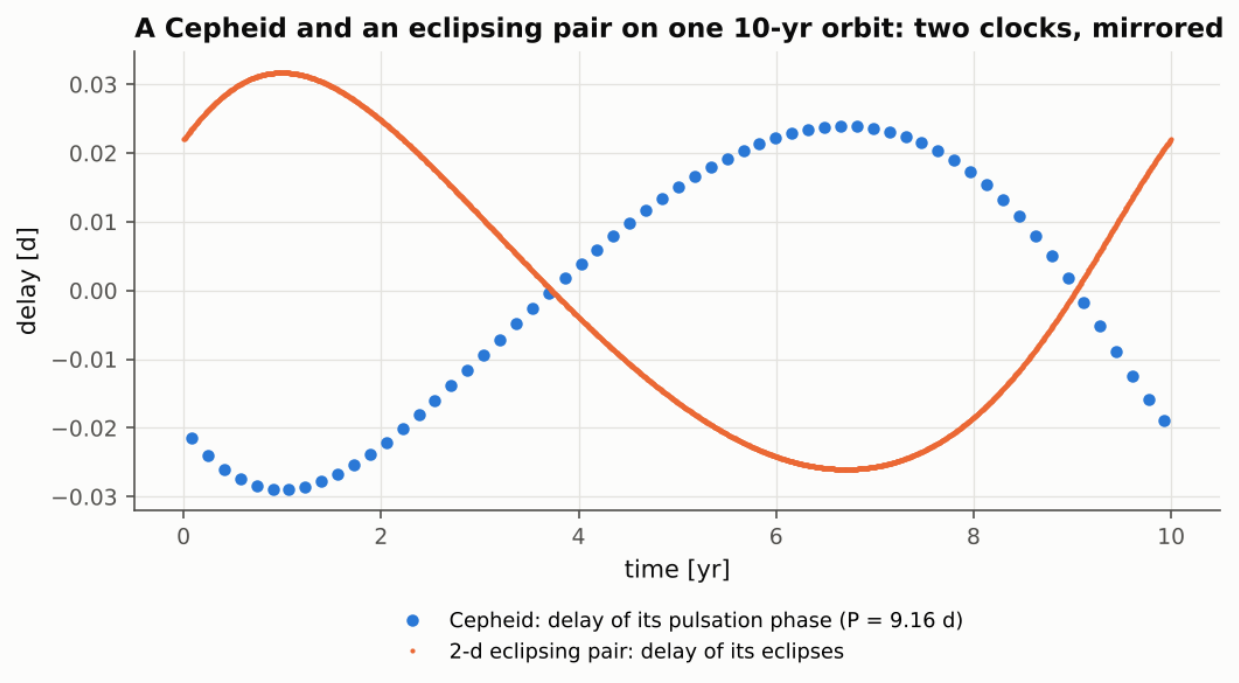



**Figure 21.** The Cepheid triple: two clocks in one system over one 10-yr outer orbit. Blue: the delay of the Cepheid's pulsation phase; orange: the delay of the eclipses of the pair. Both are light-travel-time delays across the outer orbit.

**Result (Table 42).** In Figure 20 (bottom left) the minima of the δ Sct triple scatter by about ±2 min around zero – the pulsations under the eclipses disturb each measurement – but show no trend in 3 years: the long-period variable changes the total light by 50 % in 60 days, yet

the minima stay unbiased (mean offset +8.6 ± 5 s). The scatter of 136–137 s is typical of observed δ Sct eclipsing binaries (tens to hundreds of seconds). On the right, the minima of the Cepheid triple follow the light-time curve of the outer orbit (black line, Irwin 1952) over its full range of 5891 s, with the same scatter. Figure 21 shows the two clocks of the Cepheid triple: because the Cepheid and the pair are on opposite sides of the common centre of mass, one is late when the other is early, and the amplitudes are in the inverse ratio of their masses. One limitation appeared: the limb-darkening tables stop at 3500 K, and the 2867 K giant uses the nearest node; SWINGERsim warns about it (Chapter 10).

# Chapter 9

# Stability: the ghost test, criteria and classifiers

An ensemble of multiple stars contains many members that would not survive: their orbits are too close, and they break up within a few outer orbits. Before the long-term evolution of a member means anything, SWINGERsim has to decide whether the member is stable. It offers three tools of increasing cost:

1. **Algebraic criteria** applied to the nested triples of a hierarchy: MA01 (Mardling & Aarseth 2001) and the criterion of Vynatheya et al. (2022, V22), which includes the dependence on the mutual inclination. They cost microseconds and are SWINGERsim's own code.
2. **Neural classifiers** (multi-layer perceptrons, MLP) trained on 2+2 and 3+1 quadruples. SWINGERsim builds the training sets itself (N-body ghost tests of $10^4$ quadruples per hierarchy) and trains the network with the scikit-learn library; the network library is external, the training data and the use are SWINGERsim's.
3. The **ghost test**, the definition of stability of Vynatheya et al. (2023, Sect. 2): the system and a "ghost" copy whose inner semi-major axis is larger by $10^{-6}$ au are integrated with N-body for 100 outer orbits (Newtonian point masses); the system is unstable if it becomes unbound or if the inner semi-major axes of the two copies differ by more than 1 %. This is the reference.

## 9.1 Reproduction of Vynatheya et al. (2023)

**Reference.** Vynatheya et al. (2023) trained MLP classifiers on $4{\cdot}10^5$ quadruples labelled by the ghost test and published the **scores** – the fraction of test members classified correctly – of MA01 and of their MLP for 2+2 and 3+1 quadruples (their Tables 3–4). Two further measures appear below: the **recall** of the stable class (the fraction of truly stable members the classifier calls stable) and the **precision** (the fraction of members called stable that are truly stable).

**Input.** Two training sets of $10^4$ quadruples each, drawn from the prior of the paper (their Sect. 3.1: masses, period ratios, eccentricities, inclinations), classified by the ghost test with a 30-min limit per member; an MLP with the 11 input parameters and the architecture of the paper (4 hidden layers of 50 logistic units, Adam optimiser, L2 regularisation $10^{-4}$; their Sect. 3.2), five networks averaged, trained on 80 % of the members and scored on the remaining 20 %. A third set of $10^4$ 3+1 quadruples was drawn, as in the paper, only near the MA01 stability boundary.

**Table 43.** Vynatheya et al. (2023): scores against the ghost test. These four benchmarks are manual; the MLP is class ext (the scikit-learn network on SWINGERsim's training sets), MA01 class own. Columns as defined in How to read this report.

| Benchmark | Class | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|---|
| `vyn23-2p2-ma01-score` | own | 0.83 | 0.869 | +0.039 | 0.05 | PASS |
| `vyn23-2p2-mlp-score` | ext | 0.94 | 0.9375 | −0.0025 | 0.03 | PASS |
| `vyn23-3p1-ma01-score` | own | 0.56 | 0.7896 | +0.23 | 0.05 | **EXPLAINED** |
| `vyn23-3p1-mlp-score` | ext | 0.93 | 0.9195 | −0.0105 | 0.03 | PASS |

**Table 44.** The three training sets. Columns 2–4: the fraction of members that are stable by the ghost test, by MA01 and by V22; columns 5–7: the scores of MA01, V22 and the MLP against the ghost test.

| Set | Stable (ghost) | Stable (MA01) | Stable (V22) | Score MA01 | Score V22 | Score MLP |
|---|---|---|---|---|---|---|
| 2+2, prior of the paper | 28.2 % | 39.3 % | 34.9 % | 0.869 | 0.895 | 0.938 |
| 3+1, prior of the paper | 9.5 % | 28.5 % | 26.3 % | 0.790 | 0.812 | 0.920 |
| 3+1, near the MA01 boundary | 21.0 % | – | – | 0.509 | 0.568 | – |

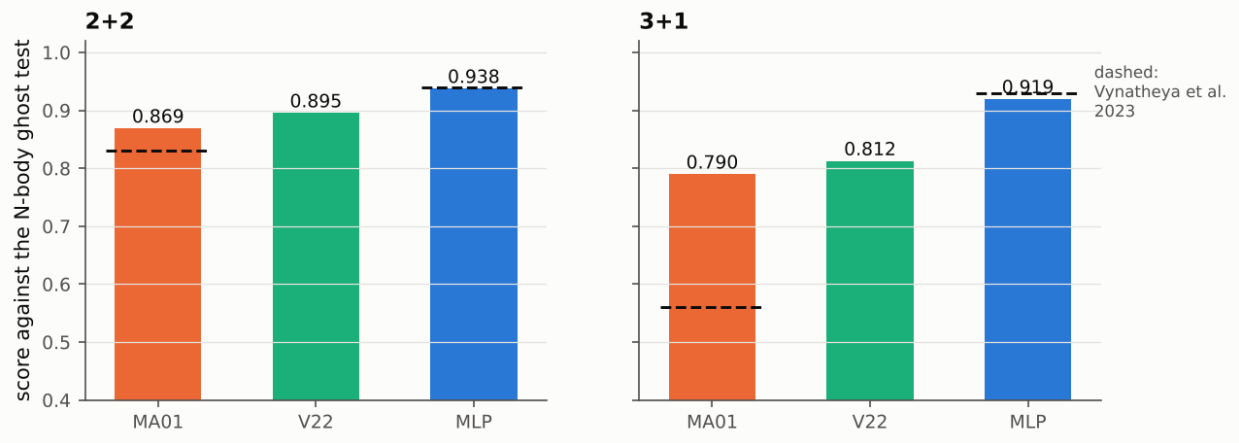



**Figure 22.** Scores of the three classifiers – MA01 (orange), V22 (green) and the neural network MLP (blue) – against the N-body ghost test, for the 2+2 set (left) and the first 3+1 set (right). The vertical axis is the score, the fraction of test quadruples whose stability is predicted correctly (1 = perfect); the number above each bar repeats it. The black dashed lines are the scores published by Vynatheya et al. (2023), who give none for V22.

**Result (Tables 43 and 44).** In Figure 22 the blue bars reach the dashed published lines in both sets: the MLP scores 0.938 (2+2) and 0.920 (3+1) against the published 0.94 and 0.93 – within 0.01, although it was trained on 40 times fewer systems. MA01 scores 0.869 on the 2+2 set against the published 0.83 (+0.04, within the tolerance of 0.05 set by the difference between training sets). On the first 3+1 set, however, MA01 scores 0.79 against 0.56. This benchmark is EXPLAINED by the sampling of the test set, for three reasons. First, with only 9.5 % stable systems a classifier that calls *everything* unstable already scores 0.905; MA01 calls 28.5 % stable – too many – but recognises most of the many unstable members, so its score is high. Second, the paper drew its members close to the MA01 boundary, and its own Table 4 (precision and recall) implies a score of 0.69, not 0.56. Third, the set drawn near the boundary as in the paper gives 0.51, close to the published 0.56 (and V22 0.57). This window set has fewer stable members (21 %) than the about 36 % that the published precisions and recalls imply, which lowers the precision of the stable class (0.29 against the published 0.54; recall 0.94 against 0.95). A score depends on how the test set is drawn and is meaningful only together with the sampling. No member of the $3{\cdot}10^4$ ghost tests reached the time limit.

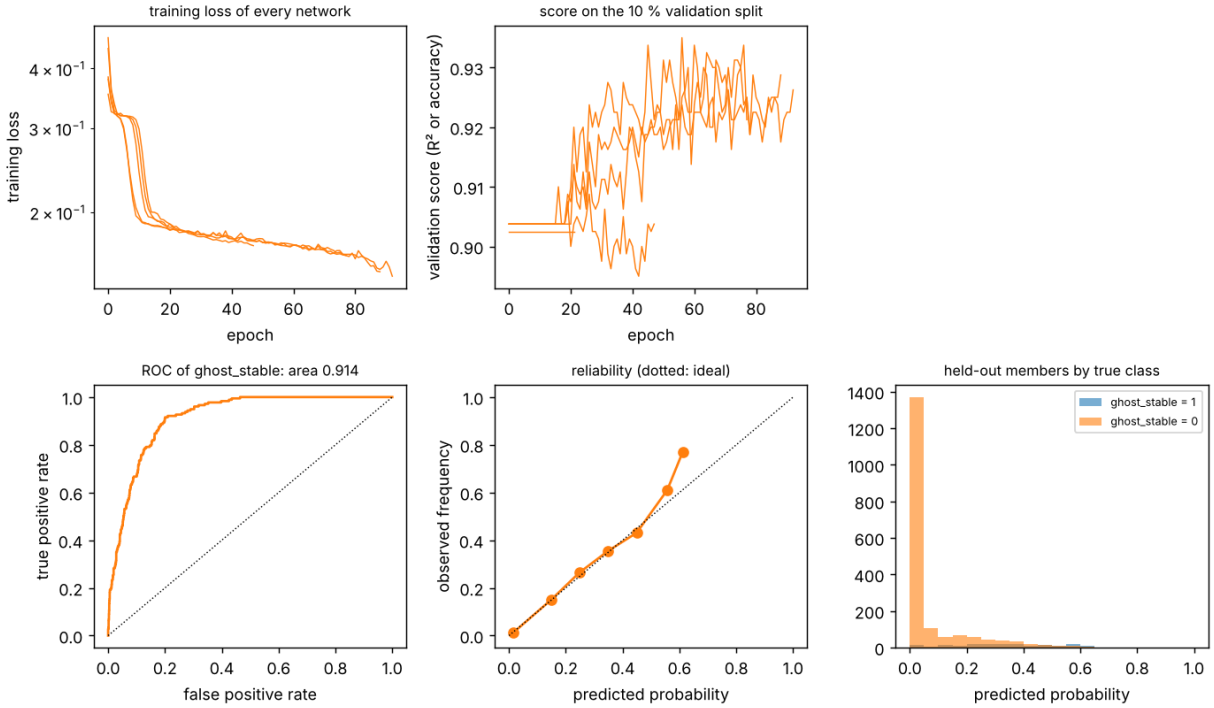



**Figure 23.** Diagnostics of the 3+1 neural network. *Top left:* training loss of each of the five networks against the training epoch. *Top middle:* score on the validation members kept aside during training against the epoch. *Bottom left:* ROC curve – for every threshold of the predicted probability, the fraction of truly stable members called stable (vertical) against the fraction of unstable members called stable (horizontal); dotted: a random classifier; the area under the curve is 0.914. *Bottom middle:* reliability – of the members with a predicted probability $p$, the fraction that is really stable (dotted: ideal). *Bottom right:* histogram of the predicted probabilities of the held-out members by their true class (blue stable, orange unstable).

**Calibration.** Figure 23 shows that the 3+1 network has learnt (the loss falls and levels off; the validation score rises from 0.90, the score of calling everything unstable, to 0.92–0.93) and that its probabilities can be taken at face value: the reliability points lie on the diagonal. The histogram shows the limitation: the few stable 3+1 systems rarely get a probability above 0.5, so the recall of the stable class at the threshold 0.5 is only 0.20 (precision 0.65; for 2+2: recall 0.90, precision 0.88). For 3+1 systems the probability itself should be used, not the label at 0.5.

**Findings about the criteria.** On the broad sets MA01 and V22 call more systems stable than the ghost test (2+2: 39.3 % and 34.9 % against 28.2 %; 3+1: 28.5 % and 26.3 % against 9.5 %). They are optimistic: a system they call stable is not necessarily stable for 100 outer orbits. In ensembles SWINGERsim therefore reports the algebraic verdicts as diagnostics and uses the N-body backend (or the ghost test) where stability matters, as in Chapter 6.

## 9.2 Chaos maps and simulation-based inference

**Chaos maps.** SWINGERsim computes MEGNO on a grid of two orbital parameters to map regular and chaotic regions. The benchmark is the inner stability limit of a circumbinary orbit around an equal-mass circular binary, published as a grid by Quarles et al. (2018): 2.27 binary separations (the fit of Holman & Wiegert 1999 gives 2.39). SWINGERsim, requiring all of 8 orbital phases to be regular, finds 2.30 (`chaosmap-circumbinary-limit`, class R; tolerance 0.06, the grid step of 0.05 plus the coarser sampling of 8 phases and $10^4$ binary orbits instead of $10^5$).

**Simulation-based inference.** For posteriors that an ensemble cannot reach by weighting, SWINGERsim can train a neural posterior estimator with the external `sbi` package (Tejero-Cantero et al. 2020) on its own simulations. The benchmark is a linear Gaussian model with

an analytic posterior: with 2500 training simulations the posterior width is 0.291 against the exact 0.309 (−5.8 %, tolerance 10 %; `sbi-linear-gaussian-std`, class ext). This benchmark checks that SWINGERsim passes its simulations to the package correctly; the inference algorithm itself is external.

# Chapter 10

# Problems found during testing and how they are handled

Running the benchmarks, the catalogue and ensembles of up to $5 \cdot 10^4$ members tested the software as much as the physics. This chapter collects what the tests revealed about the numerics and the practical use of SWINGERsim, written as properties of the software that a user needs to know: what can go wrong, how SWINGERsim detects or prevents it, and what the user should do. A second group of findings concerns the published references themselves; it is listed at the end, because it affects how any code should be compared with the literature.

Table 45 summarises the findings; each row is detailed in the section named in its last column.

**Table 45.** Summary of the findings. Column 2 says what can happen, column 3 how SWINGERsim handles it, column 4 where in this report the finding appears.

| Area | What can happen | How SWINGERsim handles it | Seen in |
|---|---|---|---|
| unstable members | a close encounter makes the time step collapse and a run never ends | collision search valid in both directions of time; wall-clock limit checked after every step; fate `wall_time` | Ch. 6 |
| short evolutionary phases | segments of the evolving backend step over thermal pulses | segment boundaries at the nodes of the tracks, refined to 1 % in radius | Ch. 8 |
| osculating elements | osculating periods are not the periods of eclipses | mean motion over an orbit for light-curve chunks; anchoring of eclipse periods in fits | Ch. 5 |
| mass transfer in eccentric orbits | a tertiary pumps the eccentricity of a transferring pair | warning; tides circularise; onset check allows for forced eccentricity | Ch. 8 |
| chaotic and ill-conditioned runs | two correct integrations diverge | compare statistics of ensembles, not single members | Ch. 2 |
| secular backend near the stability limit | orbit averaging fails | stability diagnostics in every run; N-body check of the best members | Ch. 6 |
| ensemble weights | one member carries the whole weight | effective number of members reported with every evidence | Ch. 6 |

| Area | What can happen | How SWINGERsim handles it | Seen in |
|---|---|---|---|
| undefined quantities | apsidal period of a nearly circular orbit | reported as not measurable | Ch. 7 |
| stellar tracks | a present mass is reached by two tracks; a mass outside the grid; a star overfilling its lobe at the start | warning and the least evolved track; member refused with the reason | Ch. 5 |
| tables of atmospheres | temperatures or gravities outside the limb-darkening grids | nearest node with a warning | Ch. 8 |
| conserved quantities | the Newtonian energy is not conserved with relativity | errors measured against the 1PN Hamiltonian | Ch. 2 |

## 10.1 Unstable members and wall-clock limits

An ensemble drawn from a broad prior always contains members close to or beyond the stability limit. In such a member two stars can fall towards each other. If the collision is detected, the run ends with the fate *collision*. The tests showed that REBOUND's default ("direct") collision search misses pairs that approach each other when the integration runs backward in time; two stars then pass through each other, the adaptive time step of IAS15 shrinks without bound (to $10^{-48}$ yr) and the run does not finish; with the direct search, eleven of the $10^4$ runs of the CzeV4315 ensemble with tides behave like this. SWINGERsim therefore uses a collision search that works in both directions of time ("line" search) for backward runs, and checks a wall-clock limit (`run.max_wall_s`) after *every* integration step, not only between output times. A member that reaches the limit ends with the fate `wall_time` and is reported as such; it is not silently dropped. Integrated with the line search, all eleven of these members end in a collision, which is their correct fate. The ghost tests of Chapter 9 have their own limit of 30 minutes per member; none of the $3{\cdot}10^4$ ghost tests reached it. What a user should do: keep the limit (it protects the ensemble, not the physics), read the fates, and treat `wall_time` members as unresolved, not as stable.

## 10.2 Step size of the evolving backend

The evolving backend advances the orbits in segments within which the stellar radii change by less than 1 %. On the thermally pulsing AGB a star's radius jumps by a factor of 2–4 at each pulse for only a few hundred years, every $10^4$–$10^5$ yr. A uniform time grid steps over the pulses: on such a grid the giant tertiary of Chapter 8 reaches at most 313 $R_\odot$ and 73 % of its Roche lobe and ends at the end of its track, while the MIST track itself reaches about 500 $R_\odot$. SWINGERsim therefore puts the segment boundaries at the nodes of the MIST track and refines them until no radius changes by more than 1 % between two boundaries; with this the same giant overflows its lobe at 452 $R_\odot$ on a pulse peak. The same rule protects every short evolutionary phase (the Hertzsprung gap, helium flashes). The price is run time on the AGB; there is no switch to turn the refinement off.

## 10.3 Osculating elements and eclipse periods

SWINGERsim's orbital elements are **osculating**: computed from the instantaneous positions and velocities. In a multiple system, and in a binary of tidally distorted stars, the osculating period is not the period with which eclipses repeat. Two consequences appeared in the tests. (i) The PHOEBE hybrid computes the light of distorted stars in chunks of the N-body orbit; if each chunk used the osculating period of a tidally distorted binary (137 s per orbit off for QZ Car Ac), its minima would drift by tens of seconds. SWINGERsim uses the mean motion over

one orbit, and the PHOEBE minima agree with the spherical model to a few seconds (Chapter 5). (ii) Published elements entered literally into an N-body model give eclipses that drift by hours within years; Part II shows how SWINGERsim *anchors* the period and phase of each eclipsing pair to the observed minima. A related point: the light travel time must be applied to the positions consistently. With exact retarded positions the minima of QZ Car follow the barycentric conjunctions displaced only by the Rømer delay inside the binary (±16 s); a first-order correction of the positions would shift them by about 200 s. The Rømer benchmarks of Chapter 4 guard this.

## 10.4 Mass transfer in an eccentric orbit

SWINGERsim's mass transfer is formulated for circular orbits, as in BSE. A tertiary can, however, force a small eccentricity on a transferring pair, or pump it by ZLK cycles. Two effects followed. First, a star in a slightly eccentric orbit fills its lobe at periastron earlier than in a circular one (in the Algol triple by 11 kyr); the onset check allows for the forced eccentricity, so the run does not stop before the transfer. Second, without tides the tertiary of the Algol triple pumps the eccentricity of the transferring pair to 0.44 (0.33 at the end of the run), outside the validity of the circular formulation; SWINGERsim prints a warning that names the binary and the eccentricity. With tides switched on, as in the run of Chapter 8, the pair stays circular ($e < 4{\cdot}10^{-4}$) and the masses at the end are the same (0.454 and 3.172 $M_\odot$). What a user should do: switch on tides for any binary that transfers mass in a multiple system.

## 10.5 Chaotic runs and ill-conditioned starts

In chaotic configurations two correct integrations diverge after a finite time: the N-body and secular runs of the Fig. 4 set-up of Vokrouhlický (2016) after about 4 kyr (the paper reports the same for its two integrators), SWINGERsim and MSE after three ZLK cycles of the 2+2, and individual members of the Pejcha ensembles. A similar sensitivity appears for regular orbits started next to a separatrix: a ZLK system started at $e_0 = 0.01$ differs by 10 % in the timing of its cycles after two cycles between the N-body and the secular backend (0.5 % for constant masses away from the separatrix). Neither is an error. What a user should do: compare chaotic systems statistically (fractions of ensembles, as in Chapter 2, where the two backends agree within the errors), and avoid comparing single members or times beyond the Lyapunov time.

## 10.6 The secular backend near the stability limit

The secular backends average over the orbits and therefore assume a stable hierarchy with well-separated periods. Near the stability limit this fails: in ASASSN-V J075203 (period ratio 6–9) the N-body apsidal period of A is about half the secular one, and in CzeV4315 half of the secular 2+2 solutions turn out to be chaotic or to collide in N-body (Chapter 6). A secular run cannot see instability; it reports the MA01 and V22 ratios and the octupole and "single-averaging" parameters of every nested triple in its summary and marks mild hierarchies ("secular estimates approximate"). What a user should do: use the secular backend to pre-select, then check the members that matter with the N-body backend and MEGNO.

## 10.7 What an ensemble weight of one member means

The evidence of a model is the mean weight of its members, and its reliability is measured by the effective number of members $n_{\mathrm{eff}}$ (Chapter 6). When the observations are precise and the prior broad, almost all weight can fall on one member. ASASSN-V J075203 with an isotropic prior gives $n_{\mathrm{eff}} = 1.0$: one member out of 20 000 carries everything, so the "evidence" is the weight of a single lucky draw and says nothing about the model. The 2+2 of CzeV4315

conditioned on the eclipses of A has $n_{\mathrm{eff}} = 4.8$, and a comparison of models with the same 119-d prior had $n_{\mathrm{eff}} = 1.6$ and is therefore only a noisy estimate. SWINGERsim prints $n_{\mathrm{eff}}$ next to every evidence and every weighted fraction. What a user should do: trust an evidence only with $n_{\mathrm{eff}}$ of tens or more; otherwise run more members, focus the prior on the region that fits (as the targeted ensemble of J075203 did, $n_{\mathrm{eff}} = 53$), or use an emulator or simulation-based inference.

## 10.8 Quantities that are not defined

Some quantities a user may ask for do not exist for a given orbit. The apsidal period of a nearly circular orbit is the clearest case: in TIC 120362137 the tertiary forces the eccentricity of the 3.3-d pair to oscillate between 0.0001 and 0.016, the osculating $\omega$ jumps, and a "measured" apsidal period is meaningless (the paper gives only a theoretical value). The same holds for Algol ($e \approx 0.001$) and for orbits with $e < 10^{-6}$. SWINGERsim reports such a period as "not measurable in the run" rather than printing a number. Long-term extremes of strongly interacting members, in which small numerical differences grow, are similarly ill-defined and are not used as results.

## 10.9 Stellar tracks at their limits

Three situations occur when a configuration asks the stellar tracks for something they cannot give. (i) For very massive stars with strong winds the present mass of a star at a given age can be reached by two tracks of different initial mass (for example 79 $M_\odot$ at 2.5 Myr by tracks of about 108 and 139 $M_\odot$ in QZ Car ensembles); SWINGERsim takes the least evolved star and warns. (ii) A mass that does not exist at the requested age on the chosen grid (Aa1 of QZ Car above 73.3 $M_\odot$ on the rotating tracks at 2.5 Myr) cannot be started; the member ends with fate *error* and the reason. (iii) A star that already overfills its Roche lobe at the start is refused with the reason. The refused members are information, not failures: in QZ Car they bound the mass of Ac1 from above (Chapter 5). Errors are counted separately in every ensemble summary, and a user can rerun them after a change with `--resume --retry-errors`.

## 10.10 Tables of model atmospheres

The limb-darkening tables of Claret & Bloemen (2011) stop at 3500 K and cover $\log g$ from 0 to 5. A 2867 K long-period variable or a supergiant with $\log g < 0$ uses the nearest node, with a warning that names the star and the values. For the relative light of such a star in a multiple system this is adequate; for its own eclipse profile it is not, and tables for cooler stars are outside the scope of SWINGERsim.

## 10.11 Conserved quantities as diagnostics

Every run reports its relative energy and angular-momentum errors. With relativity the conserved quantity is the 1PN Hamiltonian, not the Newtonian energy; SWINGERsim measures the error against the Hamiltonian of REBOUNDx's `gr_full`, otherwise a correct relativistic run would show a spurious drift. With dissipation or mass loss the energy is not conserved by the physics, and the diagnostic is then the adiabatic invariant ($aM$ for winds, tested to $4{\cdot}10^{-9}$ in Chapter 3). Time symmetry is a further diagnostic of conservative dynamics: failures forward and backward in time agree to 0.02 % in the CzeV4315 ensembles (Chapter 6).

## 10.12 Problems in the published references

Tracing every difference back to its paper found problems in the references themselves. They are listed here because any code compared with these papers will meet them.

Table 46 lists them; each row names the reference, what is wrong or ambiguous in it, and how the benchmark deals with it.

**Table 46.** Problems in the published references found while reproducing them. Column 2 describes the problem, column 3 what SWINGERsim's benchmark uses.

| Reference | Problem | What the benchmark uses |
|---|---|---|
| Naoz et al. (2013), Fig. 4 | the caption (angles 6.75° and 63.25°) and the quoted $a_2$ = 50 au imply different angular-momentum ratios (7.60 and 6.79); the published curve belongs to 7.60 | $a_2$ = 39.9 au, which gives 7.60 |
| Veras et al. (2011), eq. 17 | denominator $1 - e_0 \cos f$ does not follow from their eq. 4 | the form derived from eq. 4, $1 + e_0 \cos f$ |
| Antognini & Thompson (2016), Table 3 | stated normalisation $\pi(a_1^2 + a_2^2)$; the numbers correspond to $\pi a^2$ | $\pi a^2$ |
| Zasche et al. (2023), V994 Her C | quoted relativistic share 6 %; their Table 3 gives 8.1 % | EXPLAINED with 8.1 % |
| Hurley et al. (2002), Algol example | the published history would need the orbit to gain angular momentum in the Hertzsprung gap | EXPLAINED; the conservative giant phase is a separate benchmark |
| Portegies Zwart & Meinen (1993) | capture radii quoted from Ray et al. (1987) belong to other conventions | EXPLAINED; McMillan et al. (1987) as the reference |
| Vynatheya et al. (2023), MA01 for 3+1 | the quoted score 0.56 does not follow from their own Table 4 (0.69) | EXPLAINED; set drawn near the boundary |
| Pejcha et al. (2013), Vokrouhlický (2016) | the fractions grow with the run length, which is not stated precisely enough | N-body PASS at 3σ; secular EXPLAINED |
| Kemp et al. (2024), KIC 4150611 | the dynamical mass ratio contradicts the size and period ratios of the same paper | masses consistent with the ratios |
| Pribulla et al. (2020), VW LMi | the quoted 78.6-yr apsidal period is a fit to three years of data; the paper's own models give 95.5–96.2 yr | comparison with the paper's N-body model |
| MSE (Hamers et al. 2021) | the argument of periastron is lost at an inclination of exactly 0° | inclinations of 0.5–2° |

The lesson is simple but important: a published number is a measurement of something, and before a difference is called an error of either side, one has to establish what exactly was computed, from which input, with which definition, and over which time.

# Chapter 11

# Conclusions

## 11.1 What is validated, and to which accuracy

SWINGERsim carries 219 benchmarks, each with its source and a tolerance fixed before the comparison. With the final code, 205 pass and 14 lie outside their tolerance for a documented reason that is not an error of SWINGERsim; none fails. Three quarters of them (165) test code written for SWINGERsim itself, not only the external N-body integrator. Table 47 summarises the accuracy reached for each group of physics; the individual numbers are in Chapters 2–4 and in Appendix A.

**Table 47.** Validated physics. Column 2 names the references, column 3 the accuracy reached (the largest difference among the passing benchmarks of the group, or the typical one where stated), column 4 the class of the tested code (Table 1).

| Physics | References | Accuracy | Class |
|---|---|---|---|
| Newtonian and 1PN N-body dynamics | Mercury, ZLK limits, Söderhjelm, MA01, MEGNO | $10^{-4}$ (relativity), 0.3 % (ZLK, with the expected short-period excess) | R |
| secular backends (triple, 2+2, nested) | analytic limits, Naoz et al. (2013), Vokrouhlický (2016), MSE, N-body of a sextuple | $10^{-4}$ in analytic limits; 0.04° on published curves; $10^{-4}$ in $e_{\max}$ against MSE; 1.2 % in the ZLK period against N-body | own |
| ensembles of chaotic systems | Pejcha et al. (2013), Antognini & Thompson (2016) | triple fractions and enhancement agree; quadruple fractions depend on the run length (see below); cross-sections 1–2 % (exchange 23 %, 1.8σ) | R, own |
| tides and apsidal motion | Sterne (1939), Hut (1981), Leconte et al. (2010), CzeV343 | $10^{-4}$ (apsidal), 0.3 % (dissipative rates), 3 % (observed apsidal period) | R, own |
| stellar structure constants and tracks | Brooker & Olle, Claret, Tout et al., MIST, Geneva, the Sun, white dwarfs | last digit of tables; 5 % for the Sun; 4–12 % between model families | own/tab |
| inertial waves and pre-main-sequence circularisation | Ogilvie, Barker, Zahn & Bouchet, Meibom & Mathieu | 0.03 % (solver), 1 % (pre-main sequence) | own |
| mass loss | Jeans, Hadjidemetriou, Veras et al., Blaauw/Hills, Perets & Kratter | $4{\cdot}10^{-9}$ (adiabatic invariant), 0.24 % (impulsive), 6.5 % (instability onset) | R, own |

| Physics | References | Accuracy | Class |
|---|---|---|---|
| mass transfer and common envelope | Soberman et al., Postnov & Yungelson, BSE, COSMIC, Tout et al. | $3{\cdot}10^{-13}$ (closed forms); 0.02 $M_\odot$ in the stripped donor; 0.5–9 % in post-common-envelope periods | own |
| magnetism, Applegate mechanism | Matt et al. (2015), Völschow et al. (2016) | 0.7 % (solar spin), 6 % (energy ratios) | own |
| environment and gravitational waves | Grishin & Perets, Hamers & Samsing, PSR B1913+16 | $3{\cdot}10^{-5}$ (Galactic tide), 0.7 % (passing stars), 0.02 % (orbital decay) | own, R |
| light curves on spheres and meshes, contact binaries | batman, ellc, PHOEBE 2.5 | $10^{-8}$ (occultation), $4{\cdot}10^{-4}$ (PHOEBE, spheres), 0.003 mag (contact binary) | own |
| dust discs | exact geometry, Farago & Laskar, ε Aur, HD 98800 | exact geometry; 14 % in the ε Aur duration (sharp edge); 2.5° in the HD 98800 disc | own |
| eclipse timings and the stage-1 fit | Irwin, Kaplan, Giménez & Bastero, OCFit | $10^{-4}$ (LTTE), 0.3–0.6 % (Rømer delay), 0.2 % ($a \sin i$ from N-body minima) | own |
| pulsation timing | Murphy et al. (2014), δ Sct and Cepheid relations | $3{\cdot}10^{-7}$ (analytic), 1.3 % (KIC 9651065), 2 % (catalogue pulsators) | own |
| stability classifiers | Vynatheya et al. (2023), Quarles et al. (2018) | MLP scores to 0.01; MA01 to 0.05 with the sampling of the paper | own, ext |

Beyond the benchmarks, SWINGERsim reproduces the published model of QZ Car (orbit sizes to 0.07 %, light shares to 0.001), the precession, timing and pulsation-timing amplitudes of the catalogue systems (within the published uncertainty or about 10 % wherever the input is consistent), and the observed ZLK phases of σ Ori and HD 98800. It produced new, testable results: the 2+(2+1) hierarchy of CzeV4315 is favoured by a factor of about 70 over a 2+2, and tides halve its failure rate; a 2+2 explains ASASSN-V J075203 only in a narrow regular island with two predicted signatures; the eclipse-timing variations of the compact 2+2 TIC 454140642 run in phase; HD 181068 ends in a triple common envelope; and the precession of the discs of EE Cep and KH 15D constrains their size.

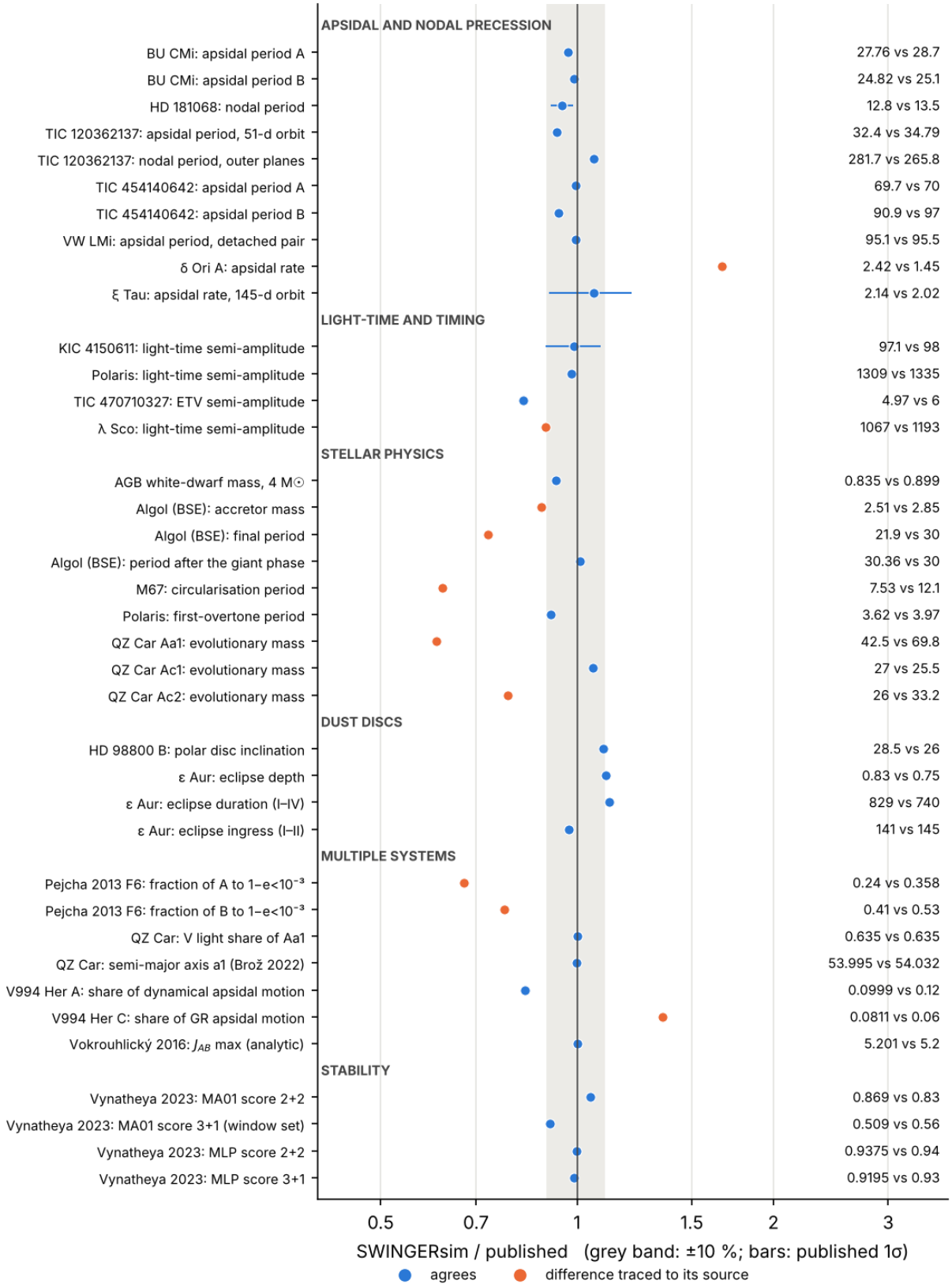



**Figure 24.** Every quantitative comparison of SWINGERsim with a published value in this report, on one scale. Each row is one number; the horizontal position of the point is SWINGERsim divided by the published value (logarithmic axis; 1 = the same number, 0.5 = half, 2 = double). The grey band marks ±10 %; horizontal bars are the published 1σ uncertainties divided by the published value. Blue: agrees; orange: the difference has been traced to its source. The column on the right gives the two numbers, SWINGERsim first.

Figure 24 puts all comparisons of whole systems and published simulations on one scale. Read it row by row: a blue point inside the grey band is an agreement to better than 10 %; a blue point outside the band with a bar that reaches the vertical line is an agreement within the published error. Of the 38 rows, 28 are blue. Each of the 10 orange rows has a traced cause: the input of the configuration or of the paper (δ Ori A, λ Sco, the QZ Car gainers), published numbers that are inconsistent (the BSE Algol, V994 Her C), the length of the published runs (Pejcha et al.), or an open point of SWINGERsim's tidal model (M67).

## 11.2 The fourteen explained differences

**Table 48.** The 14 EXPLAINED benchmarks. Column 2 gives SWINGERsim's value, column 3 the published one, column 4 the reason, column 5 the kind of reason.

| Benchmark | SWINGERsim | Published | Reason | Kind |
|---|---|---|---|---|
| `v994s-gr-C` | 0.081 | 0.06 | the paper's own Table 3 gives 8.1 %; A and B, computed the same way, match | slip in the paper |
| `mt-bse-algol-accretor` | 2.51 $M_\odot$ | 2.85 $M_\odot$ | the published history would need the orbit to gain angular momentum in the Hertzsprung gap; the conservative giant phase is reproduced to 1.2 % | published numbers inconsistent |
| `mt-bse-algol-period` | 21.9 d | 30 d | as above | published numbers inconsistent |
| `qzcar-evolutionary-mass-ac2` | 26.0 $M_\odot$ | 33.2 $M_\odot$ | Ac2 is a likely mass gainer (its fitted rotation is 410 km/s); single-star tracks cannot describe it | input / definition |
| `qzcar-evolutionary-mass-aa1` | 42.5 $M_\odot$ | 69.8 $M_\odot$ | Brož et al. suggest past transfer Aa2 → Aa1; a rejuvenated gainer looks younger and lighter | input / definition |
| `vok16-pejcha-fA` | 0.21 | 0.31 | fractions grow with the run length; at 50 kyr they are 0.36; three independent integrations agree | length of the published runs |
| `vok16-pejcha-fB` | 0.40 | 0.481 | as above (0.57 at 50 kyr) | length of the published runs |
| `vyn23-3p1-ma01-score` | 0.79 | 0.56 | the score depends on the sampling; the paper's own Table 4 implies 0.69; drawn near the boundary as in the paper: 0.51 | sampling of the published set |
| `tidal-capture-radius-n3` | 2.37 | 2.72 | the quoted radii belong to other conventions; the same fits agree with McMillan et al. (1987) | definition |
| `tidal-capture-radius-n15` | 3.33 | 3.84 | as above (3.33 against about 3.4) | definition |

| Benchmark | SWINGERsim | Published | Reason | Kind |
|---|---|---|---|---|
| `tides-circularization-m67` | 3.8 d | 12.1 d | the equilibrium tide alone is too weak on the main sequence; Barker (2022) explains M67 with inertial waves, which SWINGERsim's two-layer estimate with SSE/MIST structure and evolving spins does not reproduce (stellar structure and spin history) | open point of the tidal model |
| `tides-circularization-m67-iw` | 7.5 d | 12.1 d | as above, with inertial waves | open point of the tidal model |
| `tides-circularization-m67-final` | 7.5 d | 12.1 d | as above, with the pre-main sequence (which is reproduced: 6.6 d against 7.1 ± 1.2 d) | open point of the tidal model |
| `eecep-disc-precession` | 183 yr | 65 yr | a rigid disc of the illustrative size precesses too slowly; 65 yr needs an outer radius of about 1.6 au | constraint on the disc |

None of the fourteen is an error of SWINGERsim; the three M67 benchmarks remain an open point of its tidal model. Four are problems of the published numbers (a slip, an inconsistent history that affects two benchmarks, and a score that the paper's own table does not support), two concern stars that are not single-star models, two a different definition, two the length of published runs, three an open point of SWINGERsim's tidal model, and one turns a difference into a constraint on an object.

## 11.3 What is not validated

The following parts of SWINGERsim, or regimes of its physics, have no independent reference in this report. They should be used with that in mind.

- **Main-sequence tidal circularisation:** the 12-d circularisation of M67 is not reproduced by SWINGERsim. Barker (2022) explains it with inertial waves computed from full stellar profiles; SWINGERsim's frequency-averaged two-layer estimate with SSE and MIST structure gives 7.5 d. The difference lies in the stellar structure and the spin history and is an open point.
- **The streams of mass transfer** are prescriptions (BSE rates, accretion fractions of Hamers et al. 2022), not hydrodynamics; the accreted fraction during thermal-time-scale transfer is the least certain number of Algol evolution and is a user's choice. Tertiary mass transfer and triple common envelopes have no quantitative benchmark; they are checked for conservation and against the expected outcome.
- **Late phases of very massive stars:** from the late main sequence on, the inflated envelopes of 40–70 $M_\odot$ stars are not resolved by the MIST grid (radius errors up to 20–40 %), and the masses at the end of the main sequence differ between model families by up to a third.
- **Contact binaries** have one common temperature and no thermal relaxation oscillations; the merger is tested only against the expected behaviour.
- **Dust discs** have no gravity acting on the orbits, and a star inside a disc does not add its light.
- **Atmospheres below 3500 K** and with $\log g < 0$ use the nearest node of the limb-darkening tables.
- **Stability classifiers for more than four stars** apply the 2+2 and 3+1 networks to sub-quadruples, an approximation outside their training sets; the algebraic criteria for every nested triple and the ghost test are the reference.

- **Comparisons not yet made:** the 1:1 resonance of the binaries of a 2+2 (Breiter & Vokrouhlický 2018), secular chaos in 2+2 and 3+1 systems (Hamers & Lai 2017), fates of hierarchical systems with stellar evolution against MSE, fractions of triples destabilised by winds and mass transfer (Toonen et al. 2022), and a wider sample of observed systems such as the Multiple Star Catalog (Tokovinin 2018).

Outside the scope of SWINGERsim by design are relativistic pulsar timing, compact remnants and supernova kicks, Wolf–Rayet stars to core collapse, colliding winds, starspots and flares in the light curves, non-radial pulsation modes, and the selection effects of surveys.

## 11.4 Summary

Every physics module of SWINGERsim reproduces independent published results: analytic limits to $10^{-4}$ or better, published simulations and other codes to a fraction of a degree or a few per cent, and observed systems within their uncertainties. Where a difference remains, it has been traced to the published number, its definition or the length of the published runs, or it is stated as an open point (the main-sequence circularisation of M67). Most of this evidence concerns code written for SWINGERsim – the secular backends, the coupling of evolution, tides and mass transfer to the orbits, the light curves, timings, fits and ensembles. Parts II–IV test the same code on real observations of SQUADRA systems.

# Appendix A

# All 219 benchmarks

This appendix lists every benchmark registered in SWINGERsim with its result from the final code, by level of evidence. Each level has two tables with the same rows in the same order: the first says what each benchmark compares and which code it tests (Tables A1, A3, A5, A7), the second gives its result (Tables A2, A4, A6, A8); a benchmark is found in both by its Id. Within a level the benchmarks appear in the order in which the software registers them, so that a row described as "the same for …" follows the row it refers to. The columns are:

- **Id** – the identifier; `swingersim validate <Id>` runs the benchmark and prints its source (paper and equation, figure or table), its inputs and the reason for its tolerance;
- **Code** – the class of code it tests (Table 1): *own* – equations and numerics written for SWINGERsim; *own/tab* – SWINGERsim's interpolation of published tables; *R* – REBOUND IAS15 with REBOUNDx effects, driven by SWINGERsim's set-up; *ext* – an external library through a SWINGERsim wrapper. A benchmark of own code is the stronger evidence, because nothing outside SWINGERsim could make it pass;
- **Quantity** – what is compared, as the benchmark describes it (symbols as in the source code: `e` the eccentricity, `i0` an initial inclination, `MA01` the Mardling–Aarseth criterion, `tier A` the spherical light-curve model);
- **Expected** – the reference value; **Computed** – the value SWINGERsim computes, with all printed digits;
- **Difference** – SWINGERsim minus expected, relative (%) or absolute, as defined by the benchmark;
- **Tolerance** – the largest difference that counts as agreement, fixed before the comparison;
- **Verdict** – PASS, or **EXPLAINED** (outside the tolerance for a documented reason; Table 48).

To read a row, compare the column *Difference* with the column *Tolerance*: a PASS has an absolute difference not larger than the tolerance. Where *Expected* is 0 or 1, the benchmark compares a ratio or a difference with its ideal value. The fast benchmarks come from the last run of `swingersim validate`, the slow ones from a full run of `swingersim validate --all` on the production machine, and the nine manual ones (`phoebe-*`, `sbi-*`, `pejcha13-*`, `vyn23-*`) from separate runs; all with the final code. Totals: 93 benchmarks of level A (93 PASS), 60 of level B (55 PASS, 5 EXPLAINED), 31 of level C (29 PASS, 2 EXPLAINED) and 35 of level D (28 PASS, 7 EXPLAINED); 165 of the 219 test code of the classes own and own/tab.

## A.1 Level A: analytic results

93 benchmarks: 93 PASS, 0 EXPLAINED.

**Table A1.** Level A: what each benchmark compares and which code it tests (columns *Id*, *Code*, *Quantity*). The results follow in Table A2.

| Id | Code | Quantity |
|---|---|---|
| `gr-mercury` | R | 1PN perihelion advance of Mercury |

| Id | Code | Quantity |
|---|---|---|
| zlk-emax-65 | R | ZLK maximum eccentricity, i0 = 65 deg, test particle |
| zlk-emax-80 | R | ZLK maximum eccentricity, i0 = 80 deg, test particle |
| zlk-imin-65 | R | mutual inclination at maximum eccentricity, i0 = 65 deg |
| zlk-below-critical | R | maximum eccentricity below the critical angle, i0 = 30 deg |
| zlk-kozai-constant | R | secular drift of sqrt(1-e^2) cos i during ZLK cycles, i0 = 65 deg |
| stability-unstable | R | 2+2 at 0.5 x Mardling-Aarseth limit changes hierarchy (1 = yes) |
| stability-stable | R | 2+2 at 2.5 x Mardling-Aarseth limit keeps hierarchy (0 = no change) |
| quad-tp-emax | own | secular quadrupole code: ZLK e_max, test particle, i0 = 65 deg |
| zlk-period-nbody | own | ZLK period, N-body relative to the quadrupole secular period (i0 = 65 deg) |
| megno-regular | R | MEGNO of a regular (quasi-periodic) hierarchical triple |
| megno-chaotic | R | MEGNO > 4 for a 2+2 inside the unstable region (1 = yes) |
| brouwer-law | R | growth exponent of the IAS15 energy error (Brouwer's law) |
| nodal-precession | R | nodal precession period, circular inner orbit, i = 20 deg |
| polytrope-k2-n1.5 | own | apsidal-motion constant of a polytrope n = 1.5 |
| polytrope-k2-n3 | own | apsidal-motion constant of a polytrope n = 3 |
| roche-eggleton-q1 | own | Roche-lobe radius / separation, q = 1 (Eggleton formula vs exact) |
| roche-eggleton-q0.1 | own | Roche-lobe radius / separation, q = 0.1 (Eggleton formula vs exact) |
| apsidal-sterne-static | R | apsidal period from the tidal distortion, e = 0.3, no rotation |
| apsidal-sterne-rotating | R | apsidal period from tidal + rotational distortion, e = 0.3, Omega = n |
| tides-leconte-a | R | tidal da/dt, N-body / theory, e = 0.3, non-rotating stars |
| tides-leconte-e | R | tidal de/dt, N-body / theory, e = 0.3, non-rotating stars |
| tides-leconte-spin | R | tidal spin-up dOmega/dt, N-body / theory, e = 0.6, Omega = 1.2 n |
| tides-leconte-e-eccentric | R | tidal de/dt, N-body / theory, e = 0.6, Omega = 1.2 n |
| secular-kozai-emax | own | secular backend: ZLK maximum eccentricity, i0 = 65 deg, test particle |
| secular-nodal-precession | own | secular backend: nodal precession period, i = 20 deg [yr] |
| secular-gr-apsidal | own | secular backend: relativistic apsidal period, a = 0.05 au, e = 0.3 |
| secular-3p1-kozai-emax | own | 3+1 secular backend: ZLK e_max of the inner orbit, test particle, i0 = 65 deg |
| gas-migration-a | own | gas phase: semi-major axis after 3000 yr with tau_a = 5000 yr [au] |
| gas-damping-e | own | gas phase: eccentricity after 3000 yr with tau_e = 2000 yr |
| gas-alignment-i | own | gas phase: inclination to the disc after 3000 yr, i0 = 40 deg, tau_i = 1500 yr |
| scatter-hyperbola-deflection | own | scattering set-up: deflection angle of a 1 + 0.5 $M_\odot$ hyperbolic passage, v_inf = 5 km/s, b = 3 au |
| escape-theory-energy-slope-L0 | own | statistical escape theory, equal masses, L0 = 0: slope of dP/dE_B for /E_B/ » /E0/ |
| escape-theory-energy-slope-highL | own | statistical escape theory, L0 = L_c: slope of dP/dE_B |
| escape-theory-mean-e-L0 | own | statistical escape theory, L0 = 0: mean eccentricity of the binary |
| escape-theory-mean-e-highL | own | statistical escape theory, L0 = L_c: mean eccentricity of the binary |
| timing-ltte-amplitude | own | synthetic O-C: light-time semi-amplitude of binary A, circular edge-on outer orbit, a = 10 au, equal masses |
| phenom-romer-delay | own | phenomenological minima: delay of the secondary minimum after half the period, 2 + 1 $M_\odot$ ZAMS, P = 5 d, circular, i = 89 deg |
| phoebe-romer-delay | ext | PHOEBE hybrid: delay of the secondary minimum after half the period, 2 + 1 $M_\odot$, P = 5 d, circular (with minus without light time) |
| pulsation-time-delay-circular | own | amplitude of the light-travel-time delays of a pulsating star in a circular binary (1.7 + 0.5 $M_\odot$, P = 100 d, i = 60 deg), measured from the synthetic light curve like Kepler data (30-min cadence, pulsation phases in 2-d segments) |

| Id | Code | Quantity |
|---|---|---|
| mt-soberman-general | own | separation after transfer with constant fractions (donor 3 -> 1.5 $M_\odot$, accretor 2 $M_\odot$; 30 % donor wind, 40 % isotropic re-emission, 30 % accreted): numerical orbit response against the closed form |
| mt-conservative | own | separation after conservative transfer (donor 3 -> 1 $M_\odot$, accretor 2 -> 4) |
| mt-isotropic-reemission | own | separation after transfer with complete isotropic re-emission (donor 3 -> 1 $M_\odot$, accretor 2 $M_\odot$) |
| ce-impulsive-outer-a | R | outer semi-major axis after a common envelope inside a triple (N-body) relative to before, divided by the impulsive-mass-loss prediction for a circular orbit |
| ce-impulsive-outer-e | R | outer eccentricity after the common envelope of the same triple minus the impulsive-mass-loss prediction |
| matt15-unsaturated-powerlaw | own | Omega(4.55 Gyr) / Omega(1.14 Gyr) of the braked solar model |
| galactic-tide-emax-60 | own | largest eccentricity of a 2e4 au orbit inclined by 60 deg to the Galactic disc (secular backend, physics.galactic_tide) |
| flyby-first-order-kick | own | relative difference of the eccentricity-vector change of a binary in a parabolic passage at Q = 20 a: first-order formula against direct N-body |
| sbi-linear-gaussian-std | ext | SBI (neural posterior estimation): posterior width of a linear Gaussian model |
| quint-out-zlk-emax | R | 2+2+1: ZLK e_max of the orbit of the two binaries driven by a fifth star, i0 = 65 deg |
| mist-grid-tams-age-1 | own/tab | MIST grid: age of 1 $M_\odot$ at the end of the main sequence (EEP 454), [Fe/H] = 0, v/vcrit = 0 |
| mist-interpolation | own/tab | MIST: radius of 1.14 $M_\odot$ at the TAMS interpolated from 1.12 and 1.16 $M_\odot$ / tabulated |
| claret2023-k2-2msun | own/tab | Claret (2023) grid: log k2 of 2.0 $M_\odot$, [Fe/H] = 0, log g = 4.069 |
| claret2023-beta-1msun | own/tab | Claret (2023) grid: radius of gyration beta of 1.0 $M_\odot$, [Fe/H] = 0, log g = 4.216 |
| sse-zams-radius-1 | own/tab | SSE track grid: ZAMS radius of 1 $M_\odot$, Z = 0.02 |
| sse-zams-radius-2 | own/tab | SSE track grid: ZAMS radius of 2.5 $M_\odot$, Z = 0.02 |
| ml-adiabatic-invariant | R | max abs(a M / (a M)_0 - 1) while m1 loses 0.3 $M_\odot$ slowly (Psi = 5.6e-6), N-body |
| ml-osculating-e-03 | R | max abs(e - e_theory) / Psi over two orbits, e0 = 0.3, Psi = 5.6e-5, N-body |
| ml-osculating-e-07 | R | max abs(e - e_theory) / Psi over two orbits, e0 = 0.7, Psi = 5.6e-5, N-body |
| ml-circular-emax | R | largest e / Psi of an initially circular binary under slow mass loss |
| ml-impulsive-a | R | a'/a after a circular binary loses 15 % of its mass within 1e-3 of a period |
| ml-impulsive-e | R | eccentricity after a circular binary loses 15 % of its mass within 1e-3 P |
| ml-zlk-nbody-secular | own | time of the 4th ZLK rise (e > 0.6) while the perturber loses half its mass, N-body / secular quadrupole |
| mist-interpolation-massive-r | own/tab | MIST: radius of 60 $M_\odot$ in the middle of the main sequence (EEP 330) interpolated from 55 and 65 $M_\odot$ / tabulated |
| mist-interpolation-massive-m | own/tab | MIST: current mass of 100 $M_\odot$ at the TAMS interpolated from 95 and 105 $M_\odot$ / tabulated |
| mist-vink2001 | own/tab | MIST wind of 40 $M_\odot$ at 30 % of its main sequence / Vink et al. (2001) eq. 24 |
| claret2004-log-k2 | own/tab | Claret (2004): log k2 of 39.8 $M_\odot$ at log g = 3.416 (3.65 Myr) |
| claret2004-beta | own/tab | Claret (2004): gyration radius beta of 39.8 $M_\odot$ at log g = 3.416 |
| ld-tlusty-row | own/tab | TLUSTY limb darkening: a1 of OG40000g400v10 in Bessell V |
| ocfit-gimenez-bastero | own | apsidal term of the times of minima, e = 0.3: largest difference between the exact conjunction times and the series of Gimenez & Bastero (1995) used by OCFit |
| ocfit-lite-amplitude | own | half peak-to-peak light-time effect over one orbit / K of Irwin (1952), e3 = 0.55 |
| ocfit-nbody-asini-A | own | stage-1 fit of N-body eclipse timings of a 2+2 (outer P = 1500 d, e = 0.3, i = 60 deg): fitted / Keplerian a sin i of pair A |
| ocfit-nbody-asini-B | own | the same for pair B (anti-phase, shared outer orbit) |
| ocfit-nbody-P3 | own | the same fit: fitted / configured outer period |

| Id | Code | Quantity |
|---|---|---|
| wind-transfer-jeans | own | 25 + 18 $M_\odot$ MIST binary with winds and mass transfer: largest relative change of a (M1 + M2) before the transfer starts |
| secular-nested-tides-hut81 | own | 3+1 in the nested secular backend: tidal de/dt of the inner binary (P = 4 d, e = 0.05, synchronous) / Hut (1981) eq. 10 |
| galactic-tide-emax-60-nbody | R | largest eccentricity of a 2e4 au binary inclined by 60 deg to the disc, N-body backend with the Galactic tide (from e = 0.01) |
| gw-secular-circular-decay | own | circular 1.4 + 1.4 $M_\odot$ binary in a triple, secular backend with physics.gw: a / a0 (1 - t/tau)^(1/4) at 90 % of the merger time |
| gw-nbody-decay | R | the same binary with e = 0.3, N-body backend: change of a over 10 yr / Peters rate x 10 yr |
| disc-strip-uniform | own | opaque edge-on slab (half thickness 0.3 R) across a uniform star: transmitted fraction |
| disc-face-on-tau | own | face-on disc of optical depth 0.7 in front of a star: transmitted fraction |
| disc-inclined-tau | own | disc of optical depth 0.7 inclined 60 deg in front of a star: transmitted fraction |
| roche-lobe-volume | own | volume-equivalent radius of the Roche lobe (q = M2/M1 = 0.3) from the surface mesh of the Roche module / Eggleton's formula |
| rotating-star-critical | own | equatorial / polar radius of a Roche-model star at critical rotation |
| rotating-star-equator | own | R_eq/R_pol - (1 + w^2/2) with w = Omega / Omega_K(R_eq), Roche model, Omega / Omega_crit = 0.6 |
| mesh-sphere-eclipse | own | largest difference of the light curves of two spheres through both eclipses: surface meshes (the mesh module) against the exact tier-A eclipses |
| disc-ring-precession | own | precession rate of a narrow rigid ring at 4 a_b around a circular binary / test-particle rate |
| disc-polar-critical | own | critical inclination [deg] for polar libration of a disc around a binary with e = 0.5, from integrations of SWINGERsim's rigid-disc equation |
| rejuvenation-hurley | own | main-sequence age of a 2 $M_\odot$ accretor after gaining 1 % of its mass / (t'_MS / t_MS)(M / M') t |
| k2-rotation | own | rotational correction of k2 (2 $M_\odot$, 2 $R_\odot$, V = 200 km/s) / 10^(-0.87 × 2 V^2 / 3 g R) (Claret 1999) |
| contact-jdot | own | angular-momentum loss time [Gyr] of a 1.0 + 0.5 $M_\odot$ contact binary, P = 0.35 d (SWINGERsim units vs cgs) |
| pdot-parabola | R | quadratic coefficient of the eclipse times of a binary with dP/dt = 4e-6 d/yr over 60 yr (N-body with period kicks) / analytic (1/2) P dP/dt |
| dynamic-onset-circularisation | own | Roche-lobe filling of the donor after the circularisation of a Kozai-Lidov triple at the contact (e_c = 0.89), times (1 + e_c) |

**Table A2.** Level A: results (columns *Expected*, *Computed*, *Difference*, *Tolerance*, *Verdict*), in the same order as Table A1.

| Id | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|
| gr-mercury | 42.98 ″/century | 42.9803 ″/century | +0.00 % | 0.5 % | PASS |
| zlk-emax-65 | 0.838047 | 0.84023 | +0.26 % | 1 % | PASS |
| zlk-emax-80 | 0.974548 | 0.976155 | +0.16 % | 1 % | PASS |
| zlk-imin-65 | 39.2315° | 39.3441° | +0.113° | 0.5° | PASS |
| zlk-below-critical | 0 | 0.00173037 | +0.00173 | 0.01 | PASS |
| zlk-kozai-constant | 0 | $5.16811\cdot10^{-5}$ | $+5.17\cdot10^{-5}$ | 0.001 | PASS |
| stability-unstable | 1 | 1 | +0 | 0 | PASS |
| stability-stable | 0 | 0 | +0 | 0 | PASS |
| quad-tp-emax | 0.838047 | 0.838082 | +0.00 % | 0.1 % | PASS |
| zlk-period-nbody | 0 | 0.00660493 | +0.0066 | 0.02 | PASS |
| megno-regular | 2 | 1.99959 | −0.00041 | 0.05 | PASS |
| megno-chaotic | 1 | 1 | +0 | 0 | PASS |
| brouwer-law | 0.5 | 0.59301 | +0.093 | 0.15 | PASS |
| nodal-precession | 11351.4 yr | 11339.8 yr | −0.10 % | 1 % | PASS |

| Id | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|
| polytrope-k2-n1.5 | 0.1433 | 0.143279 | −0.01 % | 0.2 % | PASS |
| polytrope-k2-n3 | 0.0144 | 0.014443 | +0.30 % | 0.5 % | PASS |
| roche-eggleton-q1 | 0.379861 | 0.378921 | −0.25 % | 1 % | PASS |
| roche-eggleton-q0.1 | 0.2054 | 0.206773 | +0.67 % | 1 % | PASS |
| apsidal-sterne-static | 318.227 yr | 318.201 yr | −0.01 % | 0.5 % | PASS |
| apsidal-sterne-rotating | 291.213 yr | 291.185 yr | −0.01 % | 0.5 % | PASS |
| tides-leconte-a | 1 | 0.999285 | −0.07 % | 1 % | PASS |
| tides-leconte-e | 1 | 0.999356 | −0.06 % | 1 % | PASS |
| tides-leconte-spin | 1 | 1.00113 | +0.11 % | 1 % | PASS |
| tides-leconte-e-eccentric | 1 | 0.996923 | −0.31 % | 1 % | PASS |
| secular-kozai-emax | 0.838047 | 0.838082 | +0.00 % | 0.5 % | PASS |
| secular-nodal-precession | 11351.4 yr | 11351.4 yr | −0.00 % | 0.5 % | PASS |
| secular-gr-apsidal | 7113.77 yr | 7113.77 yr | −0.00 % | 0.5 % | PASS |
| secular-3p1-kozai-emax | 0.838047 | 0.838082 | +0.00 % | 0.5 % | PASS |
| gas-migration-a | 5.48812 au | 5.48812 au | −0.00 % | 0.0001 % | PASS |
| gas-damping-e | 0.111565 | 0.111565 | +0.00 % | 0.0001 % | PASS |
| gas-alignment-i | 5.63999° | 5.63943° | −0.000561° | 0.01° | PASS |
| scatter-hyperbola-deflection | 173.548° | 173.548° | $+1.36\cdot10^{-12}$° | $1\cdot10^{-6}$° | PASS |
| escape-theory-energy-slope-L0 | −3.5 | −3.5026 | −0.0026 | 0.05 | PASS |
| escape-theory-energy-slope-highL | −4 | −4.02479 | −0.0248 | 0.1 | PASS |
| escape-theory-mean-e-L0 | 0.8133 | 0.812817 | −0.000483 | 0.01 | PASS |
| escape-theory-mean-e-highL | 0.7 | 0.710776 | +0.0108 | 0.02 | PASS |
| timing-ltte-amplitude | 41.5837 min | 41.5863 min | +0.01 % | 0.1 % | PASS |
| phenom-romer-delay | 27.4517 s | 27.3578 s | −0.34 % | 1.5 % | PASS |
| phoebe-romer-delay | 27.452 s | 27.287 s | −0.60 % | 2 % | PASS |
| pulsation-time-delay-circular | 53.859 s | 53.859 s | +0.00 % | 0.01 % | PASS |
| mt-soberman-general | 1.29573 a/a0 | 1.29573 a/a0 | −0.00 % | $1\cdot10^{-5}$ % | PASS |
| mt-conservative | 2.25 a/a0 | 2.25 a/a0 | −0.00 % | $1\cdot10^{-5}$ % | PASS |
| mt-isotropic-reemission | 2.03003 a/a0 | 2.03003 a/a0 | −0.00 % | $1\cdot10^{-5}$ % | PASS |
| ce-impulsive-outer-a | 1 | 1.01274 | +1.27 % | 3 % | PASS |
| ce-impulsive-outer-e | 0 | 0.00442155 | +0.00442 | 0.02 | PASS |
| matt15-unsaturated-powerlaw | 0.5 | 0.491949 | −0.00805 | 0.02 | PASS |
| galactic-tide-emax-60 | 0.829156 | 0.829181 | $+2.5\cdot10^{-5}$ | 0.01 | PASS |
| flyby-first-order-kick | 0 | 0.00594917 | +0.00595 | 0.03 | PASS |
| sbi-linear-gaussian-std | 0.3092 | 0.291 | −5.8 % | 10 % | PASS |
| quint-out-zlk-emax | 0.81272 | 0.8152 | +0.31 % | 1.0 % | PASS |
| mist-grid-tams-age-1 | 9919.42 Myr | 9919.42 Myr | +0.00 % | 0.001 % | PASS |
| mist-interpolation | 1 | 1.01068 | +1.07 % | 2 % | PASS |
| claret2023-k2-2msun | −2.478 | −2.47794 | $+6\cdot10^{-5}$ | 0.002 | PASS |
| claret2023-beta-1msun | 0.2265 | 0.226499 | $-6.34\cdot10^{-7}$ | 0.0005 | PASS |
| sse-zams-radius-1 | 0.88825 $R_\odot$ | 0.888312 $R_\odot$ | +0.01 % | 0.1 % | PASS |
| sse-zams-radius-2 | 1.79434 $R_\odot$ | 1.79435 $R_\odot$ | +0.00 % | 0.1 % | PASS |
| ml-adiabatic-invariant | 0 | $3.56804\cdot10^{-9}$ | $+3.57\cdot10^{-9}$ | $1\cdot10^{-6}$ | PASS |
| ml-osculating-e-03 | 0 | 0.0126471 | +0.0126 | 0.05 | PASS |
| ml-osculating-e-07 | 0 | 0.0123538 | +0.0124 | 0.05 | PASS |
| ml-circular-emax | 2 | 2.00265 | +0.13 % | 2 % | PASS |
| ml-impulsive-a | 1.21429 | 1.21491 | +0.05 % | 1 % | PASS |
| ml-impulsive-e | 0.176471 | 0.176892 | +0.24 % | 1 % | PASS |
| ml-zlk-nbody-secular | 1 | 1.0021 | +0.21 % | 1.0 % | PASS |
| mist-interpolation-massive-r | 1 | 0.999431 | −0.06 % | 1 % | PASS |
| mist-interpolation-massive-m | 1 | 1.00107 | +0.11 % | 3 % | PASS |
| mist-vink2001 | 1 | 1.05511 | +5.51 % | 10 % | PASS |

| Id | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|
| claret2004-log-k2 | −2.81523 | −2.81569 | −0.000463 | 0.002 | PASS |
| claret2004-beta | 0.161706 | 0.161699 | $-6.61\cdot10^{-6}$ | 0.001 | PASS |
| ld-tlusty-row | 1.18262 | 1.18262 | +0 | 0.0001 | PASS |
| ocfit-gimenez-bastero | 0 s | 0.000629837 s | +0.00063 s | 0.01 s | PASS |
| ocfit-lite-amplitude | 1 | 1 | −0.00 % | 0.0001 % | PASS |
| ocfit-nbody-asini-A | 1 | 0.99864 | −0.14 % | 1.0 % | PASS |
| ocfit-nbody-asini-B | 1 | 1.0024 | +0.24 % | 1.0 % | PASS |
| ocfit-nbody-P3 | 1 | 0.99996 | −0.00 % | 0.5 % | PASS |
| wind-transfer-jeans | 0 | $1.68718\cdot10^{-11}$ | $+1.69\cdot10^{-11}$ | $1\cdot10^{-8}$ | PASS |
| secular-nested-tides-hut81 | 1 | 1 | −0.00 % | 0.0001 % | PASS |
| galactic-tide-emax-60-nbody | 0.82916 | 0.82908 | −0.01 % | 1.0 % | PASS |
| gw-secular-circular-decay | 1 | 1 | −0.00 % | 0.01 % | PASS |
| gw-nbody-decay | 1 | 1.0003 | +0.03 % | 1.0 % | PASS |
| disc-strip-uniform | 0.623838 | 0.62475 | +0.000912 | 0.005 | PASS |
| disc-face-on-tau | 0.496585 | 0.496585 | +0 | $1\cdot10^{-9}$ | PASS |
| disc-inclined-tau | 0.246597 | 0.246597 | $-4.16\cdot10^{-16}$ | $1\cdot10^{-9}$ | PASS |
| roche-lobe-volume | 1 | 1.00509 | +0.51 % | 1 % | PASS |
| rotating-star-critical | 1.5 | 1.5 | −0.00 % | 0.0001 % | PASS |
| rotating-star-equator | 0 | $-2.22045\cdot10^{-16}$ | $-2.22\cdot10^{-16}$ | $1\cdot10^{-9}$ | PASS |
| mesh-sphere-eclipse | 0 | 0.000259643 | +0.00026 | 0.001 | PASS |
| disc-ring-precession | 1 | 1.00018 | +0.02 % | 0.1 % | PASS |
| disc-polar-critical | 37.7612° | 37.7612° | $+6.29\cdot10^{-9}$° | 0.2° | PASS |
| rejuvenation-hurley | 1 | 0.999174 | −0.08 % | 0.2 % | PASS |
| k2-rotation | 1 | 1 | +0.00 % | $1\cdot10^{-7}$ % | PASS |
| contact-jdot | 2.18893 Gyr | 2.18898 Gyr | +0.00 % | 0.1 % | PASS |
| pdot-parabola | 1 | 1.00005 | +0.00 % | 0.1 % | PASS |
| dynamic-onset-circularisation | 1 | 1 | −0.00 % | 0.01 % | PASS |

## A.2 Level B: published numerical results

60 benchmarks: 55 PASS, 5 EXPLAINED.

**Table A3.** Level B: what each benchmark compares and which code it tests (columns *Id*, *Code*, *Quantity*). The results follow in Table A4.

| Id | Code | Quantity |
|---|---|---|
| naoz13-fig4-imin | R | Naoz et al. 2013 Fig. 4 (PSR B1620-26 setup): minimum mutual inclination |
| naoz13-fig4-imax | R | Naoz et al. 2013 Fig. 4: maximum mutual inclination (orbit flip to retrograde) |
| naoz13-fig3-flip-time | R | Naoz et al. 2013 Fig. 3 (eccentric ZLK): time of the first flip through 90 deg |
| naoz13-fig3-imax | R | Naoz et al. 2013 Fig. 3: maximum inclination after the flip |
| quad-naoz13-fig4-imin | own | secular quadrupole code: Naoz et al. 2013 Fig. 4, minimum inclination |
| quad-naoz13-fig4-imax | own | secular quadrupole code: Naoz et al. 2013 Fig. 4, maximum inclination |
| vok16-v994-JA | own | Vokrouhlicky 2016, V994 Her setup: initial tilt J_A of binary A to the outer orbit |
| vok16-v994-JB | own | Vokrouhlicky 2016, V994 Her setup: initial tilt J_B of binary B to the outer orbit |
| vok16-jab-max-analytic | own | Vokrouhlický 2016, analytic limit J_AB^max for V994 Her (Fig. 2 setup) |
| vok16-fig2-jab-min | own | secular backend: Vokrouhlický 2016 Fig. 2, minimum J_AB over 100 kyr |
| vok16-fig2-jab-max | own | secular backend: Vokrouhlický 2016 Fig. 2, maximum J_AB over 100 kyr |
| vok16-fig2-jab-1myr | own | secular backend: Fig. 2 setup, maximum J_AB over 1 Myr |

| Id | Code | Quantity |
|---|---|---|
| vok16-fig3-jab-max | own | secular backend: Vokrouhlický 2016 Fig. 3, maximum J_AB over 20 kyr |
| vok16-eclipse-prob-A | own | V994 Her: chance that binary A eclipses, isotropic orientation |
| vok16-eclipse-prob-B | own | V994 Her: chance that binary B eclipses, isotropic orientation |
| secular-naoz13-fig3-flip-time | own | 3+1 secular backend: Naoz et al. 2013 Fig. 3, time of the first flip [yr] |
| secular-naoz13-fig3-imax | own | 3+1 secular backend: Naoz et al. 2013 Fig. 3, maximum inclination after the flip |
| at16-binbin-single-ionization | R | binary-binary scattering, equal masses, a1 = a2, e = 0, v_inf = v_crit: cross-section of 'single ionization' / (pi a^2) |
| at16-binbin-exchange-ionization | R | binary-binary scattering, equal masses, a1 = a2, e = 0, v_inf = v_crit: cross-section of 'exchange + ionization' / (pi a^2) |
| at16-binbin-exchange | R | binary-binary scattering, equal masses, a1 = a2, e = 0, v_inf = v_crit: cross-section of 'exchange' / (pi a^2) |
| chaosmap-circumbinary-limit | R | MEGNO map: inner stability limit of a circumbinary orbit, equal-mass circular binary, all of 8 phases regular [a_bin] |
| qzcar-broz-a1 | own | QZ Car: semi-major axis of Ac1+Ac2 (P = 6.0 d) from the Table 2 elements and masses |
| qzcar-broz-a2 | own | QZ Car: semi-major axis of Aa1+Aa2 (P = 20.7 d) from the Table 2 elements and masses |
| qzcar-broz-a3 | own | QZ Car: semi-major axis of Ac+Aa (P = 14 728 d) from the Table 2 elements and masses |
| mt-zeta-lobe-q2 | own | response zeta_L = dln R_L / dln M_d of the donor's Roche lobe (Eggleton 1983) to conservative transfer at q = 2 |
| ce-tout97-period | own | orbital period after the common envelope of the CV progenitor of Tout et al. (1997): 5.36 $M_\odot$ AGB star (1.014 $M_\odot$ core) with a 1 $M_\odot$ companion, alpha_CE = 1 |
| ce-bse-algol-second-period | own | orbital period after the second common envelope of the BSE Algol example (2.85 $M_\odot$ giant with a 0.416 $M_\odot$ core, 0.417 $M_\odot$ white dwarf, P = 18 d, alpha_CE = 3) |
| matt15-sun-period | own | rotation period of a solar model at 4.55 Gyr with the wind torque of Matt et al. (2015), started at P = 1 d |
| applegate-huaqr | own | energy needed by the Applegate mechanism over the energy radiated in one modulation period, HU Aqr (thin-shell model) |
| applegate-nnser | own | energy needed by the Applegate mechanism over the energy radiated in one modulation period, NN Ser (thin-shell model) |
| applegate-v471tau | own | energy needed by the Applegate mechanism over the energy radiated in one modulation period, V471 Tau (thin-shell model) |
| vyn23-2p2-ma01-score | own | overall score of the MA01 criterion against the N-body ghost test on SWINGERsim's 2+2 training set (sampled as in Vynatheya et al. 2023, Sect. 3.1) |
| vyn23-2p2-mlp-score | ext | held-out score of SWINGERsim's MLP stability emulator (2+2; 11 parameters, 4 x 50 neurons) against the N-body ghost test |
| vyn23-3p1-ma01-score | own | overall score of the MA01 criterion against the N-body ghost test on SWINGERsim's 3+1 training set (sampled as in Vynatheya et al. 2023, Sect. 3.1) |
| vyn23-3p1-mlp-score | ext | held-out score of SWINGERsim's MLP stability emulator (3+1; 11 parameters, 4 x 50 neurons) against the N-body ghost test |
| escape-theory-vs-nbody-light-escaper | own | resonant binary-single encounters (1 + 0.6 $M_\odot$ binary, 0.8 $M_\odot$ star): N-body fraction in which the 0.6 $M_\odot$ star escapes vs the statistical theory (expected value: SWINGERsim's own evaluation of the theory of Stone & Leigh 2019, not a published number) |
| czev343-apsidal-tidal | own | CzeV343 binary A: tidal apsidal rate [rad/day] |
| czev343-apsidal-rotational | own | CzeV343 binary A: rotational apsidal rate, pseudo-synchronous spins [rad/day] |
| czev343-apsidal-gr | own | CzeV343 binary A: relativistic apsidal rate [rad/day] |
| vok16-pejcha-fA | own | Pejcha-like 2+2 population, secular: binary A reaches 1 - e < 1e-3 |
| vok16-pejcha-fB | own | Pejcha-like 2+2 population, secular: binary B reaches 1 - e < 1e-3 |

| Id | Code | Quantity |
|---|---|---|
| pejcha13-F6-fA | R | Pejcha et al. F6 population, N-body: binary A reaches 1 - e < 1e-3 |
| pejcha13-F6-fB | R | Pejcha et al. F6 population, N-body: binary B reaches 1 - e < 1e-3 |
| mist-ms-lifetime-1 | own/tab | main-sequence lifetime of 1 $M_\odot$ (ZAMS to TAMS), SSE vs MIST |
| mist-ms-lifetime-2 | own/tab | main-sequence lifetime of 2 $M_\odot$ (ZAMS to TAMS), SSE vs MIST |
| mist-tams-radius-2 | own/tab | radius of 2 $M_\odot$ at the end of the main sequence, SSE vs MIST |
| iw-two-layer | own | inertial-wave dissipation of a two-layer star (alpha 0.6, beta 0.95): impulsive formalism / closed formula |
| tides-circularization-pms | own | period below which twin 1 $M_\odot$ binaries (e0 = 0.3) are circular when they reach the main sequence (MIST pre-main sequence from 1 Myr to 50 Myr; equilibrium tide and inertial waves) |
| ml-tedi-onset | R | onset of the inner instability (e_in > 0.1) / crossing of the MA01 limit, inner primary losing 3 of 5 $M_\odot$ in 8000 yr |
| mist-geneva-tauh-15-0 | own/tab | hydrogen-burning lifetime of 15 $M_\odot$, non-rotating, MIST vs Geneva |
| mist-geneva-tauh-15-4 | own/tab | hydrogen-burning lifetime of 15 $M_\odot$, rotating, MIST vs Geneva |
| mist-geneva-tauh-60-0 | own/tab | hydrogen-burning lifetime of 60 $M_\odot$, non-rotating, MIST vs Geneva |
| mist-geneva-tauh-60-4 | own/tab | hydrogen-burning lifetime of 60 $M_\odot$, rotating, MIST vs Geneva |
| mist-geneva-tauh-120-0 | own/tab | hydrogen-burning lifetime of 120 $M_\odot$, non-rotating, MIST vs Geneva |
| mist-geneva-tauh-120-4 | own/tab | hydrogen-burning lifetime of 120 $M_\odot$, rotating, MIST vs Geneva |
| flyby-nbody-backend | R | parabolic passage at Q = 20 a integrated by the N-body backend (physics.encounters): relative difference of the eccentricity kick from the first-order result of Hamers & Samsing (2019) |
| tidal-capture-radius-n3 | own | largest periastron distance for tidal capture / R, two n = 3 stars (0.6 $M_\odot$, 0.65 $R_\odot$) at 10 km/s |
| tidal-capture-radius-n15 | own | the same for two n = 1.5 stars |
| tidal-capture-radius-n15-mcmillan | own | largest periastron distance for tidal capture / R, two equal n = 1.5 stars at 10 km/s |
| mt-bse-algol-giant-phase | own | period at the end of the BSE Algol example's transfer, computed from its state at the end of the Hertzsprung gap (0.53 + 2.74 $M_\odot$, 17 d; conservative) |

**Table A4.** Level B: results (columns *Expected*, *Computed*, *Difference*, *Tolerance*, *Verdict*), in the same order as Table A3.

| Id | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|
| naoz13-fig4-imin | 57.5° | 57.2391° | −0.261° | 0.5° | PASS |
| naoz13-fig4-imax | 106.7° | 106.93° | +0.23° | 0.5° | PASS |
| naoz13-fig3-flip-time | $3.8\cdot10^6$ yr | $3.9868\cdot10^6$ yr | +4.92 % | 10.0 % | PASS |
| naoz13-fig3-imax | 142.6° | 143.01° | +0.408° | 2° | PASS |
| quad-naoz13-fig4-imin | 57.5° | 57.5376° | +0.0376° | 0.1° | PASS |
| quad-naoz13-fig4-imax | 106.7° | 106.721° | +0.0207° | 0.1° | PASS |
| vok16-v994-JA | 11.6° | 11.6462° | +0.0462° | 0.1° | PASS |
| vok16-v994-JB | 15.8° | 15.7932° | −0.00678° | 0.1° | PASS |
| vok16-jab-max-analytic | 5.2° | 5.20074° | +0.000736° | 0.05° | PASS |
| vok16-fig2-jab-min | 3.5° | 3.47152° | −0.0285° | 0.15° | PASS |
| vok16-fig2-jab-max | 5.2° | 5.20069° | +0.000687° | 0.15° | PASS |
| vok16-fig2-jab-1myr | 5.2° | 5.2006° | +0.000633° | 0.15° | PASS |
| vok16-fig3-jab-max | 6° | 6.20874° | +0.209° | 0.5° | PASS |
| vok16-eclipse-prob-A | 0.366 | 0.369265 | +0.89 % | 2 % | PASS |

| Id | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|
| vok16-eclipse-prob-B | 0.397 | 0.399075 | +0.52 % | 2 % | PASS |
| secular-naoz13-fig3-flip-time | $3.8{\cdot}10^{6}$ yr | $4.04981{\cdot}10^{6}$ yr | +6.57 % | 10 % | PASS |
| secular-naoz13-fig3-imax | 142.6° | 142.556° | −0.0435° | 2° | PASS |
| at16-binbin-single-ionization | 2.645 | 2.6873 | +1.60 % | 10.0 % | PASS |
| at16-binbin-exchange-ionization | 3.259 | 3.2305 | −0.88 % | 10.0 % | PASS |
| at16-binbin-exchange | 0.302 | 0.37165 | +23.06 % | 30.0 % | PASS |
| chaosmap-circumbinary-limit | 2.27 | 2.3 | +0.03 | 0.06 | PASS |
| qzcar-broz-a1 | 54.032 $R_\odot$ | 53.9946 $R_\odot$ | −0.07 % | 0.2 % | PASS |
| qzcar-broz-a2 | 136.132 $R_\odot$ | 136.081 $R_\odot$ | −0.04 % | 0.2 % | PASS |
| qzcar-broz-a3 | 13053 $R_\odot$ | 13046 $R_\odot$ | −0.05 % | 0.2 % | PASS |
| mt-zeta-lobe-q2 | 2.59 | 2.60707 | +0.0171 | 0.05 | PASS |
| ce-tout97-period | 35 h | 34.84 h | −0.46 % | 10 % | PASS |
| ce-bse-algol-second-period | 0.04 d | 0.0365638 d | −8.59 % | 15 % | PASS |
| matt15-sun-period | 27.97 d | 27.7635 d | −0.74 % | 2 % | PASS |
| applegate-huaqr | 0.1 | 0.105659 | +5.66 % | 10 % | PASS |
| applegate-nnser | 3.3 | 3.30338 | +0.10 % | 10 % | PASS |
| applegate-v471tau | 0.014 | 0.0143688 | +2.63 % | 10 % | PASS |
| vyn23-2p2-ma01-score | 0.83 | 0.869 | +0.039 | 0.05 | PASS |
| vyn23-2p2-mlp-score | 0.94 | 0.9375 | −0.0025 | 0.03 | PASS |
| vyn23-3p1-ma01-score | 0.56 | 0.7896 | +0.23 | 0.05 | **EXPLAINED** |
| vyn23-3p1-mlp-score | 0.93 | 0.9195 | −0.0105 | 0.03 | PASS |
| escape-theory-vs-nbody-light-escaper | 0.769 | 0.75369 | −0.0153 | 0.07 | PASS |
| czev343-apsidal-tidal | 0.000414 rad/d | 0.000422272 rad/d | +2.00 % | 3 % | PASS |
| czev343-apsidal-rotational | $6.4{\cdot}10^{-5}$ rad/d | $6.32533{\cdot}10^{-5}$ rad/d | −1.17 % | 5 % | PASS |
| czev343-apsidal-gr | $1.47{\cdot}10^{-5}$ rad/d | $1.47874{\cdot}10^{-5}$ rad/d | +0.59 % | 2 % | PASS |
| vok16-pejcha-fA | 0.31 | 0.21 | −0.1 | 0.07 | **EXPLAINED** |
| vok16-pejcha-fB | 0.481 | 0.4025 | −0.0785 | 0.07 | **EXPLAINED** |
| pejcha13-F6-fA | 0.358 | 0.24 | −0.118 | 0.14 | PASS |
| pejcha13-F6-fB | 0.530 | 0.41 | −0.12 | 0.15 | PASS |
| mist-ms-lifetime-1 | 9878 Myr | 11003.5 Myr | +11.39 % | 15 % | PASS |
| mist-ms-lifetime-2 | 1054 Myr | 1164.17 Myr | +10.45 % | 15 % | PASS |
| mist-tams-radius-2 | 3.913 $R_\odot$ | 3.56583 $R_\odot$ | −8.87 % | 15 % | PASS |
| iw-two-layer | 1 | 1.00031 | +0.03 % | 0.5 % | PASS |
| tides-circularization-pms | 7.85 d | 7.9307 d | +1.03 % | 10 % | PASS |
| ml-tedi-onset | 1 | 1.06475 | +6.47 % | 15 % | PASS |
| mist-geneva-tauh-15-0 | 11.015 Myr | 12.3538 Myr | +12.15 % | 15 % | PASS |
| mist-geneva-tauh-15-4 | 13.447 Myr | 12.6825 Myr | −5.69 % | 15 % | PASS |
| mist-geneva-tauh-60-0 | 3.53 Myr | 3.71791 Myr | +5.32 % | 15 % | PASS |
| mist-geneva-tauh-60-4 | 4.465 Myr | 3.94657 Myr | −11.61 % | 15 % | PASS |
| mist-geneva-tauh-120-0 | 2.672 Myr | 2.78283 Myr | +4.15 % | 15 % | PASS |
| mist-geneva-tauh-120-4 | 3.178 Myr | 3.33128 Myr | +4.82 % | 15 % | PASS |
| flyby-nbody-backend | 0 | 0.00738403 | +0.00738 | 0.05 | PASS |
| tidal-capture-radius-n3 | 2.72 | 2.36599 | −13.01 % | 5 % | **EXPLAINED** |
| tidal-capture-radius-n15 | 3.84 | 3.33454 | −13.16 % | 5 % | **EXPLAINED** |
| tidal-capture-radius-n15-mcmillan | 3.4 | 3.33454 | −1.93 % | 5 % | PASS |
| mt-bse-algol-giant-phase | 30 d | 30.3561 d | +1.19 % | 3 % | PASS |

## A.3 Level C: other codes

31 benchmarks: 29 PASS, 2 EXPLAINED.

**Table A5.** Level C: what each benchmark compares and which code it tests (columns *Id*, *Code*, *Quantity*). The results follow in Table A6.

| Id | Code | Quantity |
|---|---|---|
| `sextuple-secular-emax` | own | 2+2+2 sextuple: maximum eccentricity of A, secular backend vs N-body |
| `sextuple-secular-zlk` | own | 2+2+2 sextuple: ZLK period of A, secular backend vs N-body |
| `sextuple-nbody-emax` | R | 2+2+2 sextuple: maximum eccentricity of A, N-body (reproducibility) |
| `mse-3p1-inner-emax` | own | 3+1 (a = 1, 15, 300 au, i_mid = 85 deg): maximum e of the inner binary in 1e5 yr |
| `mse-3p1-inner-t-emax` | own | 3+1: time of the maximum e of the inner binary |
| `mse-3p1-mid-emax` | own | 3+1: maximum e of the middle orbit |
| `mse-2p2-A-emax` | own | 2+2 (a = 1, 1.5, 40 au): maximum e of binary A in 8e4 yr |
| `mse-2p2-B-emax` | own | 2+2: maximum e of binary B in 8e4 yr |
| `mse-3p1-rlof-time` | own | 3+1 (a = 0.3, 5, 200 au, i = 89 deg): onset of Roche-lobe overflow of the inner binary driven by ZLK cycles |
| `phoebe-sphere-vs-tier-a` | own | largest difference of the normalised light curves, tier A vs PHOEBE 2.5 with the same physics (spheres, black bodies, Claret limb darkening), e = 0.1 |
| `qzcar-broz-lv-ac1` | own | QZ Car: share of Ac1 in the V-band light from its radius and Teff |
| `qzcar-broz-lv-ac2` | own | QZ Car: share of Ac2 in the V-band light from its radius and Teff |
| `qzcar-broz-lv-aa1` | own | QZ Car: share of Aa1 in the V-band light from its radius and Teff |
| `qzcar-broz-lv-aa2` | own | QZ Car: share of Aa2 in the V-band light from its radius and Teff |
| `cepheid-period-radius-6` | own/tab | log R of a 6 $M_\odot$ MIST Cepheid (blue loop, middle of the strip, period from Bono et al. 2000) minus the period-radius relation of rotating Geneva Cepheid models |
| `cepheid-period-radius-7` | own/tab | as cepheid-period-radius-6 for a 7 $M_\odot$ Cepheid |
| `mt-bse-algol-donor` | own | donor mass at the end of the first mass transfer of the BSE Algol example (2.9 + 0.9 $M_\odot$, Z = 0.02, started at P = 8 d, e = 0.7; from the onset of overflow, circular, P = 2.7 d) |
| `mt-bse-algol-accretor` | own | accretor mass at the end of the first mass transfer of the BSE Algol example |
| `mt-bse-algol-period` | own | orbital period at the end of the first mass transfer of the BSE Algol example |
| `mt-cosmic-algol-donor` | own | donor mass when the first mass transfer of the BSE Algol example ends, against the BSE code in COSMIC 4.2.1 (2.9 + 0.9 $M_\odot$, circular 2.7 d, Z = 0.02, acc_lim = -1) |
| `lc-occultation-small` | own | flux of a limb-darkened star behind a dark disk, rp = 0.1, central |
| `lc-occultation-limb` | own | flux of a limb-darkened star behind a dark disk, rp = 0.3 at the limb (z = 0.9) |
| `lc-occultation-large` | own | flux of a limb-darkened star behind a larger dark disk, rp = 1.2, z = 0.5 |
| `lc-detached-primary` | own | primary eclipse depth of a detached binary of spheres |
| `lc-detached-secondary` | own | secondary eclipse depth of a detached binary of spheres |
| `lc-ellipsoidal` | own | semi-amplitude of the ellipsoidal variation, R/a = 0.15, q = 0.8, i = 60 deg |
| `ld-tlusty-vs-atlas` | own/tab | disk-averaged intensity /I(1) in V at 25 000 K, log g 4: TLUSTY non-LTE / ATLAS LTE |
| `sse-wind-rgb-1msun` | own/tab | mass lost by a 1 $M_\odot$ star (Z = 0.02) on the red giant branch with the Reimers wind, eta = 0.1: SSE track with the SSE winds against the MIST v1.2 (MESA) model at the tip of the giant branch |
| `contact-vs-phoebe` | own | V amplitude of a contact binary (q = 0.42, fill-out 0.3, i = 80 deg): SWINGERsim mesh minus PHOEBE 2.5 with the same physics |
| `rv-light-vs-phoebe` | own | centre-of-light minus centre-of-mass velocity of the secondary of a contact binary (q = 0.42, fill-out 0.3, i = 80 deg): largest difference SWINGERsim - PHOEBE / its amplitude (52 km/s) |
| `dynamic-onset-nbody-vs-secular` | own | time of the Roche-lobe contact driven by Kozai-Lidov cycles (first maximum, ~61 yr): N-body / secular (double-averaged, octupole) |

**Table A6.** Level C: results (columns *Expected*, *Computed*, *Difference*, *Tolerance*, *Verdict*), in the same order as Table A5.

| Id | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|
| sextuple-secular-emax | 0.75207 | 0.751821 | −0.03 % | 2 % | PASS |
| sextuple-secular-zlk | 70.2 yr | 71.04 yr | +1.20 % | 5 % | PASS |
| sextuple-nbody-emax | 0.75207 | 0.75207 | +0.00 % | 0.2 % | PASS |
| mse-3p1-inner-emax | 0.9707 | 0.970714 | $+1.37\cdot10^{-5}$ | 0.002 | PASS |
| mse-3p1-inner-t-emax | 49325 yr | 49322.8 yr | −0.00 % | 0.2 % | PASS |
| mse-3p1-mid-emax | 0.2245 | 0.224461 | $-3.86\cdot10^{-5}$ | 0.002 | PASS |
| mse-2p2-A-emax | 0.9494 | 0.949492 | $+9.2\cdot10^{-5}$ | 0.002 | PASS |
| mse-2p2-B-emax | 0.7758 | 0.776227 | +0.000427 | 0.002 | PASS |
| mse-3p1-rlof-time | 380.03 yr | 380.1 yr | +0.02 % | 0.2 % | PASS |
| phoebe-sphere-vs-tier-a | 0 | 0.00037 | +0.00037 | 0.001 | PASS |
| qzcar-broz-lv-ac1 | 0.22 | 0.220775 | +0.000775 | 0.005 | PASS |
| qzcar-broz-lv-ac2 | 0.133 | 0.133856 | +0.000856 | 0.005 | PASS |
| qzcar-broz-lv-aa1 | 0.635 | 0.635024 | $+2.36\cdot10^{-5}$ | 0.005 | PASS |
| qzcar-broz-lv-aa2 | 0.01 | 0.0103458 | +0.000346 | 0.005 | PASS |
| cepheid-period-radius-6 | 0 dex | −0.00765757 dex | −0.00766 dex | 0.05 dex | PASS |
| cepheid-period-radius-7 | 0 dex | 0.00392438 dex | +0.00392 dex | 0.05 dex | PASS |
| mt-bse-algol-donor | 0.42 $M_\odot$ | 0.404336 $M_\odot$ | −0.0157 $M_\odot$ | 0.05 $M_\odot$ | PASS |
| mt-bse-algol-accretor | 2.85 $M_\odot$ | 2.51105 $M_\odot$ | −0.339 $M_\odot$ | 0.15 $M_\odot$ | **EXPLAINED** |
| mt-bse-algol-period | 30 d | 21.9301 d | −26.90 % | 15 % | **EXPLAINED** |
| mt-cosmic-algol-donor | 0.401 $M_\odot$ | 0.404336 $M_\odot$ | +0.00334 $M_\odot$ | 0.03 $M_\odot$ | PASS |
| lc-occultation-small | 0.987891 | 0.987891 | $-1.92\cdot10^{-12}$ | $1\cdot10^{-7}$ | PASS |
| lc-occultation-limb | 0.945236 | 0.945236 | $+3.66\cdot10^{-9}$ | $1\cdot10^{-7}$ | PASS |
| lc-occultation-large | 0.128979 | 0.128979 | $-5.52\cdot10^{-9}$ | $1\cdot10^{-7}$ | PASS |
| lc-detached-primary | 0.380821 | 0.380868 | $+4.72\cdot10^{-5}$ | 0.0001 | PASS |
| lc-detached-secondary | 0.144658 | 0.144654 | $-3.85\cdot10^{-6}$ | 0.0001 | PASS |
| lc-ellipsoidal | 0.0027949 | 0.00260619 | −6.75 % | 10 % | PASS |
| ld-tlusty-vs-atlas | 1 | 0.997587 | −0.24 % | 2 % | PASS |
| sse-wind-rgb-1msun | 0.04158 $M_\odot$ | 0.0421789 $M_\odot$ | +1.44 % | 10 % | PASS |
| contact-vs-phoebe | 0 mag | 0.0030377 mag | +0.00304 mag | 0.03 mag | PASS |
| rv-light-vs-phoebe | 0 | 0.022028 | +0.022 | 0.05 | PASS |
| dynamic-onset-nbody-vs-secular | 1 | 1.0173 | +1.73 % | 3.0 % | PASS |

## A.4 Level D: observed systems

35 benchmarks: 28 PASS, 7 EXPLAINED.

**Table A7.** Level D: what each benchmark compares and which code it tests (columns *Id*, *Code*, *Quantity*). The results follow in Table A8.

| Id | Code | Quantity |
|---|---|---|
| v994s-dyn-A | R | V994 Her sextuple: share of the apsidal motion of A caused by B (and C) |
| v994s-dyn-B | R | V994 Her sextuple: share of the apsidal motion of B caused by A (and C) |
| v994s-dyn-C | R | V994 Her sextuple: share of the apsidal motion of C caused by A and B |
| v994s-gr-A | R | V994 Her: relativistic share of the apsidal motion of A |
| v994s-gr-B | R | V994 Her: relativistic share of the apsidal motion of B |
| v994s-gr-C | R | V994 Her: relativistic share of the apsidal motion of C |
| qzcar-evolutionary-mass-ac1 | own/tab | QZ Car: present mass of the single-star MIST model with the radius and Teff of Ac1, against its dynamical mass |

| Id | Code | Quantity |
|---|---|---|
| qzcar-evolutionary-mass-ac2 | own/tab | the same for Ac2 |
| qzcar-evolutionary-mass-aa1 | own/tab | the same for Aa1 |
| agb-final-mass-1p5 | own/tab | white-dwarf mass (core at the end of the AGB) of a 1.5 $M_\odot$ MIST star (solar [Fe/H]) against the white-dwarf initial-final mass relation |
| agb-final-mass-2p5 | own/tab | white-dwarf mass (core at the end of the AGB) of a 2.5 $M_\odot$ MIST star (solar [Fe/H]) against the white-dwarf initial-final mass relation |
| agb-final-mass-3p0 | own/tab | white-dwarf mass (core at the end of the AGB) of a 3.0 $M_\odot$ MIST star (solar [Fe/H]) against the white-dwarf initial-final mass relation |
| agb-final-mass-4p0 | own/tab | white-dwarf mass (core at the end of the AGB) of a 4.0 $M_\odot$ MIST star (solar [Fe/H]) against the white-dwarf initial-final mass relation |
| murphy2014-kic9651065-a1 | own | first orbital harmonic A1 of the time delays of KIC 9651065 (delta Sct in a 272.7-d eccentric orbit) from its orbit (a1 sin i = 0.37 au, e = 0.468, varpi = 2.01) |
| murphy2014-kic9651065-a2-a1 | own | ratio of the second to the first orbital harmonic of the time delays of KIC 9651065 (sets the eccentricity; e = 0.468, varpi = 2.01 rad) |
| delta-scuti-pl-1p8 | own/tab | M_V of a 1.8 $M_\odot$ MIST star at Teff = 7500 K on the main sequence minus M_V of the observed fundamental-mode delta Scuti P-L relation at its period from Q = 0.0329 d |
| delta-scuti-pl-2p0 | own/tab | as delta-scuti-pl-1p8 for a 2.0 $M_\odot$ star |
| cmdra-a-radius-excess-solar | own/tab | observed radius of CM Dra A over the MIST radius of its mass (4 Gyr, [Fe/H] = 0), minus 1 |
| cmdra-a-radius-excess-lowz | own/tab | the same for [Fe/H] = -0.3 (the metallicity of Terrien et al. 2012) |
| czev343-apsidal-period | R | CzeV343: observed apsidal period of binary A, full N-body quadruple [yr] |
| sse-sun-radius | own/tab | Sun: radius of a 1 $M_\odot$, Z = 0.02 SSE star at 4.57 Gyr |
| sse-sun-luminosity | own/tab | Sun: luminosity of a 1 $M_\odot$, Z = 0.02 SSE star at 4.57 Gyr |
| sse-sun-convection-zone | own/tab | Sun: base of the convective envelope r_bcz/R (BSE envelope model) |
| iw-q-solar | own | log10 Q' of inertial waves, 1 $M_\odot$ main-sequence star, P_rot = 10 d |
| iw-q-m-dwarf | own | log10 Q' of inertial waves, fully convective 0.3 $M_\odot$ star, P_rot = 10 d |
| tides-circularization-m67 | own | circularization period of solar-type binaries at 4 Gyr (M67), Hurley et al. (2002) tides (observed: 12.1 +1.0/−1.5 d, Meibom & Mathieu 2005, Table 3) |
| tides-circularization-m67-iw | own | circularization period at 4 Gyr (M67) with inertial waves, main sequence only (equilibrium tide only: 3.8 d; observed as above) |
| gw-hulse-taylor-pdot | own | orbital decay dP/dt of PSR B1913+16 by gravitational waves (masses 1.438 and 1.390 $M_\odot$, P = 0.323 d, e = 0.617) |
| epsaur-mid-eclipse | own | epsilon Aur: mid-eclipse of 2009-2011 from the spectroscopic orbit |
| epsaur-duration | own | epsilon Aur: duration of the eclipse, contacts I-IV |
| epsaur-depth | own | epsilon Aur: depth of the eclipse in V (opaque disc, 0.6 au thick) |
| hd98800-polar-disc | own | HD 98800 B: inclination to the sky of a polar disc (along the eccentricity vector of the B orbit) |
| eecep-disc-precession | own | precession period [yr] of the EE Cep disc from the torque of the B star on a rigid disc (illustrative disc of configs/catalog/ee_cep.yaml, r_out 0.8 au) |
| tides-circularization-pms-population | own | circularization period (final period, e < 0.01) of twin 1 $M_\odot$ binaries at 3 Myr (MIST pre-main sequence from 1 Myr, equilibrium tide with linear damping and inertial waves; observed pre-main-sequence binaries: 7.1 ± 1.2 d, Meibom & Mathieu 2005, Table 3) |
| tides-circularization-m67-final | own | circularization period (final period) of twin 1 $M_\odot$ binaries at 4 Gyr (MIST from 1 Myr, equilibrium tide with linear damping and inertial waves; observed M67: 12.1 +1.0/−1.5 d) |

**Table A8.** Level D: results (columns *Expected*, *Computed*, *Difference*, *Tolerance*, *Verdict*), in the same order as Table A7.

| Id | Expected | Computed | Difference | Tolerance | Verdict |
|---|---|---|---|---|---|
| v994s-dyn-A | 0.12 | 0.0999365 | −16.72 % | 25 % | PASS |
| v994s-dyn-B | 0.05 | 0.0519361 | +3.87 % | 25 % | PASS |
| v994s-dyn-C | 0 | $1.23366\cdot10^{-6}$ | $+1.23\cdot10^{-6}$ | 0.01 | PASS |
| v994s-gr-A | 0.1 | 0.0938886 | −6.11 % | 10 % | PASS |
| v994s-gr-B | 0.08 | 0.0748582 | −6.43 % | 10 % | PASS |
| v994s-gr-C | 0.06 | 0.0811201 | +35.20 % | 10 % | **EXPLAINED** |
| qzcar-evolutionary-mass-ac1 | 25.5 $M_\odot$ | 27.0258 $M_\odot$ | +5.98 % | 15 % | PASS |
| qzcar-evolutionary-mass-ac2 | 33.2 $M_\odot$ | 26.0243 $M_\odot$ | −21.61 % | 15 % | **EXPLAINED** |
| qzcar-evolutionary-mass-aa1 | 69.8 $M_\odot$ | 42.5172 $M_\odot$ | −39.09 % | 15 % | **EXPLAINED** |
| agb-final-mass-1p5 | 0.609 $M_\odot$ | 0.565426 $M_\odot$ | −0.0436 $M_\odot$ | 0.1 $M_\odot$ | PASS |
| agb-final-mass-2p5 | 0.689 $M_\odot$ | 0.610396 $M_\odot$ | −0.0786 $M_\odot$ | 0.1 $M_\odot$ | PASS |
| agb-final-mass-3p0 | 0.745 $M_\odot$ | 0.653775 $M_\odot$ | −0.0912 $M_\odot$ | 0.1 $M_\odot$ | PASS |
| agb-final-mass-4p0 | 0.899 $M_\odot$ | 0.835342 $M_\odot$ | −0.0637 $M_\odot$ | 0.1 $M_\odot$ | PASS |
| murphy2014-kic9651065-a1 | 165.57 s | 167.767 s | +1.33 % | 6 % | PASS |
| murphy2014-kic9651065-a2-a1 | 0.231986 | 0.219651 | −0.0123 | 0.025 | PASS |
| delta-scuti-pl-1p8 | 0 mag | −0.0251584 mag | −0.0252 mag | 0.25 mag | PASS |
| delta-scuti-pl-2p0 | 0 mag | −0.115268 mag | −0.115 mag | 0.25 mag | PASS |
| cmdra-a-radius-excess-solar | 0.035 | 0.0335989 | −0.0014 | 0.015 | PASS |
| cmdra-a-radius-excess-lowz | 0.06 | 0.0641756 | +0.00418 | 0.015 | PASS |
| czev343-apsidal-period | 33.4 yr | 34.491 yr | +3.27 % | 6.0 % | PASS |
| sse-sun-radius | 1 $R_\odot$ | 0.984843 $R_\odot$ | −1.52 % | 5 % | PASS |
| sse-sun-luminosity | 1 $L_\odot$ | 0.95786 $L_\odot$ | −4.21 % | 5 % | PASS |
| sse-sun-convection-zone | 0.713 | 0.733903 | +0.0209 | 0.03 | PASS |
| iw-q-solar | 7 | 7.0122 | +0.0122 | 0.3 | PASS |
| iw-q-m-dwarf | 7 | 6.94697 | −0.053 | 0.3 | PASS |
| tides-circularization-m67 | 12.1 d | 3.7847 d | −68.72 % | 30.0 % | **EXPLAINED** |
| tides-circularization-m67-iw | 12.1 d | 7.4572 d | −38.37 % | 30 % | **EXPLAINED** |
| gw-hulse-taylor-pdot | $-2.40263\cdot10^{-12}$ | $-2.4021\cdot10^{-12}$ | +0.02 % | 0.1 % | PASS |
| epsaur-mid-eclipse | $2.45543\cdot10^{6}$ JD | $2.45541\cdot10^{6}$ JD | −15 JD | 30 JD | PASS |
| epsaur-duration | 740 d | 638 d | −13.78 % | 15 % | PASS |
| epsaur-depth | 0.75 mag | 0.83134 mag | +0.0813 mag | 0.15 mag | PASS |
| hd98800-polar-disc | 26° | 28.4735° | +2.47° | 3° | PASS |
| eecep-disc-precession | 65 yr | 183.163 yr | +181.79 % | 15 % | **EXPLAINED** |
| tides-circularization-pms-population | 7.1 d | 6.5831 d | −0.517 d | 1.2 d | PASS |
| tides-circularization-m67-final | 12.1 d | 7.5337 d | −4.57 d | 1.2 d | **EXPLAINED** |

# References


- Akeson R. et al. (2021). Precision millimeter astrometry of the α Centauri AB system. *AJ* 162, 14.
- Anderson R. I., Saio H., Ekström S., Georgy C., Meynet G. (2016). On the effect of rotation on populations of classical Cepheids II. Pulsation analysis for metallicities 0.014, 0.006, and 0.002. *A&A* 591, A8. doi:10.1051/0004-6361/201528031
- Antognini J. M. O., Thompson T. A. (2016). Dynamical formation and scattering of hierarchical triples: cross-sections, Kozai–Lidov oscillations, and collisions. *MNRAS* 456, 4219. doi:10.1093/mnras/stv2938
- Barker A. J. (2020). Tidal dissipation in evolving low-mass and solar-type stars with predictions for planetary orbital decay. *MNRAS* 498, 2270. arXiv:2008.03262
- Barker A. J. (2022). Tidal dissipation due to inertial waves can explain the circularization periods of solar-type binaries. *ApJL* 927, L36. doi:10.3847/2041-8213/ac5b63
- Baron F. et al. (2012). Imaging the Algol triple system in the H band with the CHARA interferometer. *ApJ* 752, 20. doi:10.1088/0004-637X/752/1/20
- Blaauw A. (1961). On the origin of the O- and B-type stars with high velocities (the "run-away" stars), and some related problems. *BAN* 15, 265.
- Bono G., Castellani V., Marconi M. (2000). Classical Cepheid pulsation models. III. The predictable scenario. *ApJ* 529, 293. arXiv:astro-ph/9908014
- Borkovits T. et al. (2013). HD 181068: a red giant in a triply eclipsing compact hierarchical triple system. *MNRAS* 428, 1656. doi:10.1093/mnras/sts146
- Borkovits T., Rappaport S. A., Chen H.-L. et al. (2026). Discovery of the most compact 3+1-type quadruple star system TIC 120362137. *Nat. Commun.* 17, 1859. doi:10.1038/s41467-026-69223-4
- Breger M. (1979). Delta Scuti and related stars. *PASP* 91, 5. doi:10.1086/130433
- Breiter S., Vokrouhlický D. (2018). Resonant dynamics of gravitationally bound pair of binaries: the case of 1:1 resonance. *MNRAS* 475, 5215. doi:10.1093/mnras/sty132
- Breivik K. et al. (2020). COSMIC variance in binary population synthesis. *ApJ* 898, 71. doi:10.3847/1538-4357/ab9d85
- Brooker R. A., Olle T. W. (1955). Apsidal-motion constants for polytropic models. *MNRAS* 115, 101. doi:10.1093/mnras/115.1.101
- Brož M. et al. (2022). Towards a consistent model of the hot quadruple system HD 93206 = QZ Carinæ. II. N-body model. *A&A* 666, A24. arXiv:2206.12160
- Choi J., Dotter A., Conroy C., Cantiello M., Paxton B., Johnson B. D. (2016). Mesa Isochrones and Stellar Tracks (MIST). I. Solar-scaled models. *ApJ* 823, 102. doi:10.3847/0004-637X/823/2/102
- Christensen-Dalsgaard J., Gough D. O., Thompson M. J. (1991). The depth of the solar convection zone. *ApJ* 378, 413. doi:10.1086/170441
- Cincotta P. M., Simó C. (2000). Simple tools to study global dynamics in non-axisymmetric galactic potentials – I. *A&AS* 147, 205. doi:10.1051/aas:2000108
- Claret A. (1999). *A&A* 350, 56.
- Claret A. (2000). A new non-linear limb-darkening law for LTE stellar atmosphere models. *A&A* 363, 1081.
- Claret A. (2004). New grids of stellar models including tidal-evolution constants up to carbon burning. I. From 0.8 to 125 $M_{\odot}$ at Z = 0.02. *A&A* 424, 919. doi:10.1051/0004-6361:20040470
- Claret A. (2023). Theoretical tidal evolution constants for stellar models from the pre-main sequence to the white dwarf stage. *A&A* 674, A67. doi:10.1051/0004-

6361/202346250
- Claret A., Bloemen S. (2011). Gravity and limb-darkening coefficients for the Kepler, CoRoT, Spitzer, uvby, UBVRIJHK, and Sloan photometric systems. *A&A* 529, A75. doi:10.1051/0004-6361/201116451
- Cummings J. D., Kalirai J. S., Tremblay P.-E., Ramirez-Ruiz E., Choi J. (2018). The white dwarf initial–final mass relation for progenitor stars from 0.85 to 7.5 $M_\odot$. *ApJ* 866, 21. arXiv:1809.01673
- de Vries N., Portegies Zwart S., Figueira J. (2014). The evolution of triples with a Roche lobe filling outer star. *MNRAS* 438, 1909. arXiv:1309.1475
- Dotter A. (2016). MESA Isochrones and Stellar Tracks (MIST) 0: methods for the construction of stellar isochrones. *ApJS* 222, 8. doi:10.3847/0067-0049/222/1/8
- Eggleton P. P. (1983). Approximations to the radii of Roche lobes. *ApJ* 268, 368. doi:10.1086/160960
- Eggleton P. P., Kiseleva L. G., Hut P. (1998). The equilibrium tide model for tidal friction. *ApJ* 499, 853. doi:10.1086/305670
- Eisner N. L. et al. (2022). Planet Hunters TESS IV: a massive, compact hierarchical triple star system TIC 470710327. *MNRAS* 511, 4710. doi:10.1093/mnras/stab3619
- Ekström S. et al. (2012). Grids of stellar models with rotation. I. Models from 0.8 to 120 $M_\odot$ at solar metallicity (Z = 0.014). *A&A* 537, A146. doi:10.1051/0004-6361/201117751
- Evans N. R. et al. (2024). The orbit and dynamical mass of Polaris: observations with the CHARA Array. *ApJ* 971, 190. doi:10.3847/1538-4357/ad5e7a
- Fabrycky D., Tremaine S. (2007). Shrinking binary and planetary orbits by Kozai cycles with tidal friction. *ApJ* 669, 1298. arXiv:0705.4285
- Fang X., Thompson T. A., Hirata C. M. (2018). Dynamics of quadruple systems composed of two binaries: stars, white dwarfs, and implications for Ia supernovae. *MNRAS* 476, 4234. doi:10.1093/mnras/sty472
- Farago F., Laskar J. (2010). High-inclination orbits in the secular quadrupolar three-body problem. *MNRAS* 401, 1189. doi:10.1111/j.1365-2966.2009.15711.x
- Feiden G. A., Chaboyer B. (2014). Magnetic inhibition of convection and the fundamental properties of low-mass stars. II. Fully convective main-sequence stars. *ApJ* 789, 53. arXiv:1405.1767
- Gajdoš P., Parimucha Š. (2019). OCFit (software for fitting O−C diagrams). *Open Eur. J. Var. Stars* 197, 71.
- Gałan C. et al. (2012). *A&A* 544, A53 (EE Cep).
- Gazeas K., Stępień K. (2008). Angular momentum and mass evolution of contact binaries. *MNRAS* 390, 1577. doi:10.1111/j.1365-2966.2008.13844.x
- Giménez A., Bastero M. (1995). An analytical formula for the times of minima of eccentric eclipsing binaries. *Ap&SS* 226, 99. doi:10.1007/BF00626903
- Grishin E., Perets H. B. (2022). Chaotic dynamics of wide triples induced by galactic tides: a novel channel for producing compact binaries, mergers, and collisions. *MNRAS* 512, 4993. doi:10.1093/mnras/stac706
- Hadjidemetriou J. D. (1963). Two-body problem with variable mass: a new approach. *Icarus* 2, 440.
- Hamers A. S. et al. (2015). Secular dynamics of hierarchical quadruple systems: the case of a triple system orbited by a fourth body. *MNRAS* 449, 4221.
- Hamers A. S., Portegies Zwart S. F. (2016). Secular dynamics of hierarchical multiple systems composed of nested binaries (SecularMultiple). *MNRAS* 459, 2827.
- Hamers A. S., Lai D. (2017). Secular chaotic dynamics in hierarchical quadruple systems, with applications to hot Jupiters in stellar binaries and triples. *MNRAS* 470, 1657. doi:10.1093/mnras/stx1319
- Hamers A. S., Samsing J. (2019). Analytic computation of the secular effects of encounters on a binary: features arising from second-order perturbation theory. *MNRAS* 487, 5630. arXiv:1904.09624
- Hamers A. S. et al. (2021). Multiple Stellar Evolution (MSE). *MNRAS* 502, 4479. doi:10.1093/mnras/stab287
- Hamers A. S., Glanz H., Neunteufel P. (2022). A statistical view of the stable and unstable Roche lobe overflow of a tertiary star onto the inner binary in triple systems. *ApJS* 259, 25. doi:10.3847/1538-4365/ac49e7
- Hayashi T., Trani A. A., Suto Y. (2022). Dynamical disruption timescales and chaotic

behavior of hierarchical triple systems. *ApJ* 939, 81. doi:10.3847/1538-4357/ac8f48
- Heisler J., Tremaine S. (1986). The influence of the Galactic tidal field on the Oort comet cloud. *Icarus* 65, 13.
- Hełminiak K. G. et al. (2017). *A&A* 602, A30. doi:10.1051/0004-6361/201630379
- Hilditch R. W. (2001). *An Introduction to Close Binary Stars*. Cambridge University Press.
- Hills J. G. (1983). The effects of sudden mass loss and a random kick velocity produced in a supernova explosion on the dynamics of a binary star of arbitrary orbital eccentricity. *ApJ* 267, 322.
- Hoard D. W., Howell S. B., Stencel R. E. (2010). Taming the invisible monster: system parameter constraints for ε Aurigae from the far-ultraviolet to the mid-infrared. *ApJ* 714, 549. doi:10.1088/0004-637X/714/1/549
- Holman M. J., Wiegert P. A. (1999). Long-term stability of planets in binary systems. *AJ* 117, 621.
- Hurley J. R., Pols O. R., Tout C. A. (2000). Comprehensive analytic formulae for stellar evolution as a function of mass and metallicity. *MNRAS* 315, 543. doi:10.1046/j.1365-8711.2000.03426.x
- Hurley J. R., Tout C. A., Pols O. R. (2002). Evolution of binary stars and the effect of tides on binary populations. *MNRAS* 329, 897. doi:10.1046/j.1365-8711.2002.05038.x
- Hut P. (1981). Tidal evolution in close binary systems. *A&A* 99, 126.
- Irwin J. B. (1952). The determination of a light-time orbit. *ApJ* 116, 211. doi:10.1086/145604
- Jeans J. H. (1924). Cosmogonic problems associated with a secular decrease of mass. *MNRAS* 85, 2.
- Kaplan D. L. (2010). Mass constraints from eclipse timing in double white dwarf binaries. *ApJ* 717, L108. doi:10.1088/2041-8205/717/2/L108
- Kemp A. et al. (2024). *A&A* 689, A164. doi:10.1051/0004-6361/202450390
- Kennedy G. M. et al. (2019). A circumbinary protoplanetary disk in a polar configuration. *Nat. Astron.* 3, 230. doi:10.1038/s41550-018-0667-x
- Keppler M. et al. (2020). Gap, shadows, spirals, and streamers: SPHERE observations of binary-disk interactions in GG Tauri A. *A&A* 639, A62. doi:10.1051/0004-6361/202038032
- Kervella P., Thévenin F., Lovis C. (2017). Proxima's orbit around α Centauri. *A&A* 598, L7.
- Kish L. (1965). *Survey Sampling*. Wiley, New York.
- Kloppenborg B. K. et al. (2015). Interferometry of ε Aurigae: characterization of the asymmetric eclipsing disk. *ApJS* 220, 14. arXiv:1508.01909
- Koenigsberger G. et al. (2014). *AJ* 148, 62. doi:10.1088/0004-6256/148/4/62
- Köhler R. (2011). The orbit of GG Tauri A. *A&A* 530, A126. doi:10.1051/0004-6361/201016327
- Kolář J. et al. (2025). Period ratios and observation of noticeable resonance at 3:2 for 2+2 quadruple systems. *MNRAS* 538, 1160. doi:10.1093/mnras/staf364
- Kostov V. B. et al. (2021). TIC 454140642: a compact, coplanar, quadruple-lined quadruple star system consisting of two eclipsing binaries. *ApJ* 917, 93. doi:10.3847/1538-4357/ac04ad
- Kozai Y. (1962). Secular perturbations of asteroids with high inclination and eccentricity. *AJ* 67, 591. doi:10.1086/108790
- Kreidberg L. (2015). batman: BAsic Transit Model cAlculatioN in Python. *PASP* 127, 1161. doi:10.1086/683602
- Kwee K. K. (1958). Investigations of variations in the period of sixteen bright short-periodic eclipsing binary stars. *BAN* 14, 131.
- Leconte J., Chabrier G., Baraffe I., Levrard B. (2010). Is tidal heating sufficient to explain bloated exoplanets? Consistent calculations accounting for finite initial eccentricity. *A&A* 516, A64. doi:10.1051/0004-6361/201014337
- Lidov M. L. (1962). The evolution of orbits of artificial satellites of planets under the action of gravitational perturbations of external bodies. *Planet. Space Sci.* 9, 719. doi:10.1016/0032-0633(62)90129-0
- Liu B., Muñoz D. J., Lai D. (2015). Suppression of extreme orbital evolution in triple systems with short-range forces. *MNRAS* 447, 747.
- Lu T., Rein H., Tamayo D., Hadden S., Mardling R., Millholland S. C., Laughlin G.

(2023). Self-consistent spin, tidal, and dynamical equations of motion in the REBOUNDx framework. *ApJ* 948, 41. arXiv:2303.00006
- Luo L., Katz B., Dong S. (2016). Double-averaging can fail to characterize the long-term evolution of Lidov–Kozai cycles and derivation of an analytical correction. *MNRAS* 458, 3060.
- Mahy L. et al. (2018). The triple system HD 150136: from periastron passage to actual masses. *A&A* 616, A75. doi:10.1051/0004-6361/201832810
- Mandel K., Agol E. (2002). Analytic light curves for planetary transit searches. *ApJ* 580, L171. doi:10.1086/345520
- Mardling R. A., Aarseth S. J. (2001). Tidal interactions in star cluster simulations. *MNRAS* 321, 398. doi:10.1046/j.1365-8711.2001.03974.x
- Martin R. G., Lubow S. H. (2017). Polar alignment of a protoplanetary disk around an eccentric binary. *ApJL* 835, L28. doi:10.3847/2041-8213/835/2/L28
- Matt S. P. et al. (2015). The mass-dependence of angular momentum evolution in Sun-like stars. *ApJL* 799, L23. arXiv:1412.4786
- Maxted P. F. L. (2016). ellc: a fast, flexible light curve model for detached eclipsing binary stars and transiting exoplanets. *A&A* 591, A111. doi:10.1051/0004-6361/201628579
- Mazeh T. (2008). Observational evidence for tidal interaction in close binary systems. *EAS Publ. Ser.* 29, 1. arXiv:0801.0134
- McLaughlin D. B. (1924). Some results of a spectrographic study of the Algol system. *ApJ* 60, 22. doi:10.1086/142826
- McMillan S. L. W., McDermott P. N., Taam R. E. (1987). The formation and evolution of tidal binary systems. *ApJ* 318, 261.
- Meibom S., Mathieu R. D. (2005). A robust measure of tidal circularization in coeval binary populations: the solar-type spectroscopic binary population in the open cluster M35. *ApJ* 620, 970. doi:10.1086/427082
- Morris S. L. (1985). The ellipsoidal variable stars. *ApJ* 295, 143. doi:10.1086/163359
- Murphy S. J., Bedding T. R., Shibahashi H., Kurtz D. W., Kjeldsen H. (2014). Finding binaries among Kepler pulsating stars from phase modulation of their pulsations. *MNRAS* 441, 2515. arXiv:1404.5649
- Mushkin J., Katz B. (2020). Random-walk model of the energy exchange and disruption of unstable hierarchical triples. *MNRAS* 498, 665.
- Naoz S. (2016). The eccentric Kozai–Lidov effect and its applications. *ARA&A* 54, 441. doi:10.1146/annurev-astro-081915-023315
- Naoz S., Farr W. M., Lithwick Y., Rasio F. A., Teyssandier J. (2013). Secular dynamics in hierarchical three-body systems. *MNRAS* 431, 2155. arXiv:1107.2414
- Nemravová J. A. et al. (2016). ξ Tauri: a unique laboratory to study the dynamic interaction in a compact hierarchical quadruple system. *A&A* 594, A55. doi:10.1051/0004-6361/201628860
- Ogilvie G. I. (2013). Tides in rotating barotropic fluid bodies: the contribution of inertial waves and the role of internal structure. *MNRAS* 429, 613. doi:10.1093/mnras/sts362
- Oplištilová A. et al. (2023). *A&A* 672, A31. doi:10.1051/0004-6361/202245272
- Pablo H. et al. (2015). The most massive heartbeat: an in-depth analysis of ι Orionis. *ApJ* 809, 134.
- Papaloizou J. C. B., Larwood J. D. (2000). On the orbital evolution and growth of protoplanets embedded in a gaseous disc. *MNRAS* 315, 823. doi:10.1046/j.1365-8711.2000.03466.x
- Pejcha O., Antognini J. M., Shappee B. J., Thompson T. A. (2013). Greatly enhanced eccentricity oscillations in quadruple systems composed of two binaries. *MNRAS* 435, 943. doi:10.1093/mnras/stt1281
- Pejcha O., Cagaš P., Landri C., Fausnaugh M. M., De Rosa G., Prieto J. L., Henzl Z., Pešta M. (2022). The complex dynamical past and future of double eclipsing binary CzeV343: misaligned orbits and period resonance. *A&A* 667, A53. doi:10.1051/0004-6361/202244335
- Perets H. B., Kratter K. M. (2012). The triple evolution dynamical instability: stellar collisions in the field and the formation of exotic binaries. *ApJ* 760, 99. doi:10.1088/0004-637X/760/2/99
- Peters P. C. (1964). Gravitational radiation and the motion of two point masses. *Phys.*

*Rev.* 136, B1224. doi:10.1103/PhysRev.136.B1224
- Pieńkowski D. et al. (2020). International observational campaign of the 2014 eclipse of EE Cephei. *A&A* 639, A23. doi:10.1051/0004-6361/201937181
- Poro A. et al. (2024). Updated period–luminosity relations of δ Scuti stars (2375 Galactic stars, Gaia DR3 and TESS). arXiv:2401.01091
- Portegies Zwart S. F., Meinen A. T. (1993). Quick method for calculating energy dissipation in tidal interaction. *A&A* 280, 174.
- Postnov K. A., Yungelson L. R. (2014). The evolution of compact binary star systems. *Living Rev. Relativ.* 17, 3. arXiv:1403.4754
- Powell B. P. et al. (2021). TIC 168789840: a sextuply eclipsing sextuple star system. *AJ* 161, 162.
- Pribulla T. et al. (2020). *MNRAS* 494, 178. doi:10.1093/mnras/staa699
- Pribulla T., Borkovits T., Jayaraman R., Rappaport S., Mitnyan T., Zasche P. et al. (2023). BU Canis Minoris – the most compact known flat doubly eclipsing quadruple system. *MNRAS* 524, 4220. doi:10.1093/mnras/stad2015
- Prša A. et al. (2016). Physics of Eclipsing Binaries. II. Toward the increased model fidelity. *ApJS* 227, 29. doi:10.3847/0067-0049/227/2/29
- Quarles B., Satyal S., Kostov V., Kaib N., Haghighipour N. (2018). Stability limits of circumbinary planets: is there a pile-up in the Kepler CBPs? *ApJ* 856, 150.
- Ransom S. M. et al. (2014). A millisecond pulsar in a stellar triple system. *Nature* 505, 520. doi:10.1038/nature12917
- Rasio F. A. (1995). The minimum mass ratio of W Ursae Majoris binaries. *ApJ* 444, L41. doi:10.1086/187855
- Ray A., Kembhavi A. K., Antia H. M. (1987). Tidal capture in globular clusters. *A&A* 184, 164.
- Reeve D. C., Howarth I. D. (2016). Non-LTE limb-darkening coefficients for hot stars. *MNRAS* 456, 1294.
- Rein H., Liu S.-F. (2012). REBOUND: an open-source multi-purpose N-body code for collisional dynamics. *A&A* 537, A128. doi:10.1051/0004-6361/201118085
- Rein H., Spiegel D. S. (2015). IAS15: a fast, adaptive, high-order integrator for gravitational dynamics. *MNRAS* 446, 1424. doi:10.1093/mnras/stu2164
- Rossiter R. A. (1924). On the detection of an effect of rotation during eclipse in the velocity of the brighter component of beta Lyrae, and on the constancy of velocity of this system. *ApJ* 60, 15. doi:10.1086/142825
- Schaefer G. H. et al. (2016). Orbits, distance, and stellar masses of the massive triple star σ Orionis. *AJ* 152, 213. doi:10.3847/0004-6256/152/6/213
- Shibahashi H., Kurtz D. W. (2012). FM stars: a Fourier view of pulsating binary stars. *MNRAS* 422, 738.
- Soberman G. E., Phinney E. S., van den Heuvel E. P. J. (1997). Stability criteria for mass transfer in binary stellar evolution. *A&A* 327, 620. arXiv:astro-ph/9703016
- Söderhjelm S. (1975). The three-body problem and eclipsing binaries. *A&A* 42, 229.
- Stefanik R. P. et al. (2010). Epsilon Aurigae: an improved spectroscopic orbital solution. *AJ* 139, 1254. doi:10.1088/0004-6256/139/4/1254
- Stencel R. E. (2013). Results of the recent ε Aurigae eclipse campaign. arXiv:1303.7128
- Skumanich A. (1972). Time scales for Ca II emission decay, rotational braking, and lithium depletion. *ApJ* 171, 565.
- Sterne T. E. (1939). Apsidal motion in binary stars. *MNRAS* 99, 451. doi:10.1093/mnras/99.5.451
- Stone N. C., Leigh N. W. C. (2019). A statistical solution to the chaotic, non-hierarchical three-body problem. *Nature* 576, 406. doi:10.1038/s41586-019-1833-8
- Tamayo D., Rein H., Shi P., Hernandez D. M. (2020). REBOUNDx: a library for adding conservative and dissipative forces to otherwise symplectic N-body integrations. *MNRAS* 491, 2885.
- Tango W. J. et al. (2006). Orbital elements, masses and distance of λ Scorpii A and B determined with the Sydney University Stellar Interferometer and high-resolution spectroscopy. *MNRAS* 370, 884. arXiv:astro-ph/0605311
- Tejero-Cantero A. et al. (2020). sbi: a toolkit for simulation-based inference. *J. Open Source Softw.* 5, 2505. doi:10.21105/joss.02505
- Terrien R. C. et al. (2012). An H-band spectroscopic metallicity calibration for M dwarfs. *ApJL* 747, L38.

- Tokovinin A. (2018). The updated Multiple Star Catalog. *ApJS* 235, 6. doi:10.3847/1538-4365/aaa1a5
- Tokovinin A., Moe M. (2020). Formation of close binaries by disc fragmentation and migration, and its statistical modelling. *MNRAS* 491, 5158.
- Toonen S., Boekholt T. C. N., Portegies Zwart S. (2022). Stellar triples on the edge. Comprehensive overview of the evolution of destabilised triples leading to stellar and binary exotica. *A&A* 661, A61. doi:10.1051/0004-6361/202141991
- Tian Y. P., Xiang F. Y., Tao X. (2009). *Ap&SS* 319, 119 (thin-shell model of the Applegate mechanism).
- Tout C. A., Pols O. R., Eggleton P. P., Han Z. (1996). Zero-age main-sequence radii and luminosities as analytic functions of mass and metallicity. *MNRAS* 281, 257. doi:10.1093/mnras/281.1.257
- Tout C. A., Aarseth S. J., Pols O. R., Eggleton P. P. (1997). Rapid binary star evolution for N-body simulations and population synthesis. *MNRAS* 291, 732.
- Uytterhoeven K. et al. (2004). Interpretation of the variability of the β Cephei star λ Scorpii. I. The multiple character. *A&A* 427, 581. doi:10.1051/0004-6361:20041223
- Veras D., Wyatt M. C., Mustill A. J., Bonsor A., Eldridge J. J. (2011). The great escape: how exoplanets and smaller bodies desert dying stars. *MNRAS* 417, 2104. arXiv:1107.1239
- Vink J. S., de Koter A., Lamers H. J. G. L. M. (2001). Mass-loss predictions for O and B stars as a function of metallicity. *A&A* 369, 574. arXiv:astro-ph/0101509
- Vokrouhlický D. (2016). Secular dynamics of gravitationally bound pair of binaries. *MNRAS* 461, 3964. doi:10.1093/mnras/stw1596
- Völschow M. et al. (2016). Eclipsing time variations in close binary systems: planetary hypothesis vs. Applegate mechanism. *A&A* 587, A34. doi:10.1051/0004-6361/201527333
- Vynatheya P., Hamers A. S., Mardling R. A., Bellinger E. P. (2022). Algebraic and machine learning approach to hierarchical triple-star stability. *MNRAS* 516, 4146. arXiv:2207.03151
- Vynatheya P., Mardling R. A., Hamers A. S. (2023). Quadruple-star systems are not always nested triples: a machine learning approach to dynamical stability. *MNRAS* 525, 2388. arXiv:2301.09930
- Weisberg J. M., Huang Y. (2016). Relativistic measurements from timing the binary pulsar PSR B1913+16. *ApJ* 829, 55. doi:10.3847/0004-637X/829/1/55
- Will C. M. (2014). The confrontation between general relativity and experiment. *Living Rev. Relativ.* 17, 4. doi:10.12942/lrr-2014-4
- Winn J. N. et al. (2006). The orbit and occultations of KH 15D. *ApJ* 644, 510. arXiv:astro-ph/0602352
- Zahn J.-P., Bouchet L. (1989). Tidal evolution of close binary stars. II. Orbital circularization of late-type binaries. *A&A* 223, 112.
- Zanazzi J. J., Lai D. (2018). Inclination evolution of protoplanetary disks around eccentric binaries. *MNRAS* 473, 603. doi:10.1093/mnras/stx2375
- Zanazzi J. J., Wu Y. (2021). Tidal circularization of binaries by resonance locking I: the importance of the pre-main-sequence. *AJ* 161, 263. arXiv:2102.05668
- Zasche P. et al. (2023). V994 Herculis: a unique triply eclipsing sextuple star system. *MNRAS* 520, 3127. doi:10.1093/mnras/stad328
- Zúñiga-Fernández S. et al. (2026). *A&A* 712, L2. doi:10.1051/0004-6361/202661407